The minimum number of proper circles determined by a planar point set

Anonymous review manuscript

5 August 2026

Abstract

For a finite set $P \subset \mathbb{R}^2$, let $C(P)$ be the number of distinct proper Euclidean circles containing at least three points of P. We determine the minimum of $C(P)$ over all n-point sets which are neither collinear nor concyclic. The three exceptional values are

$$f(6) = 8, \qquad f(7) = 11, \qquad f(8) = 17,$$

whereas

$$f(n) = 1 + \binom{n-1}{2} - \left\lfloor \frac{n-1}{2} \right\rfloor$$

for $n = 4, 5$ and for every $n \geq 9$.

The proof is organized around unique ownership of triples by maximal line and circle blocks. Inversion converts the local block structure into real line arrangements, allowing Melchior-type inequalities and their weighted consequences. A uniform half-size cap and one nonnegative-slack identity settle all $n \geq 15$, with a separate two-case argument at $n = 14$. The finite window is reduced to named real-projective rigidity lemmas and small displayed certificates. The proof uses no computerized search through point configurations, block supports, or symmetry orbits.

1 Introduction

Erdős asked for the least number of circles determined by n planar points not all on a circle [Erd61]. If collinear sets are admitted, the minimum is trivially zero, so one must specify a nondegeneracy convention. We follow Elliott [Ell67]: the points are required to be neither all collinear nor all concyclic. A determined circle is counted once, independently of how many selected points it contains. This differs from the ordinary-circle problem, in which exactly three selected points are required [LMM+18].

Purdy and Smith corrected the previously claimed large-n formula and identified the natural extremal construction consisting of a circle and its centre [PS10]. The present argument proves the exact minimum at every finite value under Elliott's convention.

Theorem 1.1. *For every integer $n \geq 4$,*

$$(f(4), f(5), f(6), f(7), f(8)) = (3, 5, 8, 11, 17),$$

and

$$\boxed{f(n) = 1 + \binom{n-1}{2} - \left\lfloor \frac{n-1}{2} \right\rfloor} \qquad (n \geq 9).$$

Architecture of the proof

Every selected triple is contained in a unique maximal carrier: its line when it is collinear, and its proper circle otherwise. This elementary partition is the common source of all counting identities in the paper. Inverting about a selected point turns the carriers through that point into spanned lines. Melchior's inequality, applied before and after restoring the inversion centre, then yields local slack variables and two global weighted inequalities.

The large range is conceptually the shortest part. A rich-carrier pencil first bounds every block by $\lfloor n/2 \rfloor$. A single functional S_M, whose coefficients are visibly nonnegative at that cap, gives the target for every $n \geq 15$. At $n = 14$, the same functional handles the absence of a seven-point circle and one selected-circle inequality handles its presence.

For $4 \leq n \leq 13$, the universal identities leave a finite number of equality mechanisms. We isolate their geometry as reusable lemmas: the real Fano and Orchard walls, pentagon and golden-axis rigidity, circle-lift trace and conic-event bounds, and the grid and gallery obstructions. The remaining arithmetic consists of fixed coefficient identities or small structural certificates displayed in the text.

Classical and computational boundaries

The external incidence inputs are stated with their exact hypotheses at first use. They are Melchior's inequality [Mel40], the non-strict Kelly–Moser ordinary-line estimate [KM58], Langer's inequality in the form recorded by de Zeeuw [Lan03, dZ18], Lenchner's affine ordinary-point theorem [Len07], and Ungar's direction theorem [Ung82]. Inversion, the circle lift, polarity, and the local conic arguments are proved within the paper rather than treated as black boxes.

Exact-arithmetic checks used during preparation replay displayed integer identities, rational coefficients, and fixed determinants. They do not discover cases and do not supply a geometric premise. Two small finite residues remain explicit: a nine-row coefficient table in the ten-point six-circle case, and a four-state corner propagation in the twelve-line gallery. These are hand-checkable certificates, not enumerations of configurations or realizations.

Organization

Section 2 develops the block calculus, records the classical inputs and upper constructions, and proves the cases $4 \leq n \leq 9$. Section 3 proves the three most geometric finite cases $n = 10, 11, 12$. Section 4 treats $n = 13$, the two-case endpoint $n = 14$, and the uniform range $n \geq 15$, and records the fixed-certificate boundary.

2 Foundations and the cases $4 \leq n \leq 9$

This section fixes the conventions used throughout the paper, derives the universal counting rows, and proves all values up to nine. A line is used as a bookkeeping carrier, but is never counted as a circle.

2.1 The problem and the block language

Definition 2.1 (The V1 problem)**.** Let $P \subset \mathbb{R}^2$ be a set of $n \geq 4$ distinct points. We call P *admissible* if it is neither collinear nor contained in one proper Euclidean circle. A *determined circle* is a proper circle containing a noncollinear triple of P. Write $C(P)$ for the number of distinct determined circles and

$$f(n) := \min\{C(P) : |P| = n \text{ and } P \text{ is admissible}\}.$$

For a line or a proper circle X, its *full trace* is $P \cap X$. A maximal generalized s-block is a full trace of size $s \geq 3$. We write

$$L_s := \#\{s\text{-point line blocks}\}, \qquad C_s := \#\{s\text{-point circle blocks}\}, \qquad B_s := L_s + C_s.$$

Thus

$$C(P) = \sum_{s \geq 3} C_s. \tag{1}$$

For $p \in P$, let $\ell_s(p)$, $c_s(p)$, and $d_s(p) = \ell_s(p) + c_s(p)$ count, respectively, the line, circle, and all s-blocks through p.

Lemma 2.2 (Unique ownership)**.** *Every triple of P belongs to exactly one maximal generalized block. Consequently*

$$\mathsf{T} := \sum_{s \geq 3} \binom{s}{3} B_s = \binom{n}{3}, \tag{2}$$

$$\sum_{p \in P} d_s(p) = sB_s, \qquad\qquad \sum_{p \in P} \ell_s(p) = sL_s. \tag{3}$$

If L_2 denotes the number of ordinary, two-point full line traces, then

$$L_2 + \sum_{s \geq 3} \binom{s}{2} L_s = \binom{n}{2}. \tag{4}$$

Proof. A collinear triple has its unique carrier line; a noncollinear triple has its unique circumcircle. Passing to the full trace gives a unique maximal owner. Equation (2) counts triples by that owner, and (3) counts point–block incidences. Finally, every pair of distinct points has one carrier line, which proves (4). □

Lemma 2.3 (The eight-point Orchard bound)**.** *Let Q be a set of eight noncollinear points of the real projective plane, with no four collinear. If t_i denotes the number of lines containing exactly i points of Q, then*

$$t_3 \leq 7.$$

Proof. Pair counting gives $t_2 + 3t_3 = 28$. Melchior's inequality excludes $t_3 \geq 9$, so it remains only to rule out $t_3 = 8$. In that case $t_2 = 4$. At a point $v \in Q$, let r_v and s_v be the numbers of three-point and ordinary lines through v. Then

$$7 = 2r_v + s_v.$$

Thus s_v is a positive odd integer, while $\sum_v s_v = 2t_2 = 8$. Consequently $s_v = 1$ and $r_v = 3$ for every v. The four ordinary lines are therefore a perfect matching; denote its pairs by

$$A = \{A_0, A_1\}, \quad B = \{B_0, B_1\}, \quad C = \{C_0, C_1\}, \quad D = \{D_0, D_1\}.$$

Every three-point line uses three different pairs. Since the eight such lines account for all 24 cross-pair incidences, exactly two omit each of A, B, C, D. Explicitly, if n_A, n_B, n_C, n_D are the four omission counts, then the four edges between, say, A and B give $n_C + n_D = 4$; the six analogous equations force $n_A = n_B = n_C = n_D = 2$.

We first determine the resulting incidence structure. Two triples on the same three pairs cannot differ in only one entry, because then their two common points would determine both lines. They also cannot differ in exactly two entries. Indeed, after relabelling suppose the two ABC triples are

$$A_0B_0C_0, \qquad A_1B_1C_0.$$

The unused cross edges force the remaining triples to have the form

$$\begin{array}{ll} A_0B_1D_x, & A_1B_0D_y, \\ A_0C_1D_z, & A_1C_1D_{1-z}, \\ B_0C_1D_w, & B_1C_1D_{1-w}. \end{array}$$

The AD, BD, and CD edges successively give $y = z = 1 - x$ and $w = x$, after which one of the two C_1D_i edges is repeated. Hence the two triples on any fixed three pairs are complementary. After fixing the ABC pair as $000, 111$, the unused AB, AC, BC edges are the three cross-matchings. A flip of the D-labels fixes one ABD triple, and the unused AD and BD edges then force all remaining bits. Up to relabelling, the unique incidence list is

$$\begin{array}{llll} A_0B_0C_0, & A_1B_1C_1, & A_0B_1D_0, & A_1B_0D_1, \\ A_0C_1D_1, & A_1C_0D_0, & B_0C_1D_0, & B_1C_0D_1. \end{array} \tag{5}$$

Choose the projective frame

$$A_0 = (1, 0, 0), \quad A_1 = (0, 1, 0), \quad B_0 = (0, 0, 1), \quad B_1 = (1, 1, 1).$$

The first four incidences in (5) allow the normalization

$$C_0 = (1, 0, u), \quad C_1 = (1, v, 1), \quad D_0 = (w, 1, 1), \quad D_1 = (0, 1, z),$$

where the relevant parameters are nonzero. The next incidences give

$$vz = uw = vw = 1.$$

Thus $u = v$ and $w = z = 1/u$. The last incidence, $B_1C_0D_1$, now gives

$$u^2 - u + 1 = 0,$$

which has no real root. Hence $t_3 = 8$ is impossible. □

Lemma 2.4 (Radical-axis cross-block bound)**.** *Let A and B be marked points on two distinct real circles, after deleting any marked point common to the circles. A* cross-block *is a line or circle containing two points of A and two points of B. If $(|A|, |B|) = (5, 4)$, there are at most 12 cross-blocks. If $(|A|, |B|) = (4, 4)$, there are at most 10.*

Proof. For each chord of the first circle and each chord of the second, the four endpoints are concyclic precisely when the two supporting lines meet on the radical axis. A coincident pair of supporting lines gives a line cross-block and is assigned to its intersection with the same radical axis. (The common supporting line cannot itself be the radical axis, because the marked common points were deleted.)

At a point q of the radical axis, let a_q and b_q be the numbers of selected chords from A and B through q. Chords through a fixed external point form a matching, so

$$a_q \leq \lfloor |A|/2 \rfloor, \qquad b_q \leq \lfloor |B|/2 \rfloor.$$

Moreover, $\sum_q a_q = \binom{|A|}{2}$, $\sum_q b_q = \binom{|B|}{2}$, and the number of cross-blocks is

$$M = \sum_q a_q b_q.$$

For $5+4$, this gives $M \le 2\sum_q b_q = 12$.

For $4+4$, both multiplicity sums are 6, and $M \le 12$. Equality $M = 11$ is impossible: from

$$12 - M = \sum_q a_q(2 - b_q) = 1$$

there would be one point with $a_q = b_q = 1$, while every other point with $a_q > 0$ had $b_q = 2$. If r and s count the remaining points with $a_q = 1$ and $a_q = 2$, respectively, then

$$1 + r + 2s = 6, \qquad 1 + 2r + 2s \le 6,$$

forcing $r = 0$ and then $1 + 2s = 6$, a contradiction. Equality $M = 12$ would force exactly three points q, with $a_q = b_q = 2$ at each. These are the three diagonal points of each of the two complete quadrangles, yet all three lie on the radical axis. The diagonal points of a non-degenerate complete quadrangle are noncollinear (in the standard frame their determinant is -2). Thus $M \le 10$. $\square$

2.2 Sharp upper constructions

Construction 2.5 (A circle and its centre)**.** Put $r = n-1$ distinct points on a proper circle Γ and add its centre o. Choose exactly $\lfloor r/2 \rfloor$ antipodal boundary pairs; if r is odd, choose the remaining boundary point without its antipode. The boundary triples determine only Γ. A triple consisting of o and a boundary pair is collinear precisely for an antipodal pair. Every other boundary pair gives a different proper circle, since a circle different from Γ meets Γ in at most two points. Thus this admissible set determines exactly

$$F(n) := 1 + \binom{n-1}{2} - \left\lfloor \frac{n-1}{2} \right\rfloor \tag{6}$$

circles. In particular it gives the sharp upper targets $3, 5$ for $n = 4, 5$ and 25 for $n = 9$.

The three exceptional upper bounds have equally explicit fixed certificates. For points $p_i = (x_i, y_i)$, write

$$\Delta(i,j,k,l) := \det \begin{pmatrix} x_i^2 + y_i^2 & x_i & y_i & 1 \\ x_j^2 + y_j^2 & x_j & y_j & 1 \\ x_k^2 + y_k^2 & x_k & y_k & 1 \\ x_l^2 + y_l^2 & x_l & y_l & 1 \end{pmatrix}. \tag{7}$$

For a noncollinear quadruple, $\Delta = 0$ is equivalent to concyclicity.

Construction 2.6 (Six rational points and eight circles)**.** Take

i	0	1	2	3	4	5
p_i	$(0,0)$	$(4,0)$	$(1,0)$	$(1,3)$	$(\frac{2}{5}, \frac{6}{5})$	$(2,2)$.

The collinear triples are exactly

$$012, \qquad 034, \qquad 135.$$

Exact substitution in $x^2 + y^2 + Dx + Ey + F = 0$ gives the following eight circle traces and coefficient triples (D, E, F):

013	$(-4,-2,0)$
0145	$(-4,0,0)$
0235	$(-1,-3,0)$
024	$(-1,-1,0)$
1234	$(-5,-3,4)$
125	$(-5,-1,4)$
245	$(-3,-2,2)$
345	$(-2,-4,4)$.

The three four-circles own four triples each and the other five circles own one triple each. Together with the three collinear triples this accounts for all $\binom{6}{3} = 20$ triples. Thus the configuration is admissible and determines exactly 8 circles.

Construction 2.7 (Seven points and eleven circles)**.** Let $u = \sqrt{3}$ and take

i	0	1	2	3	4	5	6
p_i	$(-1, 0)$	$(1, 0)$	$(0, u)$	$(0, 0)$	$(-\frac{1}{2}, \frac{u}{2})$	$(\frac{1}{2}, \frac{u}{2})$	$(0, \frac{u}{3})$.

The collinear triples are exactly

$$013,\ 024,\ 056,\ 125,\ 146,\ 236,$$

where a string denotes the corresponding set of indices. Direct expansion of (7), using the dihedral symmetry of the equilateral triangle, gives exactly the six proper four-circles

$$0145,\ 0235,\ 0346,\ 1234,\ 1356,\ 2456.$$

Their 24 triples are disjoint. The five remaining noncollinear triples are

$$012,\ 016,\ 026,\ 126,\ 345.$$

The determinant list also shows that they have five different owners and that no five-point circle occurs. Hence the configuration determines $6 + 5 = 11$ circles.

Construction 2.8 (Eight rational points and seventeen circles)**.** Take

i	0	1	2	3	4	5	6	7
p_i	$(0, 0)$	$(1, 0)$	$(\frac{1}{3}, 0)$	$(\frac{2}{3}, 0)$	$(\frac{2}{3}, \frac{1}{3})$	$(\frac{2}{5}, \frac{1}{5})$	$(\frac{1}{3}, \frac{1}{3})$	$(\frac{3}{5}, \frac{1}{5})$.

Its collinear triples are exactly

$$012,\ 013,\ 023,\ 123,\ 045,\ 167.$$

Exact rational evaluation of (7) gives exactly the eleven proper four-circles

$$0146,\ 0157,\ 0247,\ 0256,\ 0367,\ 1245,$$
$$1347,\ 1356,\ 2346,\ 2357,\ 4567.$$

No five-point subset has all its quadruples on this list. The six remaining noncollinear triples are

$$034,\ 035,\ 126,\ 127,\ 267,\ 345.$$

Thus the eleven four-circles own 44 triples and the last six triples have six distinct owners, for a total of 17 circles.

2.3 The elementary cases $n = 4, 5$

Proposition 2.9. *One has $f(4) = 3$ and $f(5) = 5$.*

Proof. Let first $|P| = 4$. If three points lie on a line, the fourth point together with each of the three pairs on that line gives a proper circle; the three circles are distinct. If no triple is collinear, all four triples are noncollinear, and two of them could share a circle only if all four points were concyclic. Hence $C(P) \geq 3$. Construction 2.5 attains three.

Now let $|P| = 5$. A four-point line and its outsider give six distinct circles, so assume that no line has four points. Distinct three-point lines use disjoint pairs; therefore $3L_3 \leq \binom{5}{2}$

and $L_3 \le 3$. If no four-point circle occurs, every noncollinear triple has a different owner, and

$$C(P) = \binom{5}{3} - L_3 \ge 7.$$

If a four-point circle Γ occurs, it is unique. Every three-point line contains the outsider and a chord of Γ, and these chords form a matching, so there are at most two such lines. At least eight triples are noncollinear; four lie on Γ, while the other four have mutually different owners, since a repeated owner would be a second four-point circle. Thus $C(P) \ge 1+4 = 5$, attained by construction 2.5. □

We repeatedly use, without further comment, that two distinct proper circles meet in at most two points, two distinct lines in at most one point, and a line meets a proper circle in at most two points.

2.4 Classical incidence inputs and their exact scope

For later reference we record every external incidence theorem used in the proof. The hypotheses below are part of the statements; in particular, none of these results assumes general position.

Theorem 2.10 (Melchior)**.** *Let $\mathcal{A}$ be a finite arrangement of distinct real projective lines, not all concurrent, and let $t_r(\mathcal{A})$ count vertices incident with exactly r lines. Then*

$$t_2 \ge 3 + \sum_{r\ge 4}(r-3)t_r. \tag{8}$$

Equivalently, the same inequality holds for the multiplicities of lines determined by a finite set of distinct noncollinear real projective points. All projective vertices, including those at infinity, are counted.

This is the point-dual form of Melchior's theorem [Mel40]. We do not import a classification of its equality cases.

Theorem 2.11 (Kelly–Moser)**.** *If Q is a finite set of q distinct noncollinear real points and t_2 is the number of its ordinary determined lines, then*

$$3q \le 7t_2, \qquad t_2 \ge \left\lceil \frac{3q}{7} \right\rceil. \tag{9}$$

The inequality is non-strict; no equality classification is assumed.

This is Theorem 3.6 of Kelly and Moser [KM58].

Theorem 2.12 (Langer–de Zeeuw form)**.** *Let Q be a finite set of N distinct affine points over $\mathbb{C}$. Assume that every affine line contains at most $2N/3$ points of Q. If t_r counts the spanned lines containing exactly r points, then*

$$\sum_{r\ge 2} r t_r \ge \frac{N(N+3)}{3}. \tag{10}$$

In particular the result applies to real point sets. Both the occupancy cap and the conclusion are non-strict.

We use the formulation of de Zeeuw [dZ18], derived from Langer's orbifold inequality [Lan03].

Theorem 2.13 (Lenchner)**.** *Let $\mathcal{A}$ be an affine arrangement of q distinct real lines, neither all parallel nor all concurrent. Apart from one specified exceptional six-line arrangement, the number $o_{\text{fin}}(\mathcal{A})$ of finite ordinary intersections satisfies*

$$7o_{\text{fin}}(\mathcal{A}) \geq 2q - 3. \tag{11}$$

Consequently (11) *is available under the simpler guard $q \neq 6$.*

The word *finite* is essential. This is the corollary of Lenchner's Theorem 8 [Len07] used later in the paper.

Theorem 2.14 (Ungar, six-point consequence)**.** *Six distinct noncollinear real affine points determine at least six directions, where a direction is a parallelism class of determined lines.*

This is the only consequence of Ungar's direction theorem that we use [Ung82].

Theorem 2.15 (Hall)**.** *Let G be a finite bipartite graph with parts X, Y. If $|S| \leq |N(S)|$ for every $S \subseteq X$, then G has a matching saturating X.*

We use Hall's theorem [Hal35] only after the relevant neighbourhood inequalities have been established geometrically.

Inversion, projective duality, directed Menelaus, and power of a point are not treated as external incidence black boxes. The inversion statements needed for the universal rows are proved next; later geometric uses are proved at their point of application.

2.5 Inversion and the universal rows

Fix $p \in P$ and $\rho > 0$. On $\mathbb{R}^2 \setminus \{p\}$ put

$$I_{p,\rho}(x) = p + \frac{\rho}{\|x - p\|^2}(x - p).$$

Lemma 2.16 (Inversion dictionary)**.** *The map $I_{p,\rho}$ is an injective involution on $\mathbb{R}^2 \setminus \{p\}$. A line through p maps to itself, while a proper circle through p maps to a line avoiding p, and conversely. Hence every maximal s-block through p becomes a full line containing exactly $s - 1$ points of the inverted set*

$$Q_p := I_{p,\rho}(P \setminus \{p\}).$$

The set Q_p is noncollinear. It remains noncollinear after the inversion centre $o = p$ is added.

Proof. The formula immediately gives $I_{p,\rho}^2 = \text{id}$. Translating p to the origin and substituting the formula into a linear or quadratic equation gives the stated line–circle dictionary. Full traces are preserved because the map is injective.

If Q_p lay on a line through o, then all original points would lie on that line. If it lay on a line avoiding o, the inverse carrier would be a proper circle through p containing all of P. Both possibilities violate admissibility. Adding o cannot make a noncollinear set collinear. □

Corollary 2.17 (The Lenchner pivot bridge)**.** *Let $p \in P$, put $q = n - 1$, and assume $q \neq 6$. Then at least*

$$\left\lceil \frac{2q-3}{7} \right\rceil$$

proper three-point circles of P pass through p. Equivalently,

$$c_3(p) \geq \left\lceil \frac{2n-5}{7} \right\rceil.$$

Proof. Invert $P \setminus \{p\}$ at p, obtaining the q distinct, noncollinear image points Q_p from lemma 2.16. Work projectively, dualize those points, and declare the dual p^* of the inversion centre to be the line at infinity. This produces q distinct real affine lines.

The dual lines are not all parallel: otherwise the points of Q_p would be collinear on a line through p. They are not all concurrent at a finite point: otherwise Q_p would be collinear on a line avoiding p. Thus the arrangement satisfies the geometric hypotheses of theorem 2.13; the guard $q \neq 6$ also removes its sole exceptional cardinality. It consequently has at least $\lceil (2q-3)/7 \rceil$ finite ordinary intersections.

Under projective duality, such an intersection is an ordinary determined line of Q_p. Finiteness says exactly that this line avoids p: ordinary intersections on p^* correspond instead to image lines through p, and are deliberately not counted. By inversion, an ordinary image line avoiding p is a proper circle whose full trace on P consists of p and the two preimages. Distinct image lines give distinct circles, proving the claim. □

The inversion dictionary verifies every noncollinearity hypothesis in the next two applications of theorem 2.10.

Proposition 2.18 (Universal Melchior rows)**.** *For every admissible n-point set and every $p \in P$, define*

$$\sigma_p := d_3(p) - 3 - \sum_{s \geq 5} (s-4) d_s(p). \tag{12}$$

Then $\sigma_p \geq 0$, and

$$\mathsf{P} := \sum_{s \geq 3} s(4-s) B_s \geq 3n, \tag{13}$$

$$\sum_{p \in P} \sigma_p = \mathsf{P} - 3n. \tag{14}$$

If $\ell_2(p)$ counts the ordinary original lines through p, then

$$\ell_2(p) + \sum_{s \geq 3} (s-1)\ell_s(p) = n-1, \tag{15}$$

$$\sum_{s \geq 3} s\ell_s(p) \leq n - 1 + \sigma_p. \tag{16}$$

After summing (16), one obtains

$$\mathsf{D} := \sum_{s \geq 3} s(s-4) C_s + \sum_{s \geq 3} 2s(s-2) L_s \leq n(n-4). \tag{17}$$

Finally, direct application of Melchior to P gives

$$3L_3 + \sum_{s \geq 4} \left(\binom{s}{2} + s - 3 \right) L_s \leq \binom{n}{2} - 3. \tag{18}$$

Proof. Apply theorem 2.10 to the $n-1$ inverted points. By lemma 2.16, its $(s-1)$-point lines are exactly the s-blocks through p. Melchior is therefore precisely $\sigma_p \geq 0$. Summing over p and using (3) gives (13) and (14).

The original lines through p partition $P \setminus \{p\}$ into arms, which proves (15). Add the inversion centre o. The line multiplicities of the resulting n-point set are exactly

$$t_2 = c_3(p) + \ell_2(p), \qquad t_i = c_{i+1}(p) + \ell_i(p) \quad (i \geq 3).$$

The slack in Melchior is then

$$\sigma_p + \ell_2(p) - \sum_{s\geq 3} \ell_s(p) \geq 0.$$

Using (15) yields (16). Sum it over p, use (3) and (14), and subtract P from the resulting inequality. The circle coefficient is $-s(4-s) = s(s-4)$ and the line coefficient is $s^2 - s(4-s) = 2s(s-2)$, giving (17).

For the last assertion, substitute (4) in (8) applied directly to P. □

Proposition 2.19 (Conditional Langer row)**.** *Suppose that, for every pivot p, every line determined by Q_p contains at most $2(n-1)/3$ image points. Then*

$$\mathsf{L} := \sum_{s\geq 3} s(s-1)B_s \geq \frac{n(n-1)(n+2)}{3}. \tag{19}$$

Proof. Apply theorem 2.12 to the $N = n-1$ image points at each pivot. Its incidence count is $\sum_{s\geq 3}(s-1)d_s(p)$. Summing over p and using (3) proves (19). Notice that the occupancy hypothesis must be verified before this row is used. □

Equations (2), (13), (17), and, under its stated cap, (19), are the four universal rows imported by the later sections.

2.6 Two pencil bounds

Lemma 2.20 (Rich-line pencil)**.** *If a line contains $m \geq 3$ points of an admissible n-point set and $q = n - m \geq 1$ points lie off it, then*

$$C(P) \geq q\binom{m}{2} - \binom{q}{2}\left\lfloor \frac{m}{2} \right\rfloor. \tag{20}$$

Proof. For a fixed outsider, its circles with pairs on the line are proper and mutually distinct. For two fixed outsiders, common circles use disjoint pairs of line points: two pairs sharing a point would give two circles with three common points. Thus two outsider pencils overlap in at most $\lfloor m/2 \rfloor$ circles. The two-term Bonferroni inequality gives (20); higher intersections do not invalidate this lower bound. □

Lemma 2.21 (Rich-circle pencil)**.** *If a proper circle Γ contains $m \geq 3$ selected points and $q = n - m \geq 1$ points lie outside it, then*

$$C(P) \geq 1 + q\left(\binom{m}{2} - \left\lfloor \frac{m}{2} \right\rfloor\right) - \binom{q}{2}\left\lfloor \frac{m}{2} \right\rfloor. \tag{21}$$

Proof. For an outsider x, the chords of Γ passing through x form a matching on the m marked points, so at most $\lfloor m/2 \rfloor$ marked pairs are collinear with x. Every other pair gives a distinct proper circle through x. As in the line case, two outsider pencils overlap in at most $\lfloor m/2 \rfloor$ circles. Add Γ and apply Bonferroni. □

2.7 Six points

Proposition 2.22. *One has $f(6) = 8$.*

Proof. Construction 2.6 gives the upper bound. Suppose, for a contradiction, that an admissible six-point set satisfies $C(P) \le 7$, and let k be the maximum number of selected points on a proper circle.

If $k = 5$, lemma 2.21 gives $C(P) \ge 9$. If $k = 3$, every noncollinear triple has a different circle owner. Let a largest line contain m points and put $r = 6 - m$. For $m \ge 4$, a collinear triple not contained in that line uses at most one of its points, and an outsider pair can be collinear with at most one point on the line. Thus the number T_0 of collinear triples satisfies

$$T_0 \le \binom{m}{3} + \binom{r}{3} + \binom{r}{2}.$$

This leaves at least 10 noncollinear triples for $m = 5$ and at least 15 for $m = 4$. If all lines have size at most three, their collinear triples use disjoint sets of three pairs, so there are at most $\left\lfloor \binom{6}{2}/3 \right\rfloor = 5$ of them. Again at least 15 circles remain.

It remains that $k = 4$. The rich-line bound is 10 for both $m = 4$ and $m = 5$, so every line has size at most three. All maximal generalized blocks therefore have size three or four, and every four-block is a circle. Put

$$b = C_4, \qquad L = L_3.$$

Triple ownership, the pivot Melchior row, the exact circle count, and the global line row give, respectively,

$$B_3 + 4b = 20, \tag{22}$$

$$0 \le \sum_{p \in P} (d_3(p) - 3) = 42 - 12b, \tag{23}$$

$$C(P) = 20 - 3b - L, \tag{24}$$

$$L \le 4. \tag{25}$$

Hence $b \le 3$, while $C(P) \le 7$ forces

$$b = 3, \qquad L = 4. \tag{26}$$

At any $p \in P$, ownership of the ten triples containing p reads

$$d_3(p) + 3d_4(p) = 10.$$

The first Melchior row gives $d_3(p) \ge 3$, and $d_3(p) \equiv 1 \pmod 3$. Since $\sum_p d_4(p) = 4b = 12$, necessarily

$$d_3(p) = 4, \qquad d_4(p) = 2, \qquad \sigma_p = 1 \quad \text{for every } p. \tag{27}$$

The added-centre inequality (16) becomes

$$\kappa_p := 6 - 3\ell_3(p) \ge 0. \tag{28}$$

Since $L = 4$, summing (28) gives zero, so every point lies on exactly two of the four three-point lines. No three of those lines are concurrent, and their six pairwise intersections are the selected points: the line pattern is a complete quadrilateral. Label its four lines

$$012, \qquad 034, \qquad 135, \qquad 245.$$

Apply a Euclidean similarity, which preserves lines and circles, so that

$$
\begin{aligned}
p_0 = (0,0), \quad p_1 = (1,0), \quad p_3 = (u,v), \qquad v \neq 0,\\
p_2 = (a,0), \quad p_4 = b(u,v), \quad p_5 = t(1,0) + (1-t)(u,v), \qquad t = \frac{a(b-1)}{b-a}.
\end{aligned} \tag{29}
$$

Here distinctness gives $a, b \notin \{0,1\}$. Moreover $a \neq b$: if $a = b$, then the line p_2p_4 is parallel to p_1p_3 (their direction vectors differ by the nonzero scalar a), and the two lines are distinct because $a \neq 1$; they therefore cannot share the affine point p_5. Put $R = u^2 + v^2 > 0$.

By (27), the complements of the three four-circles are three disjoint pairs partitioning P. Each complement meets every one of the four collinear triples. In a complete quadrilateral the only such pairs are the three diagonal pairs, so the circle supports are forced to be

$$0145, \qquad 0235, \qquad 1234.$$

Substitution in (7) gives, in the same order,

$$
\begin{aligned}
\Delta(0,1,4,5) &= \frac{vb^2(a-1)(b-1)}{(a-b)^2}(-Rb + 2au - a),\\
\Delta(0,2,3,5) &= \frac{va^2(a-1)(b-1)}{(a-b)^2}(-Rb - a + 2bu),\\
\Delta(1,2,3,4) &= v(a-1)(b-1)(-Rb + a).
\end{aligned}
$$

All prefactors displayed here are nonzero. Thus concyclicity gives

$$a(1-2u) + bR = 0, \qquad a + b(R - 2u) = 0, \qquad a - bR = 0.$$

The third equation gives $a = bR$. Substitution in the second gives $2b(R-u) = 0$, hence $R = u$. The first equation then becomes $2bu(1-u) = 0$. Since $b \neq 0$ and $u = R > 0$, one has $u = 1$. But $R = u = 1$ and $R = u^2 + v^2$ force $v = 0$, contradicting (29). Thus $C(P) \geq 8$. □

2.8 Seven points and the planar Fano wall

Lemma 2.23 (Planar Fano wall)**.** *There do not exist seven distinct real points together with seven proper four-point circles whose three-point complements form a Steiner triple system on the seven points.*

Proof. We first derive, rather than quote, the unique form of a Steiner triple system on seven labels. Name one triple 123. The two remaining triples through 1 partition $\{4,5,6,7\}$ into two pairs, so after relabelling they are 145 and 167. The triples through 2 use neither pair 45 nor pair 67; after possibly interchanging $6, 7$, they are 246 and 257. The unused pairs on $\{4,5,6,7\}$ are then 47 and 56, forcing the final two triples through 3. Thus the system is

$$123, \quad 145, \quad 167, \quad 246, \quad 257, \quad 347, \quad 356. \tag{30}$$

Invert at point 1. The four triples in (30) avoiding 1 have complementary four-circles through 1, which become the lines

$$357, \qquad 346, \qquad 256, \qquad 247. \tag{31}$$

The first three form a nondegenerate triangle and the fourth is a transversal. Put

$$A = 3, \quad B = 5, \quad C = 6, \qquad X = 7 \in AB, \quad Y = 4 \in AC, \quad Z = 2 \in BC.$$

The complements of the three Fano triples through 1 become the proper circles

$$ABYZ, \qquad ACXZ, \qquad BCXY. \tag{32}$$

Use directed affine parameters

$$AB : A = 0, B = 1, X = x, \quad AC : A = 0, C = 1, Y = y, \quad BC : B = 0, C = 1, Z = z.$$

All six quantities $x, y, z, 1-x, 1-y, 1-z$ are nonzero. Put $a^2 = |BC|^2$, $b^2 = |CA|^2$, and $c^2 = |AB|^2$. Power of a point applied at the opposite vertices to the three circles in (32) gives

$$b^2(1-y) = a^2(1-z), \qquad c^2(1-x) = a^2 z, \qquad c^2 x = b^2 y.$$

Multiplying the resulting ratios yields

$$\frac{x}{1-x}\frac{z}{1-z}\frac{1-y}{y} = +1. \tag{33}$$

But X, Y, Z are collinear by the fourth line in (31), so directed Menelaus in triangle ABC gives the same left-hand side equal to -1. This contradiction proves the lemma. □

Proposition 2.24. *One has $f(7) = 11$.*

Proof. Construction 2.7 gives $f(7) \leq 11$. Suppose that an admissible seven-point set satisfies $C(P) \leq 10$. The two pencil bounds give

circle size m	4	5	6
lower bound	7	15	13

line size m	4	5	6
lower bound	12	18	15.

The entry $m = 4$ for circles need not be excluded; all larger circles and all lines of size at least four are excluded. Thus maximal blocks have size three or four, and every four-block is a circle.

Let $L = L_3$, let $Q = C_4$, and let $R = C_3$. Triple ownership gives

$$35 = L + 4Q + R, \qquad C(P) = Q + R = 35 - L - 3Q. \tag{34}$$

Since $21 = L_2 + 3L$, Melchior gives $L_2 \geq 3$ and hence $L \leq 6$. The complements of two different four-circles intersect in at most one point: the two circle supports intersect in at most two. Therefore no pair of labels occurs in two complementary triples, and $3Q \leq 21$, so $Q \leq 7$.

The assumption $C(P) \leq 10$ and (34) imply $L + 3Q \geq 25$. If $Q \leq 6$, the left side is at most $6 + 18 = 24$. Consequently $Q = 7$. The seven complements contain $7\binom{3}{2} = 21$ distinct pairs, hence every pair exactly once: they form a Steiner triple system. This contradicts lemma 2.23. □

2.9 Eight points

Lemma 2.25 (Even-arrangement slack)**.** *Let $\mathcal{A}$ be a non-pencil arrangement of an even number of distinct real projective lines. Its Melchior slack*

$$\kappa := t_2 - 3 - \sum_{r \geq 4} (r-3) t_r$$

cannot equal 1.

Proof. Let V, E, F be the numbers of vertices, edges, and faces of the projective cell decomposition. Euler's formula and the incidence count $E = \sum_r rt_r$ give

$$\kappa = \sum_{\text{faces } G} (|G| - 3).$$

Because the number of lines is even, the sign of the product of homogeneous linear equations for the lines is well-defined on the real projective plane and two-colours the faces. Every edge borders one face of each colour. If F_i is the number of faces of colour i, the sum of their boundary lengths is E, and their excess is

$$\kappa_i = E - 3F_i \geq 0.$$

Thus $\kappa_0 \equiv \kappa_1 \pmod 3$ and $\kappa = \kappa_0 + \kappa_1 \neq 1$. □

Proposition 2.26. *One has* $f(8) = 17$.

Proof. Construction 2.8 gives the upper bound. Suppose that an admissible eight-point set has $C(P) \leq 16$, and let k be the maximum number of selected points on a proper circle. For $k = 7, 6, 5$, lemma 2.21 gives, respectively,

$$C(P) \geq 19, \qquad 22, \qquad 19.$$

If $k = 3$, every noncollinear triple has a different owner. If a largest line has $m \geq 4$ points and $r = 8 - m$, the number of collinear triples is at most

$$\binom{m}{3} + \binom{r}{3} + \binom{r}{2} \leq 35.$$

If all lines have size at most three, their triples use disjoint triples of pairs and there are at most $\lfloor 28/3 \rfloor = 9$ of them. In either event $C(P) \geq 21$. The case $k = 8$ is inadmissible, so only $k = 4$ remains.

A line of size $m = 5, 6, 7$ gives, by lemma 2.20, respectively $24, 27, 21$ circles. Hence every maximal generalized block has size three or four. Put

$$b = B_4 = C_4 + L_4, \qquad L = L_3 + L_4, \qquad h = L_4.$$

Triple ownership, the summed pivot row, the original Melchior row, the added-centre row, and the circle count are

$$B_3 + 4b = 56, \tag{35}$$

$$K := \sum_{p \in P} \sigma_p = 144 - 12b \geq 0, \tag{36}$$

$$3L + 4h \leq 25, \tag{37}$$

$$9L + 7h \leq 200 - 12b, \tag{38}$$

$$C(P) = 56 - 3b - L. \tag{39}$$

Indeed, (38) is $9L_3 + 16L_4 \leq 56 + K$ rewritten in terms of L, h. Equations (36), (37), and (39) imply

$$b \in \{11, 12\}. \tag{40}$$

Suppose first that $b = 11$. Then $K = 12$ and (39) gives $L \geq 7$. The two line inequalities force

$$L = 7, \qquad h = 0.$$

At p, write $r_p = d_4(p)$. Ownership of triples through p and the definition of σ_p give

$$d_3(p) + 3r_p = 21, \qquad \sigma_p = 18 - 3r_p.$$

The local added-centre slack is

$$\kappa_p := 7 + \sigma_p - 3\ell_3(p).$$

It is nonnegative and congruent to 1 modulo 3, hence at least one. On the other hand,

$$\sum_{p\in P} \kappa_p = 56 + K - 9L_3 = 56 + 12 - 63 = 5,$$

which cannot be the sum of eight positive integers.

Now let $b = 12$. Then $K = 0$, so $\sigma_p = 0$ for every p, and

$$\kappa_p = 7 - 3\ell_3(p) - 4\ell_4(p) \geq 0. \tag{41}$$

The added-centre configuration has eight points and no line containing more than four of them. Its dual is therefore a non-pencil arrangement of eight real projective lines, and κ_p is its Melchior slack. By lemma 2.25,

$$\kappa_p \neq 1. \tag{42}$$

Equations (41)–(42) imply $\ell_3(p) \leq 1$: if $\ell_4(p) = 0$, the only additional nonnegative case is $\ell_3(p) = 2$, which gives slack one; if $\ell_4(p) \geq 1$, then necessarily $\ell_4(p) = 1$ and already $\ell_3(p) \leq 1$. Hence $3L_3 \leq 8$ and $L_3 \leq 2$.

Equation (39) gives $L \geq 4$. Combining this with (37), (38), and $L_3 \leq 2$ leaves

$$L_3 = 2, \qquad L_4 = 2, \qquad C(P) = 16. \tag{43}$$

Moreover,

$$\sum_p \kappa_p = 56 - 9L_3 - 16L_4 = 6.$$

The four possible local pairs and slacks are

$(\ell_3(p), \ell_4(p))$	$(0,0)$	$(1,0)$	$(0,1)$	$(1,1)$
κ_p	7	4	3	0.

Since $\sum_p \ell_4(p) = 4L_4 = 8$, every point lies on one four-line; the slack sum then forces two points of type $(0,1)$ and six of type $(1,1)$. Thus the two four-lines are disjoint and partition P. Each of the two three-lines contains three points, so two of them lie on the same four-line. The three-line and that four-line would then be the same geometric line, contrary to maximality of their different supports. This final contradiction proves $C(P) \geq 17$. □

2.10 Nine points

Proposition 2.27. *One has $f(9) = 25$.*

Proof. Construction 2.5, with four antipodal boundary pairs, gives $f(9) \leq 1 + \binom{8}{2} - 4 = 25$. For the reverse inequality, suppose that an admissible nine-point set P satisfies

$$C(P) \leq 24, \tag{44}$$

and let k be the largest number of selected points on a proper circle. Then $3 \leq k \leq 8$.

The branches $k \geq 6$. For $k = 8, 7, 6$, respectively, the rich-circle bound (21) gives

$$C(P) \geq 25, \quad 34, \quad 28.$$

Thus none of these branches is compatible with (44).

The branch $k = 5$. Fix a five-point circle Γ and put $X = P \setminus \Gamma$, so $|X| = 4$. First, lemma 2.20 excludes every line of size at least five: for line sizes $m = 5, 6, 7, 8$, its lower bound is, respectively,

$$28, \quad 36, \quad 39, \quad 28.$$

Hence all lines below have size at most four.

For a proper circle other than Γ, let $c_{r,t}$ count those having r points on Γ and t points in X. Necessarily $r \leq 2$. Let $\lambda_{r,t}$ denote the analogous number of rich lines and set

$$R := \sum_t t\lambda_{2,t}. \tag{45}$$

At each outsider, the chords of Γ passing through it form a matching on five vertices. Therefore

$$R \leq 8. \tag{46}$$

For $x \in X$, let $\mathcal{S}_x$ consist of the proper circles through x and two points of Γ. Exactly the chord-lines through x fail to yield such circles, whence

$$\sum_{x \in X} |\mathcal{S}_x| = 40 - R.$$

For an outsider pair x, y, the Γ-pairs represented in $\mathcal{S}_x \cap \mathcal{S}_y$ form a matching, so this intersection has size at most two. No circle contains two points of Γ and all four outsiders, because $k = 5$. Thus inclusion–exclusion stops after outsider triples. If c_2 counts all circles with exactly two Γ-points, then

$$c_2 \geq 40 - R - 12 + c_{2,3} = 28 - R + c_{2,3}. \tag{47}$$

Let $c_{\leq 1}$ count circles other than Γ with at most one Γ-point. Since $C(P) = 1 + c_2 + c_{\leq 1}$, (44) gives the four-outsider master inequality

$$R \geq 5 + c_{2,3} + c_{\leq 1}. \tag{48}$$

We now use only the four possible incidence types of X.

No three outsiders are collinear and the four are not concyclic. The four outsider triples determine four distinct proper circles. They are counted by $c_{2,3} + c_{\leq 1}$, so (48) gives $R \geq 9$, contradicting (46).

All four outsiders are collinear. Put $a = c_{1,2}$ and $b = c_{2,2}$. Counting the 30 triples with one Γ-point and two outsiders gives

$$a + 2b = 30. \tag{49}$$

There is no line contribution, since every outsider pair already lies on the four-outsider line; nor can a proper circle contain three of its collinear points. The pair-intersection bound above gives $b \leq 12$, and hence $a \geq 6$. Moreover, the exact membership count for the $\mathcal{S}_x$ gives $c_2 = 40 - R - b$. Therefore

$$C(P) = 1 + (40 - R - b) + a = 26 - R + \frac{3}{2}a \geq 27,$$

again a contradiction.

Exactly three outsiders are collinear. Let λ be their line and let $s \in \{0,1\}$ record whether λ contains a point of Γ. Put

$$a := c_{1,3}, \qquad h := c_{2,3}, \qquad \ell := \lambda_{1,2}.$$

The other three outsider triples determine three distinct proper circles. Thus $h + c_{\le 1} \ge 3$. Combining this with (46), (47), and (48) first gives $R = 8$ and $h + c_{\le 1} = 3$. If

$$I := \sum_{\{x,y\} \subset X} |\mathcal{S}_x \cap \mathcal{S}_y| = c_{2,2} + 3h,$$

then $I \le 12$, while exact inclusion–exclusion gives

$$C(P) = 1 + 40 - R - I + h + c_{\le 1} = 36 - I.$$

Thus (44) forces $I = 12$ and equality throughout:

$$R = 8, \qquad C(P) = 24, \qquad h + c_{\le 1} = 3, \qquad c_{2,2} + 3h = 12. \tag{50}$$

In particular those three outsider-triple circles exhaust $c_{\le 1} + h$, so $c_{1,2} = 0$. The last equality says that each outsider pair lies on exactly two circles with two Γ-points, and their two Γ-pairs are disjoint. There can be no line of type $(2,2)$: its Γ-pair would have to be disjoint from both circle pairs, which is impossible on five points.

Counting again the triples with one Γ-point and two outsiders, and then applying the global line row, gives

$$\ell + 3a + 3s = 6, \tag{51}$$
$$3\ell + 4s \le 6. \tag{52}$$

Indeed, the term already covered by the two-Γ circles is $2(c_{2,2} + 3h) = 24$; while the rich lines are the eight chord incidences, λ, and the ℓ lines of type $(1,2)$. Equations (51)–(52) yield

$$\ell = 0, \qquad a + s = 2.$$

For each outsider pair, the two disjoint Γ-pairs in (50) leave one Γ-point. Its triple with the outsider pair must be owned by one of the a three-outsider circles or, when $s = 1$, by λ. The pair-shadows of two distinct three-subsets of a four-set cover only five of its six pairs. One pair is therefore left without an owner, a contradiction.

The four outsiders are concyclic. Let Ω be their circle. It contains $\delta \in \{0,1\}$ selected points of Γ. Put

$$a := c_{1,2}, \quad b := c_{2,2}, \quad u := \lambda_{2,1}, \quad v := \lambda_{2,2}, \quad \ell := \lambda_{1,2}.$$

Apart from Γ, Ω, every rich circle or line has type $(2,1), (1,2)$, or $(2,2)$: a distinct carrier meets each of Γ, Ω in at most two points. Consequently

$$R = u + 2v, \tag{53}$$
$$C(P) = 42 - R - b + a. \tag{54}$$

Write $b = 12 - e$, where $e \ge 0$. From (44), (46), and (54),

$$R \ge 6 + e + a, \qquad e + a \le 2. \tag{55}$$

The same 30-triple count and the global line row now read

$$\ell + 2v = 6 - a + 2e - 6\delta, \tag{56}$$
$$3R + 3\ell + v \le 33. \tag{57}$$

If $\delta = 0$, all $b + v$ generalized blocks of type $(2,2)$ are cross-blocks between five points of Γ and four points of Ω. By lemma 2.4, $b + v \le 12$, hence $v \le e$. But (55)–(57) give

$$33 \ge 3R + 3\ell + v \ge 36 + 9e - 5v,$$

or $5v \ge 3 + 9e$, impossible when $v \le e$.

If $\delta = 1$, remove the common selected point of Γ and Ω. A type-$(2,2)$ block cannot use that point: a circle would share three points with Ω, while a line would contain three collinear points of Ω. The $4+4$ case of lemma 2.4 therefore gives

$$b + v \le 10, \qquad v \le e - 2.$$

Together with (55), this forces

$$e = 2, \qquad a = v = 0, \qquad R = 8.$$

Equation (56) gives $\ell = 4$, and (57) becomes $36 \le 33$, the final contradiction in the $k = 5$ branch.

The branches $k \le 4$. The same rich-line calculation excludes lines of size at least five. All maximal generalized blocks therefore have size three or four. Put

$$B := B_4 = C_4 + L_4, \qquad L := L_3 + L_4.$$

Triple ownership and the circle objective give

$$C(P) = 84 - 3B - L. \tag{58}$$

The global line row is $3L_3 + 7L_4 \le 33$, so $L \le 11$. Under (44), (58) then forces $B \ge 17$.

Fix $p \in P$ and invert the other eight points at p. They are noncollinear by lemma 2.16, and no four are collinear: such a line would invert to a five-point proper circle or a five-point line through p. A three-point image line is exactly a generalized four-block of P through p. Hence lemma 2.3 gives $d_4(p) \le 7$. Summing over all pivots,

$$4B = \sum_{p \in P} d_4(p) \le 9 \cdot 7 = 63,$$

so $B \le 15$, contrary to $B \ge 17$.

Every possible value of k has now been excluded under (44). Thus $C(P) \ge 25$, completing the proof. □

Theorem 2.28 (The small values)**.** *For the V1 problem,*

$$(f(4), f(5), f(6), f(7), f(8), f(9)) = (3, 5, 8, 11, 17, 25).$$

Proof. Combine propositions 2.9, 2.22, 2.24, 2.26 and 2.27. □

3 The finite cases $n = 10, 11, 12$

Throughout this section $C = C(P)$ and

$$F(n) := 1 + \binom{n-1}{2} - \left\lfloor \frac{n-1}{2} \right\rfloor .$$

Thus $F(10) = 33$, $F(11) = 41$, and $F(12) = 51$. We prove the corresponding lower bounds. The upper bounds are already supplied by construction 2.5. All blocks, degrees, and slack variables have the meaning fixed in section 2.

3.1 A deletion-free outer router

The following lemma removes every largest-circle size except five and six for eleven and twelve points. It is useful to record it before the local geometry, because it also supplies the block-size guards used later.

Lemma 3.1 (Outer router). *Let $n \in \{11, 12\}$ and suppose $C \le F(n) - 1$. If m is the maximum number of selected points on a proper circle, then $m \in \{5, 6\}$. Moreover, for $n = 11$ every line has at most five selected points, and for $n = 12$ every line has at most six.*

Proof. The pencil bounds lemmas 2.20 and 2.21 give

$$\begin{array}{c|c|l} n & R_n(m),\ m = 7, 8, \ldots, n-1 & \min_{s \ge s_0} L_n(s) \\ \hline 11 & 55, 61, 61, 41 & 45 \quad (s_0 = 6), \\ 12 & 61, 73, 85, 76, 51 & 55 \quad (s_0 = 7). \end{array} \tag{59}$$

Here R_n and L_n denote the right sides of the rich-circle and rich-line bounds. Every entry is at least $F(n)$, proving the asserted caps and excluding $m \ge 7$.

If $m \le 3$, the triple partition and the global line row give, respectively,

$$C \ge \binom{11}{3} - \frac{5}{6}\left(\binom{11}{2} - 3\right) = \frac{365}{3}, \qquad C \ge \binom{12}{3} - \frac{10}{9}\left(\binom{12}{2} - 3\right) = 150.$$

The constants $5/6$ and $10/9$ are the maxima of $\binom{s}{3}/(\binom{s}{2} + s - 3)$ over the allowed line sizes.

It remains to exclude $m = 4$. Fix a maximal four-circle Γ and put

$$T' = \sum_{B \ne \Gamma} \binom{|B|}{3} = \binom{n}{3} - 4.$$

Let O be the summed Lenchner row $3C_3 \ge 3n$, let G denote the left side of (18), and let K denote the summed Kelly–Moser row. At $n = 11$ it says $3B_3 \ge 55$. The fixed combination

$$1 + \frac{1}{4}T' + \frac{1}{8}O - \frac{5}{24}G + \frac{1}{8}K$$

has coefficient one on every proper circle and coefficients $0, -11/24, 0$ on lines of sizes $3, 4, 5$. Hence

$$C \ge 1 + \frac{161}{4} + \frac{33}{8} - \frac{5 \cdot 52}{24} + \frac{55}{8} = \frac{497}{12} > 41.$$

At $n = 12$ replace K by the summed pivot row $3B_3 - 5B_5 - 12B_6 \ge 36$. The combination

$$1 + \frac{1}{4}T' + \frac{2}{15}O - \frac{1}{5}G + \frac{7}{60}P$$

has line coefficients $0, -2/5, -29/60, 0$ for sizes $3, 4, 5, 6$ and coefficient one on every proper circle. Therefore

$$C \ge 1 + \frac{216}{4} + \frac{72}{15} - \frac{63}{5} + \frac{7 \cdot 36}{60} = \frac{257}{5} > 51.$$

This excludes $m = 4$ and completes the router. $\square$

3.2 Ten points

Lemma 3.2 (The maximum-four branch at ten points)**.** *If $|P| = 10$, no proper circle contains more than four selected points, and $C \le 32$, then a contradiction follows.*

Proof. A line of size $s \ge 6$ gives, by lemma 2.20, respectively $42, 54, 52, 36$ circles for $s = 6, 7, 8, 9$. Thus a five-block is a line and all other blocks have size three or four. Put $b = B_4$, $g = L_5$, $t = L_3$, and $h = L_4$. The triple partition, circle count, and global Melchior row are

$$B_3 + 4b + 10g = 120, \qquad C = 120 - 3b - 10g - t - h, \qquad 3t + 7h + 12g \le 42.$$

At every pivot, Kelly–Moser and the congruence $36 = d_3 + 3d_4 + 6d_5$ give $d_3 \ge 6$, hence $B_3 \ge 20$. It follows first that $g = 0$, then $b = 25$ and $t + h \ge 13$. The summed added-centre row

$$9L_3 + 16L_4 + 25L_5 \le 90 + 3B_3 - 30 - 5B_5$$

excludes $t + h = 14$ and forces $h = 0$ when $t + h = 13$. Therefore the only possible aggregate profile is

$$(B_3, B_4; L_3, L_4, L_5; C_3, C_4) = (20, 25; 13, 0, 0; 7, 25), \tag{60}$$

and every pivot has $(d_3, d_4) = (6, 10)$.

Invert at any pivot. The nine-point image satisfies the hypotheses of lemma 3.11; its ordinary graph is two triangles and three isolated points. Let $\mathcal{H}$ be the 20-edge three-uniform hypergraph of generalized three-blocks. The link of every vertex of $\mathcal{H}$ is two triangles. If $abc \in \mathcal{H}$, the edge bc lies in a unique link triangle, say with fourth vertex d; the same link argument at b forces all four faces of $abcd$. Thus the 20 hyperedges partition into five tetrahedron boundaries. At most one face of each boundary is a line, since two faces share two points and $L_4 = 0$. Hence $L_3 \le 5$, contrary to (60). ☐

We first isolate the local parity input and the three geometric kernels. This keeps the final numerical router transparent: it will inspect only integer moments, while every equality face is discharged by a named lemma.

Lemma 3.3 (Ten-point local parity)**.** *Assume that $|P| = 10$, that no proper circle contains more than five selected points, and that $C \le 32$. Put $r_p = d_5(p)$ and $q_p = \ell_4(p)$. Then*

$$36 = d_3(p) + 3d_4(p) + 6d_5(p), \qquad d_3(p) \in \{6, 9\}. \tag{61}$$

If $\ell_5(p) = 0$, the largest possible value of $\ell_3(p)$ is

q_p	*type*	$r_p = 0$	$r_p = 1$	$r_p = 2$	$r_p = 3$
0	$d_3 = 6$	4	3	2	3
0	$d_3 = 9$	4	4	3	4
1	$d_3 = 6$	2	1	2	1
1	$d_3 = 9$	3	2	3	2
2	$d_3 = 6$	0	1	0	–
2	$d_3 = 9$	1	2	1	0
3	$d_3 = 6$	0	–	–	–
3	$d_3 = 9$	1	0	–	0.

(62)

A dash means that the indicated local state is impossible. If $\ell_5(p) = 1$, the maximum is at least one smaller than the entry in the $q_p = 0$ row with the same values of $d_3(p)$ and r_p.

Proof. The rich-line pencil bound excludes lines of size at least six. Inversion at p gives a noncollinear nine-point set whose ordinary lines are the three-blocks through p. Kelly–Moser and the pair partition give $d_3(p) \geq 6$ and $d_3(p) \equiv 0 \pmod 3$. Deletion of p is nondegenerate: otherwise the corresponding rich line or rich circle pencil already has at least 33 proper circles. Hence theorem 2.28 gives $c_3(p) \leq C - f(9) \leq 7$, while the ray bound gives $\ell_3(p) \leq 4$. This proves (61).

Set $\sigma_p = d_3(p) - 3 - d_5(p)$. Added-centre Melchior says that

$$\kappa_p := 9 + \sigma_p - 3\ell_3(p) - 4\ell_4(p) - 5\ell_5(p) \geq 0. \tag{63}$$

Moreover $\kappa_p \neq 1$. Indeed, after inversion and restoration of the centre, dualize to an arrangement of ten real projective lines. The sign of the product of their defining forms two-colours the cells. If κ_+ and κ_- are the sums of $|F| - 3$ over the two colours, edge counting gives $\kappa_+ \equiv \kappa_- \pmod 3$; total slack one is therefore impossible. Substituting $d_3 = 6, 9$ in (63), and taking the largest integer $\ell_3 \leq 4$ for which $\kappa_p \geq 0$ and $\kappa_p \neq 1$, gives every entry of (62) directly. Setting $\ell_5 = 1$ subtracts five from the numerator in (63); for $r = 0, 1, 2, 3$, in both the low and high rows, the same integer maximization lowers ℓ_3 by at least one. This proves the final assertion. □

Lemma 3.4 (Punctured-pentagon transversal)**.** *Let five five-subsets of a ten-set have pairwise intersections at most two and point-degree multiset* $(3^5, 2^5)$*. At a degree-three point* p*, at most two four-subsets through* p *can meet every five-subset in at most two points and meet one another in at most two points.*

Proof. The degree row gives $\sum_p \binom{d_5(p)}{2} = 5\binom{3}{2} + 5\binom{2}{2} = 20 = 2\binom{5}{2}$. Since every block pair meets in at most two points, every pair therefore meets in exactly two. Every five-subset contains two degree-two and three degree-three points: if these numbers are e, t, then $e + t = 5$ and $e + 2t = 8$. On the five block labels, the supports of the degree-two points form a two-regular graph. It has no double edge, since a doubled ij edge would exhaust the intersection of blocks i, j, whereas their six required occurrences among the five degree-three supports could then not be supplied. Thus it is a five-cycle. Complements of the degree-three supports form the complementary five-cycle, because each label pair occurs altogether twice. This proves the incidence pattern uniquely, without a support catalogue.

Relabel one degree-three point as $p = 124$ and the other nine points as

$$A = 12,\ B = 134,\ C = 245, \quad D = 135,\ Q = 15,\ R = 23,\ E = 235,\ S = 34,\ T = 45.$$

The three blocks through p have pairwise vertices A, B, C and private pairs

$$\{D, Q\}, \qquad \{R, E\}, \qquad \{S, T\}. \tag{64}$$

A four-set through p cannot use A, B, or C: such a vertex fills the available intersection slot on two incident five-blocks and leaves room for at most one further point. It must therefore choose one point from each pair in (64). Membership in the two five-blocks 3 and 5 is

$$D : 35,\ Q : 5,\ R : 3,\ E : 35,\ S : 3,\ T : 5.$$

Forbidding three chosen points in either omitted block leaves exactly

$$DRT, \qquad QRS, \qquad QRT, \qquad QES. \tag{65}$$

Two residual triples are compatible precisely when they share at most one point. The compatible pairs are

$$DRT\text{–!}QRS, \qquad DRT\text{–!}QES, \qquad QES\text{–!}QRT,$$

the edges of a four-vertex path. Its clique number is two, proving the claim. □

Lemma 3.5 (Four-pentagon obstruction)**.** *Assume the hypotheses of lemma 3.3, and suppose that exactly four maximal generalized five-blocks occur. Then the configuration is impossible.*

Proof. Write the blocks as $B_1, \dots, B_4$, put $r(p) = d_5(p)$, and set

$$S = \sum_p \binom{r(p)}{2} = \sum_{i<j} |B_i \cap B_j|.$$

We have $\sum_p r(p) = 20$, $S \le 12$, and convexity gives $S \ge 10$. Moreover

$$\sum_p (r(p) - 2)^2 = 2S - 20. \tag{66}$$

Thus the only degree rows are

$$\begin{array}{c|c} S & \{r(p) : p \in P\} \\ \hline 10 & (2^{10}) \\ 11 & (1, 2^8, 3) \\ 12 & (1^2, 2^6, 3^2). \end{array} \tag{67}$$

For each five-block and four-block D introduce centred vectors

$$w_i = \mathbf{1}_{B_i} - \frac{1}{2}\mathbf{1}, \qquad v_D = \mathbf{1}_D - \frac{2}{5}\mathbf{1}.$$

Maximality gives the coefficientwise inequalities

$$\langle v_D, w_i \rangle = |D \cap B_i| - 2 \le 0. \tag{68}$$

If $S = 10$, put $e_{ij} = 2 - |B_i \cap B_j|$. The five actual pair-incidences of B_i, against the maximum six across its three block partners, give $\sum_{j \ne i} e_{ij} = 1$, so the two deficient pairs form a perfect matching, say $12, 34$. Since every point has degree two, $\sum_i w_i = 0$; equality after summing (68) forces $|D \cap B_i| = 2$ for all i. A four-block through the unique 12 point must contain the unique 34 point: the three remaining membership edges need degree sequence $(1, 1, 2, 2)$, which is impossible without the edge 34. Its last two points form $13|24$ or $14|23$. Different four-blocks cannot reuse either of them, whence $d_4(12) \le 4$; but (61) requires $d_4(12) = 5$ or 6.

If $S = 11$, there are a degree-three point T, a degree-one point U, and one deficient block pair. Comparing the incidence and intersection row at each label shows that, after relabelling, the ten supports are

$$T = 123, \quad U = 1, \quad 12, 13, 23, 14, 24, 24, 34, 34. \tag{69}$$

Here $\sum_i w_i = \mathbf{1}_T - \mathbf{1}_U$. Summing (68) shows that every four-block through T also contains U. Its last two points must be a matching between the two 24 points and the two 34 points, so $d_4(T) \le 2$, whereas (61) requires 3 or 4.

It remains that $S = 12$. Comparing the label rows now forces, up to relabelling, the support dictionary

$$\begin{array}{c|c} T, T' & 123, 124 \\ U_1, U_2 & 1, 2 \\ a, b, c, d & 13, 14, 23, 24 \\ e, f & 34, 34. \end{array} \tag{70}$$

Indeed, if t_i, s_i count triple and singleton supports at label i, the five-incidence row gives $s_i = t_i - 1$; hence the two triples meet in two labels and the singletons carry those labels. Also

$$\sum_i w_i = \mathbf{1}_T + \mathbf{1}_{T'} - \mathbf{1}_{U_1} - \mathbf{1}_{U_2}. \tag{71}$$

A four-block through T cannot contain T', and (71) forces at least one singleton. Splitting according as it contains both, only U_1, or only U_2, the four intersection caps leave precisely

$$\begin{matrix} TU_1U_2e, & TU_1de, & TU_2be, \\ TU_1U_2f, & TU_1df, & TU_2bf. \end{matrix} \tag{72}$$

Within each row the first entry conflicts with the other two, and the three same-column pairs between rows conflict. Thus at most three coexist; equality has one of the two row-switched orientations. The same holds at T'. Both points must consequently have $(d_3, d_4, d_5) = (9, 3, 3)$, and the orientations must be opposite, since equal orientations would produce two four-blocks sharing U_1, U_2 and one of e, f. Interchanging e, f if necessary, the following six blocks are forced:

$$\begin{matrix} TU_1U_2e, & TU_1df, & TU_2bf, \\ T'U_1U_2f, & T'U_1ce, & T'U_2ae. \end{matrix} \tag{73}$$

Every remaining four-block avoids T, T'. Its intersections with the four five-blocks are bounded by

$$\begin{matrix} |D \cap \{U_1, a, b\}| \le 2, & |D \cap \{U_2, c, d\}| \le 2, \\ |D \cap \{a, c, e, f\}| \le 2, & |D \cap \{b, d, e, f\}| \le 2, \end{matrix} \tag{74}$$

and compatibility with (73) forbids the six triples

$$U_1U_2e,\ U_1df,\ U_2bf,\ U_1U_2f,\ U_1ce,\ U_2ae. \tag{75}$$

According as both, one, or neither singleton is used, (74)–(75) give exactly

$$\begin{array}{r|l} U_1, U_2 & U_1U_2ac,\ U_1U_2ad,\ U_1U_2bc,\ U_1U_2bd \\ U_1 & U_1acd,\ U_1ade,\ U_1bcd,\ U_1bcf \\ U_2 & U_2abc,\ U_2abd,\ U_2adf,\ U_2bce \\ \text{neither} & abcd. \end{array} \tag{76}$$

For example, with both singletons the forbidden triples exclude e, f and the first two inequalities choose one of a, b and one of c, d; with only U_1, choosing a leaves cd or de, choosing b leaves cd or cf; the other row is symmetric. With no singleton, the last two inequalities exclude e, f, proving that the list is exhaustive.

Name the thirteen entries of (76) rowwise $A, \dots, M$. The seven sets

$$\{A, B\}, \{C, D\}, \{E, F\}, \{G, H\}, \{I, L\}, \{J, K\}, \{M\} \tag{77}$$

are conflict cliques, so at most seven residual blocks coexist. Equality uses M and therefore forces successively F, H, L, K, A, D; it is uniquely

$$U_1U_2ac,\ U_1U_2bd,\ U_1ade,\ U_1bcf,\ U_2adf,\ U_2bce,\ abcd. \tag{78}$$

Let α be the number of points with $d_3 = 9$. Summing d_3 and the triple partition gives $B_3 = 20 + \alpha$ and $B_4 = 15 - \alpha/4$. Since T, T' are high, integrality leaves $\alpha = 4$ or 8. If $\alpha = 4$, then $B_4 = 14$, but the preceding caps give at most $6 + 7 = 13$. Hence $\alpha = 8$, $B_4 = 13$, and all blocks in (73) and (78) occur.

Invert at $T = 123$. The three incident five-blocks are the sides of a nondegenerate triangle with vertices $A = T'$, $B = a$, and $C = c$. Use directed parameters

$$\begin{array}{c|l} AB & A=0,\ B=1,\ U_1=u,\ b=v \\ AC & A=0,\ C=1,\ U_2=s,\ d=t \\ BC & B=0,\ C=1,\ e=x,\ f=y. \end{array} \tag{79}$$

The first row of (73) gives the collinearities

$$U_1U_2e, \qquad U_1df, \qquad U_2bf, \tag{80}$$

while its second row gives the cyclic quadruples

$$AU_1U_2f, \qquad AU_1Ce, \qquad AU_2Be. \tag{81}$$

The nonincident five-block and the last residual block are the proper circles

$$\{A, b, d, e, f\}, \qquad \{B, b, C, d\}. \tag{82}$$

Here $u - s \neq 0$: equality would put the finite, distinct point e at the pole of the first collinearity U_1U_2e, contrary to the chosen finite side-parameter chart. No affine transformation of the ambient Euclidean plane is being used here; the parameters in (79) are directed coordinates on the three actual sides of the inverted triangle. Menelaus applied to (80) gives

$$x = \frac{s(u-1)}{u-s}, \qquad y = \frac{t(u-1)}{u-t} = \frac{s(v-1)}{v-s}. \tag{83}$$

Put $R = |AC|^2/|AB|^2 > 0$. Subtracting the circle equations in (81) from the first circle in (82), at e and f, yields

$$\begin{aligned} (v-u)(1-x) + R(t-1)x &= 0, \\ (v-1)(1-x) + R(t-s)x &= 0, \\ (v-u)(1-y) + R(t-s)y &= 0. \end{aligned} \tag{84}$$

The first two equations together with (83) give $R = -u/s$; power of A to the second circle in (82) then gives $v = Rt = -ut/s$. Substitution in (83) and (84) produces

$$\begin{aligned} s^2u + st - s - tu &= 0, \\ 2stu - st - su + t^2 - tu &= 0, \\ s^2t - s^2 + st^2 - stu + t^2u - t^2 &= 0. \end{aligned} \tag{85}$$

All cleared factors are nonzero because the named side-points are distinct. In particular $t - u \neq 0$ follows from the second collinearity, $s^2 + tu = -s(v - s) \neq 0$ from the third, and $s^2 - t \neq 0$ below would otherwise force $t = 1$. The first equation in (85) therefore gives

$$u = \frac{s(1-t)}{s^2-t}.$$

The remaining two become

$$\begin{aligned} (s+t)(s^2t - 3st + s + t^2) &= 0, \\ (s+t)(s^3t - s^3 + s^2t - 2st^2 + t^2) &= 0. \end{aligned} \tag{86}$$

If $s + t = 0$, then $v = u$, contrary to distinctness. Subtracting $(1 - 2s)$ times the first remaining factor from the second gives $s(s-1)^2(3t-1) = 0$. Hence $t = 1/3$, and the first factor becomes $3s^2 + 1 = 0$, impossible over $\mathbb{R}$. This closes the last row of (67). $\square$

Lemma 3.6 (Three-pentagon rich-line endpoint)**.** *Assume the hypotheses of lemma 3.3 and the aggregate row*

$$C = 32, \quad B_5 = 3, \quad |\{p : d_3(p) = 9\}| = 2, \quad L_5 = 0, \quad L_4 = h \in \{1, 2, 3\}, \quad L_3 = 10 - h.$$

Then the configuration is impossible.

Proof. Put $r(p) = d_5(p)$ and $q(p) = \ell_4(p)$. On each four-line D, maximality gives

$$\sum_{p \in D} r(p) = \sum_{i=1}^{3} |D \cap B_i| \le 6. \tag{87}$$

The degree sum is 15. Convexity and the pair-intersection cap give $5 \le \sum_p \binom{r(p)}{2} \le 6$, and the only degree profiles are

$$P_1 = (1^5, 2^5), \qquad P_2 = (1^6, 2^3, 3), \qquad P_3 = (0, 1^3, 2^6). \tag{88}$$

Indeed, the lower moment is the distribution $(1^5, 2^5)$; increasing it by one either replaces two 1's by $0, 2$, giving P_3, or replaces two 2's by $1, 3$, giving P_2. The $q = 0$ rows of (62) give all-low triple-line capacities

$$25, \qquad 27, \qquad 25, \tag{89}$$

and each of the two high designations raises a baseline by at most one.

For $h = 1$, (87) forces at least two degree-one points on the line in P_1, P_2; each loses two units from its $q = 0$ capacity, even if high. The total capacities are therefore at most $23, 25 < 3L_3 = 27$. In P_3 the line contains either the degree-zero point or two degree-one points. Making the zero point high reduces its line loss by one but gives no baseline gain, so the total is at most $25 < 27$.

For $h = 2$, two lines need four degree-one incidences in P_1, P_2. If their intersection is degree one, its $q = 2$ row pays two units and the two remaining incidences pay two each; otherwise the loss is larger. Starting from the high-upgraded maxima $27, 29$ leaves at most $21, 23 < 3L_3 = 24$. For P_3 the cheapest possibility is that the lines meet at the degree-zero point; (62) still leaves at most $23 < 24$.

Suppose $h = 3$. The twelve line incidences have only three possible intersection topologies: a common point of all three lines, two distinct pairwise intersections, or three distinct pairwise intersections. Indeed, without a common point their union has size $12 - j \le 10$, where j is the number of selected pairwise intersections, so $j = 2$ or 3. Measure loss from (89), already crediting the two possible high-point discounts. A degree-one point on one or two lines costs two, a degree-two point costs zero on one line and two on two, and a degree-zero point costs two on one line and four on two or three. Directly from these four costs and (87), the losses are

$$\begin{array}{c|ccc} & \text{common point} & \text{two intersections} & \text{triangle} \\ \hline P_1 & 10 & 8 & 6 \\ P_2 & 14 & 12 & 10 \\ P_3 & 8 & 6 & 6. \end{array} \tag{90}$$

Here is a derivation rather than a case catalogue. In the common-point topology that point cannot have degree two; the first two rows cost $3 + 4 \cdot 2 - 1 = 10$ and $3 + 5 \cdot 2 + 2 - 1 = 14$, while the third costs at least 8. With two intersections, let x be the number of degree-one double points and z indicate that the degree-zero point is double. Before the two discounts the first two losses are at least $4 + 2(5 - x)$ and $4 + 2(6 - x) + 2$; for P_3 they are

$4z+2(2-z)+2(1-z)+2(3-x)-2=10-2x\geq 6$. In the triangle topology let x count degree-one vertices and let ϵ indicate whether the off-line point has degree one or zero. The first row costs at least $6+2(5-x-\epsilon)-2\geq 6$; the analogous P_2 expression has the extra degree-three deficit and is at least 10. For P_3 the exact loss is

$$12-2x-2\epsilon. \tag{91}$$

It is at least six except when the three triangle vertices are exactly the three degree-one points and the off-line point is the degree-zero point. In that sole exception each side contains two degree-two points and both high discounts are attained; the loss is four. All other entries of (90) leave capacity below $3L_3=21$.

Let p be the off-line degree-zero point in the equality pattern. It is low, $\ell_3(p)=4$, and (61) gives

$$(d_3(p),d_4(p),d_5(p))=(6,10,0).$$

After inversion at p, the other nine points have line profile $(t_2,t_3,t_4)=(6,10,0)$. By lemma 3.11, their six ordinary lines form $C_3\sqcup C_3$. The four original three-lines through p would give four pairwise disjoint ordinary edges, but $\nu(C_3\sqcup C_3)=2$. This final contradiction closes the endpoint. ☐

Lemma 3.7 (Disjoint golden-link endpoint)**.** *Assume the hypotheses of lemma 3.3 and the aggregate row*

$$C=32,\quad B_3=B_4=20,\quad B_5=2,\quad L_3=10,\quad L_4=L_5=0,\quad d_3(p)=6 \textit{ for every } p. \tag{92}$$

Then the configuration is impossible.

Proof. Let the two five-blocks meet in $m=0,1,$ or 2 selected points. Their degree profiles are, respectively,

$$(1^{10}),\qquad (0,1^8,2),\qquad (0^2,1^6,2^2). \tag{93}$$

The $q=0$, low row of (62) gives total triple-line capacity exactly 30 in each profile, equal to $\sum_p \ell_3(p)=3L_3$. Thus every point attains its local maximum. If $m>0$, choose a degree-zero point p. Then $(d_3,d_4,d_5,\ell_3)(p)=(6,10,0,4)$. Inversion at p gives the nine-point line profile $(t_2,t_3,t_4)=(6,10,0)$, whose ordinary graph is $C_3\sqcup C_3$ by lemma 3.11. The four original lines through p would be a four-edge matching in this graph, contradicting $\nu(C_3\sqcup C_3)=2$. Hence $m=0$.

The blocks are therefore disjoint proper circles; call their point sets A,B. Every four-block has two points on each, since three on a base circle would repeat an owned triple. Fixing an A-pair, at most two four-blocks contain it, because their B-pairs must be disjoint. There are ten A-pairs and exactly twenty four-blocks, so equality holds: every A-pair has exactly two B partners. Symmetrically every B-pair has exactly two A partners.

Use the quadratic lift and lemma 3.12. No lift of an A-point lies in Π_B, since Π_B would then cut the quadric in a generalized block containing the five points of B and that sixth selected point; symmetrically no B-point lies in Π_A. Thus the base planes are distinct and meet in a projective line ℓ. A $2A+2B$ block plane cuts them in two chords whose chord lines meet ℓ at the same centre; conversely, an A-chord and a B-chord with the same centre span a block plane. That plane cannot acquire a fifth selected point, since it is different from the two base planes and (92) contains only the two five-blocks A,B. At one centre the chords on either base form a matching. If it carries α A-chords and β B-chords, all $\alpha\beta$ cross-pairs are four-blocks; the exact partner degree two of every chord gives

$$\alpha=\beta=2. \tag{94}$$

Thus the edges of each K_5 split into five two-edge matchings, in common centre correspondence. By lemma 3.13(P2), after labelling the B-vertices $0, \dots, 4$ their matching rows are

$$M_0 = 13|24,\ M_1 = 02|34,\ M_2 = 03|14,\ M_3 = 04|12,\ M_4 = 01|23. \tag{95}$$

Choose $p \in A$. Row 0 is the unique A-matching omitting p. For $i = 1, \dots, 4$, let q_i be the other endpoint of the A-chord through p whose centre corresponds to M_i. After inversion at p, the four q_i lie on the image line of A, and q_i is the intersection of the two B-chords in M_i. Normalize the five B-points as

$$b_0 = (1,0,0),\ b_1 = (0,1,0),\ b_2 = (0,0,1),\ b_3 = (1,1,1),\ b_4 = (a,b,1). \tag{96}$$

The four relevant centres are

$$q_1 = (a-b, 0, 1-b), \quad q_2 = (a,1,1), \quad q_3 = (0,b,1), \quad q_4 = (-1,-1,0). \tag{97}$$

Their collinearity gives, exactly as in lemma 3.13(P3),

$$a = 1 - b = b^2, \qquad b^2 + b - 1 = 0. \tag{98}$$

The eight four-blocks through p give, at q_i, the two three-lines prescribed by M_i. These cover every q_iB pair except q_ib_i and every internal B-pair except the two edges of M_0. Hence the six ordinary edges in the inverted link are exactly

$$q_ib_i\ (1 \le i \le 4), \qquad b_1b_3, \qquad b_2b_4, \tag{99}$$

namely the two paths

$$q_1 - b_1 - b_3 - q_3, \qquad q_2 - b_2 - b_4 - q_4.$$

In the disjoint profile every point has $r = 1$; the local table and $\sum_p \ell_3(p) = 30$ force $\ell_3(p) = 3$ everywhere. Thus the three original lines through p give a concurrent three-edge matching in these two paths. Such a matching takes both end edges of one path and one edge of the other. The automorphism $\rho = (1234)$ of (95) interchanges the paths, while ρ^2 reverses both. Consequently there are only two orbits, represented by

$$\{q_1b_1, q_3b_3, q_2b_2\}, \qquad \{q_1b_1, q_3b_3, b_2b_4\}. \tag{100}$$

Writing a line as the cross product of its endpoint vectors, their concurrency determinants are

$$\begin{aligned} \det(q_1 \times b_1, q_3 \times b_3, q_2 \times b_2) &= -(a-1)(ab - a + b), \\ \det(q_1 \times b_1, q_3 \times b_3, b_2 \times b_4) &= a^2b - a^2 + ab - b^2. \end{aligned} \tag{101}$$

On the golden wall (98) these reduce to $4 - 6b$ and $11b - 7$. Their possible roots $2/3$ and $7/11$ do not satisfy $b^2 + b - 1 = 0$. Neither matching in (100) is concurrent, the required contradiction. □

Lemma 3.8 (Ten-point five-block kernel)**.** *Assume $|P| = 10$, the largest proper circle has size five, and $C \le 32$. Put*

$$x = B_3, \quad b = B_4, \quad k = B_5, \quad L = L_3 + L_4 + L_5, \quad h = L_4, \quad g = L_5.$$

Then no such configuration exists.

Proof. The rich-line bound excludes lines of size at least six. Triple ownership, the circle objective, and the two Melchior rows give

$$x + 4b + 10k = 120, \tag{102}$$

$$C = 30 + \frac{3}{4}x - \frac{3}{2}k - L, \tag{103}$$

$$15x - 34k + 28h + 64g \le 36C - 840. \tag{104}$$

By lemma 3.3, $d_3(p) \in \{6, 9\}$. Let α be the number of high points, those with $d_3 = 9$. Summing d_3 gives

$$x = 20 + \alpha. \tag{105}$$

For $r_p = d_5(p)$ we also have

$$\sum_p r_p = 5k, \qquad \sum_p \binom{r_p}{2} \le 2\binom{k}{2}. \tag{106}$$

We first remove the large values of k. The centred-vector argument and the real triangle-power obstruction in lemma 3.13(P1) give $k \le 5$. If $k = 5$, convexity and (106) force $(r_p) = (3^5, 2^5)$ and equality in every pair intersection. At a degree-three point, lemma 3.4 gives $d_4 \le 2$, whereas (61) gives $d_4 = 3$ or 4. Thus $k \ne 5$; lemma 3.5 removes $k = 4$. We have proved

$$1 \le k \le 3. \tag{107}$$

Values $C \le 27$ disappear before any further routing. Indeed, $d_3(p) = c_3(p) + \ell_3(p) \le (C - 25) + 4 = C - 21$, while $d_3(p) \ge 6$. At $C = 27$ equality would force $\ell_3(p) = 4$ at all ten points, so $3L_3 = 40$, impossible.

For $C \ge 28$, equations (102)–(104) and (105) become the exact scalar router

$$\begin{aligned} \alpha + 2k &\equiv 0 \pmod 4, \\ L &= 45 - C + \frac{3\alpha - 6k}{4}, \\ 15\alpha - 34k + 28h + 64g &\le 36C - 1140. \end{aligned} \tag{108}$$

For $k = 1, 2, 3$, the least nonnegative α allowed by the congruence is $2, 0, 2$, respectively; increasing it by four raises the last left side by 60. Comparing with the five right sides $-132, -96, -60, -24, 12$ for $C = 28, \ldots, 32$ therefore gives, without a configuration search, the complete list

C	(k, α, L)	(h, g)
$28, 29$	none	–
30	$(2, 0, 12), (3, 2, 12)$	$(0, 0)$
31	$(2, 0, 11), (3, 2, 11)$	$(0, 0), (1, 0)$
32	$(1, 2, 13), (2, 4, 13), (3, 6, 13)$	$(0, 0)$
	$(2, 0, 10)$	$(0, 0), (1, 0), (2, 0), (0, 1)$
	$(3, 2, 10)$	$(0, 0), (1, 0), (2, 0), (3, 0), (0, 1).$

(109)

For example, at $C = 30$ the two base values in the last inequality are $-68, -72$, leaving only $8, 12$ units and hence $h = g = 0$; the other line-state columns follow by the same displayed costs $28, 64$.

It remains to apply the local table, and we record every capacity comparison. For two five-blocks the three degree profiles (93) have all-low capacity 30; for three five-blocks the profiles (88) have all-low capacities $25, 27, 25$. Each high point adds at most one. Thus at $C = 30$ the two rows in (109) have capacities at most $30, 29$, below $3L_3 = 36$. At $C = 31$, the three-block row has capacity at most 29, below both required totals $33, 30$. In the two-block row, $h = 0$ gives $30 < 33$; on a four-line D one has $\sum_{p\in D} r_p \le 4$, and the weak loss $2 - r_p$ sums to at least four, leaving at most $26 < 30$ when $h = 1$.

At $C = 32$, the three $L = 13$ rows have capacities at most

$$37, \qquad 34, \qquad 33, \tag{110}$$

against the required 39. For the first number, the five points of the unique five-block contribute 3 each, its outsiders 4 each, and at most two high points on the block add one; the other two numbers are $30 + 4$ and $27 + 6$.

Consider $(k, \alpha, L) = (2, 0, 10)$. A five-line loses at least one unit at each of its points, leaving at most $25 < 27$. For one or two four-lines, sum the weak loss $2 - r_p$ on each line; it is at least four because $\sum_D r_p \le 4$. This addition does not overcount at an intersection: for $r = 2, 1, 0$ the two-line losses from (62) are respectively $2, 2, 4$, at least the corresponding doubled weak weights $0, 2, 4$. Hence capacity is at most $30 - 4h < 30 - 3h = 3L_3$. Only $h = g = 0$ survives, and it is excluded by lemma 3.7.

Finally take $(k, \alpha, L) = (3, 2, 10)$. The no-rich-line state has capacity at most $29 < 30$, and a five-line leaves at most $29 - 5 < 27$. The three remaining states $h = 1, 2, 3$, $g = 0$, are precisely the hypotheses of lemma 3.6. This exhausts (109) and completes the five-circle branch. □

Lemma 3.9 (The ten-point six-circle certificate)**.** *If ten admissible points contain a proper circle with at least six selected points, then* $C \ge 33$.

Proof. The rich-circle bound gives $46, 45, 33$ for occupancies $7, 8, 9$; the rich-line bound gives $54, 52, 36$ for line occupancies $7, 8, 9$. Thus a counterexample has a maximal six-circle Γ, four outsiders, and no block larger than six.

Let L_{ij}, C_{ij} count line and circle blocks containing i points of Γ and j outsiders. We use

$$\sum ijL_{ij} \le 24, \qquad \sum C_{ij} \le 32, \qquad \sum \left(\binom{i+j}{2} + i + j - 3 \right) L_{ij} \le 42. \tag{111}$$

the U17 row

$$\sum_j \binom{j}{2} C_{2j} \le 17,$$

and, for $q = 0, 1, 2, 3$, the four exact triple rows

$$\sum_{i,j} \binom{i}{q} \binom{j}{3-q} (L_{ij} + C_{ij}) = \binom{6}{q} \binom{4}{3-q}.$$

Also $C_{60} = 1$. Multiply the three inequalities in (111), the U17 row, and the four equalities by

$$24, 84, 12, 82; \qquad -21, 1, -84, 684,$$

and add $-13764C_{60}$. A block distinct from Γ meets it in at most two points, so there are exactly nine types. Their coefficients are

(i, j)	$(0, 3)$	$(0, 4)$	$(1, 2)$	$(1, 3)$	$(1, 4)$	$(2, 1)$	$(2, 2)$	$(2, 3)$	$(2, 4)$
L_{ij}	15	0	85	138	162	0	14	21	0
C_{ij}	63	0	85	66	6	0	0	63	168.

The selected circle has coefficient zero, so the combined left side is nonnegative. The combined right side is

$$24 \cdot 24 + 84 \cdot 32 + 12 \cdot 42 + 82 \cdot 17 - 21 \cdot 4 + 36 - 84 \cdot 60 + 684 \cdot 20 - 13764 = -10,$$

a contradiction. □

Proposition 3.10. *One has $f(10) = 33$.*

Proof. Assume $C \leq 32$ and split by the maximum proper-circle size. The three possibilities at most four, exactly five, and at least six are excluded by lemmas 3.2, 3.8 and 3.9. Construction 2.5 has 33 circles. □

3.3 Real link and circle-lift modules

We next isolate the real geometry used more than once. These statements are independent of the endpoint arithmetic.

Lemma 3.11 (The real nine-point link)**.** *Let Q be nine distinct real projective points, no four collinear, with exactly ten three-point lines. Then it has six ordinary lines, whose graph is*

$$C_3 \sqcup C_3 \sqcup 3K_1.$$

Consequently ten such points, no four collinear, have at most twelve three-point lines, and eleven such points have at most sixteen.

Proof. We first need the eight-point Orchard fact: eight noncollinear real projective points, no four collinear, have at most seven three-point lines. Indeed, eight such lines would leave four ordinary lines. Local pair counting makes the ordinary lines a perfect matching and every point lie on three three-lines. Label its four pairs A, B, C, D. Pair ownership forces, up to relabelling, the eight triples

$$A_0B_0C_0,\ A_1B_1C_1,\ A_0B_1D_0,\ A_1B_0D_1,$$
$$A_0C_1D_1,\ A_1C_0D_0,\ B_0C_1D_0,\ B_1C_0D_1.$$

Normalize A_0, A_1, B_0, B_1 to a projective frame. Writing the remaining four points with parameters u, v, w, z, four of the incidences give $vz = uw = vw = 1$ and $u + u^{-1} = 1$. Thus $u^2 - u + 1 = 0$, impossible over $\mathbb{R}$.

For the nine-point set, pair counting gives $t_2 = 6$. If k_v and e_v are the three-line and ordinary degrees at v, deleting v and applying the preceding fact gives $k_v \geq 3$, while

$$8 = 2k_v + e_v.$$

Hence three points have degree pair $(4, 0)$ and six have $(3, 2)$. The ordinary graph is either $C_6 \sqcup 3K_1$ or the graph in (112). We reject the first alternative directly. Calling the isolated vertices U_6, U_7, U_8, the ten three-lines have one of the two distributions

$$(a_3, a_2, a_1, a_0) = (1, 0, 9, 0), \qquad (0, 3, 6, 1),$$

according to the number of U-vertices on a line. In the first case the nine UVV lines force the three matchings

$$02, 14, 35; \qquad 03, 15, 24; \qquad 04, 13, 25.$$

Sending the U-line to infinity and using two independent directions reduces the last required parallelism to $2/\lambda = 0$. In the second case a projective frame at U_6, U_7, U_8 reduces the last three incidences to

$$\alpha\gamma = 1, \qquad \beta\gamma = -1, \qquad \alpha\beta = 1,$$

and hence $\beta^2 = -1$. Both are impossible over $\mathbb{R}$, proving

$$\mathrm{Ord}(Q) = C_3 \sqcup C_3 \sqcup 3K_1. \tag{112}$$

If ten points had thirteen three-lines, their ordinary-degree multiset would be $(3, 1^9)$, so the ordinary graph would be $K_{1,3} \sqcup 3K_2$. Delete its degree-three vertex. The remaining ordinary graph is $C_6 \sqcup 3K_1$ and there remain ten three-lines, contrary to (112). Fourteen or more three-lines leave fewer than five ordinary pairs, contradicting the positive odd local ordinary degrees. For eleven points, seventeen three-lines leave an ordinary C_4; deleting a cycle vertex gives the forbidden ten-point case. Eighteen leave one ordinary edge although every ordinary degree is even. This proves both ceilings. □

Use the circular lift

$$v(x, y) = [x^2 + y^2 : x : y : 1] \in \mathcal{Q} : \;\; XW - Y^2 - Z^2 = 0 \subset \mathbb{P}^3(\mathbb{R}).$$

The real quadric $\mathcal{Q}$ contains no real projective line. A line or proper circle in the original plane is a plane section of $\mathcal{Q}$.

Lemma 3.12 (Trace dictionary)**.** *Let a selected generalized block have lift plane H_B, and let H be the plane of a selected proper circle Γ. Then:*

(T1) *$H_B \cap H$ is one trace line, containing every Γ-chord and every outsider-secant direction belonging to B.*

(T2) *For a fixed outsider pair, its host chords on Γ form a matching. Two hosts determine their diagonal centre on the trace.*

(T3) *Equal direction points cannot occur on adjacent outsider edges; a direction fibre is a matching.*

(T4) *The three diagonal points of a complete quadrangle on a nonsingular real conic are distinct and noncollinear (indeed, self-polar).*

In particular, a block containing one point $\gamma_i \in \Gamma$ and three outsiders cannot have all three outsider edges double-hosted by chords avoiding γ_i.

Proof. Plane sections give (T1). If two hosts shared γ, they would both own the selected triple consisting of γ and the outsider pair, proving (T2). Equal directions on xy, xz would put the three real quadric points v_x, v_y, v_z on one real line, proving (T3). In a conic frame $[t^2 : t : 1]$, the three diagonal points are the poles of the three diagonal sides, which proves (T4). The final assertion would put all three diagonal points and γ_i on the trace of one block. □

Lemma 3.13 (Pentagon, golden axis, and polarity)**.** *Let $\mathcal{F}$ be a family of five-subsets of a ten-set, any two meeting in at most two points.*

(P1) *$|\mathcal{F}| \leq 6$. At equality every point has degree three and every pair of blocks meets in two points. Up to relabelling the incidence design is*

$$123, 124, 135, 236, 146, 156, 245, 256, 345, 346. \tag{113}$$

Six such blocks cannot all be realized as generalized blocks of ten real points.

(P2) *Five two-edge matchings which partition $E(K_5)$ form, up to relabelling, the near-one-factorization*

$$13|24,\ 02|34,\ 03|14,\ 04|12,\ 01|23. \tag{114}$$

(P3) *Let five points lie on a real conic and let the centres in (114) be collinear. After a projective normalization the modulus satisfies*

$$a = 1 - b = b^2, \qquad b^2 + b - 1 = 0. \tag{115}$$

The fifth centre is then on the same line. The five opposite diagonal lines concur at its pole; no selected chord passes through the pole, and the tangent at an omitted point contains its opposite centre but no second centre omitting that point.

Proof. For the incidence vectors x_F put $y_F = x_F - \frac{1}{2}\mathbf{1}$. Then $\|y_F\|^2 = 5/2$ and $\langle y_F, y_G\rangle \le -1/2$. The squared norm of $\sum_F y_F$ gives $|\mathcal{F}| \le 6$; equality forces all intersections and degrees stated in (P1). Deleting one block, the degree-two points and the complements of the degree-three points are complementary five-cycles, which forces (113).

We make the geometric part of (P1) explicit. Invert a putative real realization at the point labelled 123. Since equality makes every pair of blocks meet in two selected points, at most one of the six generalized blocks can be an original line. The three blocks through 123 have the distinct second intersections

$$A = 124, \qquad B = 135, \qquad C = 236.$$

Their images are therefore three pairwise nonparallel lines. They cannot be concurrent, since their pairwise intersections A, B, C are distinct; the possible original line through the inversion centre causes no exception. Thus these lines form a nondegenerate triangle ABC. Every nonincident block becomes a proper circle: a proper circle missing the pivot remains proper, while the sole possible original line missing the pivot becomes a proper circle through the inversion centre. Inversion is injective away from the pivot, so the remaining nine selected points stay distinct. Write

$$X = 146, \quad Y = 156, \quad U = 245, \quad V = 256, \quad W = 345, \quad Z = 346.$$

The three nonincident circles have supports

$$A, X, U, W, Z; \qquad B, Y, U, V, W; \qquad C, X, Y, V, Z.$$

Use directed affine parameters on the triangle sides

$$\begin{array}{c|cccc} AB & A = 0 & B = 1 & X = x & Y = y \\ AC & A = 0 & C = 1 & U = u & V = v \\ BC & B = 0 & C = 1 & W = w & Z = z \end{array}$$

and put $c^2 = |AB|^2$, $b^2 = |AC|^2$, and $a^2 = |BC|^2$. Directed power at the three vertices, applied to the displayed circles, gives all six identities

$$\begin{aligned} c^2(1-x) &= a^2 wz, & b^2(1-u) &= a^2(1-w)(1-z), \\ c^2 y &= b^2 uv, & a^2(1-w) &= b^2(1-u)(1-v), \\ c^2 xy &= b^2 v, & c^2(1-x)(1-y) &= a^2 z. \end{aligned}$$

The side lengths are positive, and distinctness from the side vertices says that x, y, u, v, w, z and all six complements $1-x, 1-y, 1-u, 1-v, 1-w, 1-z$ are nonzero. Hence the indicated divisions are legal. Dividing the last-row identities by the corresponding identities above, and comparing the second and fourth identities, gives

$$x = u^{-1}, \qquad 1 - y = w^{-1}, \qquad 1 - z = (1 - v)^{-1}.$$

Set $R = b^2/c^2$ and $S = a^2/c^2$. The first and third identities now give

$$R = \frac{w - 1}{wuv}, \qquad S = \frac{(u - 1)(v - 1)}{uwv}.$$

After division by c^2, the fourth identity is

$$S(1 - w) = R(1 - u)(1 - v).$$

Its two sides are respectively $-K$ and K, where

$$K = \frac{(u - 1)(v - 1)(w - 1)}{uvw} \neq 0.$$

Thus $-K = K$ over the reals, a contradiction. This proves the geometric nonrealizability in (P1).

For (P2), fix the matching missing vertex 0; its two edges force the next unused edge at each of the other four vertices, and four successive unused edge rows give (114). For (P3), put four conic points at $[1 : 0 : 0], [0 : 1 : 0], [0 : 0 : 1], [1 : 1 : 1]$ and the fifth at $[a : b : 1]$. Collinearity of the first four matching centres gives $a = 1 - b = b^2$. Direct substitution places the fifth centre on their line. Polarity sends that line to the common point of the five opposite sides. La Hire's theorem then gives the tangent statement; a second centre on the tangent would make one self-polar diagonal triangle collinear. □

Lemma 3.14 (Five-conic trace rigidity)**.** *Let $\Gamma = \{\gamma_0, \dots, \gamma_4\}$ be five cyclically ordered points on a nonsingular real conic. Work in the trace-dictionary setting above and assume that Γ is a maximum selected generalized five-block, with lift plane Π. Thus every outsider lift lies outside Π, and no plane section of $\mathcal{Q}$ contains more than five selected lifts. For each i, let D_i be the diagonal triangle of $\Gamma - \{\gamma_i\}$, and let δ_i be the vertex opposite the unique side whose line meets the missing real arc at γ_i. Then the following hold.*

(K2.1) *A one-Γ block cannot contain an all-double outsider triangle. If exactly one edge is single and the other two are double, its host chord is one of*

$$\gamma_{i-1}\gamma_{i+1}, \qquad \gamma_{i-2}\gamma_{i+2}. \tag{116}$$

These five two-element chord classes partition the ten chords. Consequently a fixed single-hosted outsider pair occurs in at most one such one-Γ block.

(K2.4) *An all-double outsider K_5 has exactly five centres, one omitting each colour; its trace is the golden axis. The tangent separation in lemma 3.13 forbids extending it by a one-colour page with two adjacent double edges omitting that colour.*

Proof. The all-double assertion in (K2.1) is the final clause of lemma 3.12. For the remaining assertion, in the conic frame $[t^2 : t : 1]$, the three diagonal sides of four cyclic points meet the missing arc in exactly one side, and its opposite matching is (116). The five resulting two-chord classes are disjoint and exhaust the ten chords. A fixed single outsider edge has one host chord, which belongs to exactly one of these classes; once its colour i is fixed, triple ownership forbids two blocks containing the same triple $\gamma_i xy$. This proves the fixed-single consequence.

For (K2.4), the ten edges of an all-double outsider K_5 have matching direction fibres, so its trace contains at least five diagonal centres. The trace is not a Γ-chord: otherwise its block plane would contain the five outsiders and the two selected endpoints of that chord, contrary to maximality of Γ. A nonchord trace meets each of the ten Γ-chords once. Each diagonal centre uses two such chords, and two distinct centres on the trace cannot reuse one, so there are at most five. Equality partitions the ten chords into five two-edge matchings. By lemma 3.13(P2), these form the near-one-factorization (114), with exactly one matching omitting each colour; by (P3), their trace is the golden axis and $\gamma_i\delta_i$ is its polar tangent.

Now consider a one-colour page of colour i with two adjacent double outsider edges. Their direction points are distinct by the no-real-line clause of the trace dictionary. Triple ownership makes both host matchings omit i, so both directions are vertices of D_i. The direction of the edge lying in the outsider K_5 is its unique centre δ_i omitting i; the page trace is therefore the tangent $\gamma_i\delta_i$. The second direction would be a second vertex of D_i on that tangent, contrary to lemma 3.13(P3). □

Lemma 3.15 (Six-marked conic and repetition events)**.** *Let six cyclically ordered selected points lie on a proper circle Γ. For an outsider edge $e = xy$, let $q_e \leq 3$ be the number of circles through x, y and a pair of points of Γ.*

(S1) *There are at most four full signatures $q_e = 3$; the support of each is a matching. For any four outsiders Y,*

$$\sum_{e\in\binom{Y}{2}} q_e \leq 17. \tag{117}$$

On any fixed trace line lie the centres of at most three full signatures.

(S2) *Two disjoint edges carrying the same signature determine a unique maximal generalized host of their four endpoints. If such a host contains b outsiders and $R(H)$ repetition events, then*

$$R(H) \leq \min\left\{2\binom{b}{4}, 3\binom{\lfloor b/2\rfloor}{2}\right\}. \tag{118}$$

(S3) *For five outsiders put $W = \sum q_e$. A unique four-outsider host, or a five-outsider host disjoint from Γ, gives $W \leq 26$; equality has signature multiplicities $2, 2, 1, 1$ or $2, 2, 2$, respectively. A five-outsider host meeting Γ gives $W \leq 20$.*

(S4) *If J counts incidences of outsiders with original lines carrying a pair of Γ points, then $J \leq 14$; on one fixed line or circle, $J \leq 13$.*

Proof. A real projectivity of the conic preserves or reverses cyclic order. The only fixed-point-free involutions of six cyclic labels are R_3, S_1, S_3, S_5, proving the cap four. If all four occur, they generate the rotation R_1; hence the six points form one orbit of a real projectivity of order six. A projective coordinate on the conic may therefore normalize that orbit to $(0, 1, 3, \infty, -3, -1)$, which gives centres

$$(3, 0, 1), (3, 3, -1), (-3, 0, 1), (-3, 3, 1),$$

exactly three of which are collinear. This also proves the trace-line cap. Matching support follows from lemma 3.12. If all six edges of a four-set were full, a repeated signature would put the three perfect-matching centres on one trace, contradicting the noncollinearity of the diagonal triangle. Hence at least one edge has weight at most two, which is (117).

A repeated signature on xy, zw puts their two lift secants and hence all four lifts in one plane; its pullback is the unique line or circle through the four endpoints. Each four-set supports at most two events by U17, and on one trace there are at most three signatures, each supported on a matching of size at most $\lfloor b/2 \rfloor$. These are the two bounds in (118).

For five outsiders write F for the number of full edges. Then $W \leq 20 + F$. In a unique four-host, at most two signatures repeat, so $F \leq 2 + 2 + 1 + 1 = 6$. On a five-host, at most three trace centres occur and each supports two edges, again $F \leq 6$. If the host contains a selected γ_i, no full matching is possible because it would repeat a triple $\gamma_i xy$, giving $W \leq 20$. Finally, every outsider carries at most three original chord lines. Five degree-three outsiders would require five signatures, and on a fixed host at most three may have degree three. This gives $J \leq 4 \cdot 3 + 2 = 14$ and $J \leq 3 \cdot 3 + 2 \cdot 2 = 13$. ☐

3.4 The real grid and gallery modules

Lemma 3.16 (Real 3×3 grid)**.** *Let two triples of real projective lines form a 3×3 grid. There are at most two further lines meeting three grid points with one point in each row and column. If two exist, the grid is projectively the affine grid $\{-1, 0, 1\}^2$. No point outside this grid lies on three ordinary grid secants.*

Proof. Send the line through the two pencil centres to infinity. A transversal is the graph of an affine bijection between two real three-sets. After one is made the identity, every other is an affine automorphism of a real three-set; a nonconstant affine map is determined by two values, and such a set has at most the identity and its reflection. Equality normalizes both coordinate sets to $\{-1, 0, 1\}$. In this frame the twelve ordinary grid secants are

$$x \pm y = \pm 1, \qquad x \pm 2y = \pm 1, \qquad 2x \pm y = \pm 1.$$

At infinity their six directions are distinct and each occurs exactly twice, so no ideal point lies on three of them. Now let a finite point (u, v) lie on three. If $uv \neq 0$, it can lie on at most one line from each of the pairs with left sides $u \pm v$, $u \pm 2v$, and $2u \pm v$. It must therefore use one from every pair. The first two equations imply $|v| \in \{2, 2/3\}$, while the first and third imply $|v| \in \{1, 3, 1/3\}$, a contradiction. If $uv = 0$, the displayed equations show directly that three incidences are possible only at a point of $\{-1, 0, 1\}^2$. Thus an external point, finite or ideal, lies on at most two ordinary grid secants. ☐

Lemma 3.17 (Gallery skeleton and its two completions)**.** *There is no arrangement of twelve distinct real projective lines with*

$$(t_2, t_3, t_4) = (6, 14, 3) \tag{119}$$

and a distinguished line with complete word $3D + 4T$. Nor is there one with

$$(t_2, t_3, t_5) = (7, 13, 2) \tag{120}$$

and the same distinguished word.

Proof. Melchior is sharp in either case, so the arrangement is simplicial and not a near-pencil. We first make the gallery entrance explicit. Two ordinary vertices cannot be adjacent on a line ℓ: if they are $\ell \cap A$ and $\ell \cap B$, send ℓ to infinity. The two triangular chambers along the intervening edge become opposite sectors bounded by A, B at $A \cap B$. Any further arrangement line not through $A \cap B$ crosses one of those sectors (its direction cannot subdivide the chosen edge), hence cuts a chamber. Thus every further line passes through

$A \cap B$, the excluded near-pencil. The three double vertices therefore separate the four triple vertices into positive cyclic gaps of sizes $2, 1, 1$, so the word is $TTDTDTD$.

Send the distinguished line to infinity. Its four triple directions give pairs (a_i, b_i), $0 \le i \le 3$, and its three ordinary directions give c_1, c_2, c_3. At a consecutive flag $T_{i-1} - D_i - T_i$, let P^+, P^- be the other endpoints of the two edges of c_i incident with D_i. The four triangular chambers at this flag show that each $P^\pm$ lies on c_i, on one of the two lines through T_{i-1}, and on one of the two lines through T_i. On opposite sides of a line germ at a transverse triple vertex the adjacent germs are the two different remaining lines. Hence P^+, P^- define a perfect matching between the two pairs. The three flags form a path, so labels propagate without a monodromy choice and give

$$A_i = \{a_{i-1}, a_i, c_i\}, \qquad B_i = \{b_{i-1}, b_i, c_i\} \quad (1 \le i \le 3), \tag{121}$$

together with the four rungs $\{a_i, b_i\}$.

After deleting their used pairs, split the eleven labels as

$$L = \{a_0, a_1, b_0, b_1, c_1, c_2\}, \qquad R = \{a_2, a_3, b_2, b_3, c_3\}.$$

The only residual within-side edges are

$$H = \{a_0b_1, a_0c_2, a_1b_0, b_0c_2, c_1c_2, a_2b_3, a_3b_2\}. \tag{122}$$

The left part is a tree and the right part a matching; hence every residual triangle uses exactly one edge of H. If e bases in (121) are upgraded by high vertices and those vertices consume h edges of H, pair ownership gives the common budget

$$e + h \le 13 - t_3', \tag{123}$$

where t_3' is the required number of triples after deleting the distinguished line.

For (119), deleting the distinguished line leaves $(t_2', t_3', t_4') = (7, 10, 3)$. A nonupgrading quadruple is a four-clique of the residual graph. It has two labels on each side of the cut. For the right edge a_2b_3 its common left neighbours are $\{a_0, b_0, b_1, c_1\}$, containing only the H-edge a_0b_1; for a_3b_2 they are $\{a_0, a_1, b_0, c_1\}$, containing only a_1b_0. Thus the only two nonupgrading quadruples are

$$\{a_0, a_2, b_1, b_3\}, \qquad \{a_1, a_3, b_0, b_2\},$$

and each consumes two H-edges. If e of the three quadruples upgrade bases, the other $3-e$ consume at least $2(3-e)$ edges of H. Equation (123) then gives $e + 2(3-e) \le 3$, whence $e = 3$ and no H-edge is consumed.

Only A_1, B_1, A_3, B_3 can be upgraded without using H. Track exchange $a \leftrightarrow b$ and gallery reversal act transitively on these four bases, so assume B_3 is omitted and write the upgrades as $A_1 + x, B_1 + y, A_3 + z$. Pair ownership gives

$$\begin{gathered} x \in \{a_3, b_2, b_3, c_3\}, \qquad y \in \{a_2, a_3, b_3, c_3\}, \\ z \in \{a_0, b_0, b_1, c_1\}, \qquad x \ne y, \\ \neg(x \in \{a_3, c_3\} \wedge z \in \{a_0, c_1\}), \\ \neg(y \in \{a_2, a_3, c_3\} \wedge z \in \{b_0, b_1, c_1\}). \end{gathered}$$

Requiring only the two corner bundles a_0b_1 and a_3b_2 to remain nonempty gives the complete four-state propagation certificate

z	(x, y) still possible
a_0	$(b_2, a_2), (b_2, c_3)$
b_0	(c_3, b_3)
b_1	$\varnothing$
c_1	(b_2, b_3).

The three states

$$(b_2, a_2, a_0), \qquad (b_2, b_3, c_1), \qquad (c_3, b_3, b_0)$$

empty, respectively, the bundles at a_2b_3, c_1c_2, b_0c_2. Hence only $(x, y, z) = (b_2, c_3, a_0)$ survives. There all five left H-edges can be completed only through the two poles a_3, b_3. After the mandatory triangles on the two right H-edges, their residual capacities are

$$(\lfloor 6/2 \rfloor - 1) + (\lfloor 7/2 \rfloor - 1) = 4 < 5.$$

This proves the first assertion by a fixed corner certificate.

For (120), deletion leaves $(t'_2, t'_3, t'_5) = (8, 9, 2)$. For a fivefold vertex F, let $e(F)$ be the number of bases it upgrades and let $h(F)$ be the number of edges of H among its pairs. If $e(F) = 0$, all ten pairs of F lie in the residual graph. Each side of the cut in (122) is triangle-free, so F has at most two labels on either side, contrary to $|F| = 5$. Hence $e(F) \geq 1$.

Two upgraded bases must be consecutive on the same rail:

$$A_1A_2, \quad A_2A_3, \quad B_1B_2, \quad B_2B_3.$$

Indeed, these are the only base pairs whose union has size five; the apparent pair A_i, B_i repeats two rungs, every other pair has union at least six, and three bases do not fit in five labels. Thus $e(F) \leq 2$.

Any five-set has at least $\binom{2}{2} + \binom{3}{2} = 4$ within-side pairs. A single middle base A_2 or B_2 absorbs only one of them; every other within-side pair is residual by pair ownership and hence belongs to H, so $h(F) \geq 3$. An end base absorbs three. Equality $e(F) + h(F) = 2$ is possible only when its two extra labels lie on the other side and form the unique available edge of H in their common-neighbour set. For example, the common neighbours of A_1 in R are $\{a_3, b_2, b_3, c_3\}$, whose unique H-edge is a_3b_2; rail exchange and end reflection give the four end stars

$$S_A^- = A_1 \cup \{a_3, b_2\}, \quad S_B^- = B_1 \cup \{a_2, b_3\},$$
$$S_A^+ = A_3 \cup \{a_0, b_1\}, \quad S_B^+ = B_3 \cup \{a_1, b_0\}.$$

If F upgrades two bases, direct within-side pair counting gives $h(A_1 \cup A_2) = h(B_1 \cup B_2) = 2$ and $h(A_2 \cup A_3) = h(B_2 \cup B_3) = 0$. Thus the first two unions have cost four and the equality unions are

$$P_A = A_2 \cup A_3, \qquad P_B = B_2 \cup B_3.$$

Consequently every fivefold vertex satisfies

$$e(F) + h(F) \geq 2, \tag{124}$$

with equality exactly for the four end stars and P_A, P_B.

For the two fivefold vertices, pair ownership makes their upgraded bases and consumed H-edges disjoint. Since $9 - (6 - e) = 3 + e$ new triples are required and only $7 - h$ edges of H remain, (123) and (124) force equality for both vertices. Directly comparing the six structural equality supports leaves the four-edge compatibility graph

$$(P_A, S_B^-), \quad (P_B, S_A^-), \quad (S_A^-, S_B^-), \quad (S_A^+, S_B^+).$$

For (S_A^-, P_B), the a_0c_2, c_1c_2 bundles are empty and a_3b_2 is consumed, leaving four bundles for six triples; the other mixed pair is its reflection. The two end-star pairs are mirror images, so take (S_A^-, S_B^-). All five residual triangles are forced:

$$\{a_0, b_1, c_3\}, \ \{a_0, b_3, c_2\}, \ \{a_1, b_0, c_3\}, \ \{a_3, b_0, c_2\}, \ \{c_1, c_2, c_3\}.$$

The five lines of S_A^- form a pencil P, those of S_B^- a pencil Q, and their common line is $c_1 = PQ$. Send it to infinity. Then a_0, a_1, a_3, b_2 are one parallel family, say $x =$ constant, while b_0, b_1, a_2, b_3 are a second, say $y =$ constant. The final forced triple makes $c_2 : y = mx + \beta$ and $c_3 : y = mx + \beta'$ parallel. Writing x_1, x_3 for the constants of a_1, a_3 and y_2, y_0 for those of a_2, b_0, the bases and forced triples give

$$y_2 = mx_1 + \beta, \quad y_0 = mx_3 + \beta, \quad y_0 = mx_1 + \beta', \quad y_2 = mx_3 + \beta'.$$

Hence $\beta' - \beta = y_0 - y_2 = y_2 - y_0$, so $y_0 = y_2$, contradicting the distinctness of b_0, a_2. This proves the second assertion. $\square$

3.5 Eleven points: the local K1–K4 package

For a pivot p in an eleven-point configuration whose blocks have size at most five, inversion gives the exact link equation

$$d_3(p) + 3d_4(p) + 6d_5(p) = 45. \tag{125}$$

The next lemma records the real part of this link geometry.

Lemma 3.18 (K3: inversion-link rigidity)**.** *Let P be a real eleven-point configuration in which every maximal generalized block has size at most five, and let $p \in P$. After inversion at p, the other ten marked points form a real linear space with line counts $(t_2, t_3, t_4) = (d_3, d_4, d_5)$, unique pair ownership, and two distinct link lines meeting in at most one marked point. Under (125), the following assertions hold.*

(K3.1) *If $d_5 = 3$ and $d_3 \in \{6, 9\}$, two of the three four-point link lines cannot be disjoint.*

(K3.2) *If $(d_3, d_4, d_5) = (9, 4, 4)$, the four four-point link lines form a complete quadrangle. Their four residual three-lines have, up to relabelling, the private-pair graph "triangle plus pendant." The four points left on each corresponding five-block form a harmonic quadruple.*

(K3.3) *A five-block contains at most two pivots of type $(9, 4, 4)$.*

Proof. For (K3.1), three four-point link lines with one disjoint pair have intersection graph a path. Indeed, the disjoint pair uses eight of the ten marked points. The third four-point line contains at least two of those eight and meets either base in at most one marked point, so it meets both exactly once. Dualize. Their private line families have sizes $3, 2, 3$, so the three cross-edge classes have sizes $6, 9, 6$. Denote the two connector lines by x, y and the private families by A, B, D, of sizes $3, 2, 3$. The three intersections of x with the lines of D, and the three intersections of y with the lines of A, are six forced ordinary vertices. Every other intersection is represented by an edge of the complete tripartite graph on A, B, D. The link equation gives $(t_2, t_3, t_4) = (6, 7, 3)$ or $(9, 6, 3)$. When $d_3 = 6$, seven triple vertices would have to decompose all three classes equally, impossible. When $d_3 = 9$, six triples exhaust the two six-edge classes and leave three ordinary edges in the middle class. The two middle lines then give two edge-disjoint transversals of a real 3×3 grid. After one is diagonal and the coordinate set is $(0, 1, t)$, the other has determinant $t^2 - t + 1 > 0$, a contradiction.

For (K3.2), four base lines have sixteen incidences on ten points and at most six pairwise intersections. Convexity forces degrees $2^6 1^4$: the complete quadrangle with private points u_i and pair-points v_{ij}. A residual triple is either

$$T_{ijk} = \{u_i, u_j, u_k\}, \qquad S_{ij} = \{u_i, u_j, v_{kl}\}, \quad \{i, j, k, l\} = [4].$$

No two residual lines repeat a private pair. Hence at most one T-line occurs. If T_{123} occurs, the other three lines are S_{14}, S_{24}, S_{34}. With no T-line, the four distinct S-indices form a four-edge simple graph on four vertices, and pair ownership leaves only a four-cycle or a triangle with a pendant edge.

Normalize the bases to $x = 0, y = 0, z = 0, x + y + z = 0$ and put

$$u_1 = (0, 1, a),\ u_2 = (1, 0, b),\ u_3 = (1, c, 0), u_4 = (1, d, -1 - d).$$

Since every u_i is private, distinct from all three pair-intersections on its base, one has

$$a, b, c, d \notin \{0, -1\}.$$

The seven relevant determinants are

$$b + ac; \quad b - a,\ 1 - ac,\ 1 + d(1 + a),\ b - c,\ 1 + d + b,\ d - c.$$

For the T-star, the three S-rows give $d = c$, $b = -1 - c$, and $ac = b$; the T-row then becomes $2b = 0$, contrary to nondegeneracy. For a four-cycle the rows give $a = b = c = d$ and $a^2 + a + 1 = 0$, which has no real root. For the triangle-pendant representative $S_{12}, S_{13}, S_{23}, S_{14}$, they give $a = b = c$ and $a^2 = 1$. The value $a = -1$ is incompatible with the S_{14} row, while $a = 1$ forces $d = -1/2$; substitution makes the other three candidate determinants nonzero. Explicitly, at $(a, b, c, d) = (1, 1, 1, -1/2)$ the intended rows are

$$S_{12} = S_{13} = S_{23} = S_{14} = 0,$$

while

$$T_{123} = 2, \quad T_{124} = 1, \quad T_{134} = -1, \quad T_{234} = -2, \qquad S_{24} = \frac{3}{2}, \quad S_{34} = -\frac{3}{2}.$$

Thus this is the unique real survivor. Its cross-ratio on every base is -1. Inversion restricted to a generalized block through p is a projectivity onto its image line, so this harmonic cross-ratio pulls back to the four points of the five-block other than p.

For (K3.3), identify the carrier line or circle of a five-block with $\mathbb{P}^1(\mathbb{R})$. If three of its points p, q, r had harmonic complements, send the two remaining points to $0, \infty$ and scale so that $p = 1$. The three harmonicity conditions say

$$p/q, \quad p/r, \quad q/r \in S := \{-1, 2, 1/2\}.$$

Since S is closed under reciprocals, this puts the two distinct numbers q, r in S. But

$$((S/S) \setminus \{1\}) = \{-2, -1/2, 4, 1/4\}$$

is disjoint from S. This proves the cap. $\square$

Lemma 3.19 (K1: rebased five-block moment)**.** *Let $m = B_5$ and let Δ be a selected five-block. Write $n_{ij}(\Delta)$ for the number of other i-blocks meeting Δ in j points and put*

$$H_\Delta = n_{42} + 3n_{52}, \qquad q_\Delta = n_{40} + 3n_{50}.$$

If $d_3(p) = 6 + 3\omega_p$, $\Omega_\Delta = \sum_{p \in \Delta} \omega_p$, $S_{55} = \sum_p \binom{d_5(p)}{2}$, and $I_\omega = \sum_p \omega_p d_5(p)$, then

$$H_\Delta = 57 - N_4 - 2m - \Omega_\Delta + n_{52} + n_{40} + 2n_{50}, \tag{126}$$

$$\sum_\Delta H_\Delta = (58 - N_4 - 3m)m + 2S_{55} - I_\omega + \sum_\Delta q_\Delta. \tag{127}$$

For every Δ, $H_\Delta \leq 30$.

Proof. Equation (125) is equivalent to $d_4 + 2d_5 = 13 - \omega$. Sum it over the five points of Δ and group blocks by intersection size; eliminating n_{41} and n_{51} gives (126). Summing over Δ, a pair of five-blocks meeting in $j = 0, 1, 2$ points contributes through the identities $S_{55} = A_1 + 2A_2$ and $A_0 + A_1 + A_2 = \binom{m}{2}$, which gives (127). Finally, H_Δ counts hosts over the fifteen outsider pairs. Host chords form a matching on five points, so every pair has at most two hosts and $H_\Delta \le 30$. □

Lemma 3.20 (K4: rich lines and the added centre)**.** *In the $n = 11$, maximum-circle-five branch with $C = 40, L = 14$, put $b = B_5$. Then*

$$7L_4 + 16L_5 \le b + 2. \tag{128}$$

Consequently $L_5 = 0$, $L_4 \in \{0, 1\}$, and every row except $b = 7, 8$ is impossible by local moments and Kelly–Moser.

Proof. For a maximal m-line with $q = 11 - m$ outsiders, let x_s count proper circle owners whose outsider support has size $s = 1, 2, 3$. For a fixed outsider pair, its pairs on the m-line form a matching, and therefore

$$x_1 + 2x_2 + 3x_3 = \binom{m}{2}q, \qquad Q := x_2 + 3x_3 \le \lfloor m/2 \rfloor \binom{q}{2}.$$

The $D = x_1 + x_2 = \binom{m}{2}q - Q$ owners are distinct, since a proper circle cannot contain two pairs from the same line. Hence

$$C \ge D \ge q\binom{m}{2} - \binom{q}{2}\lfloor m/2 \rfloor .$$

For $m = 6, 7, 8, 9, 10$ this is $45, 66, 72, 68, 45$, so all blocks have size at most five.

We also print the five-block energy. Deletion to $f(10) = 33$ and the five disjoint ray pairs give $d_3(p) \le 7 + 5 = 12$. If five four-point link lines existed at a pivot, their twenty incidences on ten marked points would force the complete-pentagon incidence skeleton. Every further line contains at most two of its pair-labelled points, whereas the link equation $d_3 + 3d_4 + 6d_5 = 45$ and $d_3 \le 12$ require a three-point link line. Six four-point lines are excluded by convexity, since their twenty-four incidences give at least eighteen pairwise meetings, more than $\binom{6}{2}$. Thus $d_5(p) \le 4$ at every pivot, and

$$5b = \sum_p d_5(p) \le 44,$$

so $b \le 8$.

The restored eleven-point set at a pivot is noncollinear: a carrier line through the restored centre would invert to an original line through the pivot, while one avoiding it would invert to a generalized circle through the pivot; either would contain all original points. Thus restoration of the inversion centre and Melchior give

$$3\ell_3(p) + \sum_{s \ge 4} s\ell_s(p) \le 10 + \sigma_p. \tag{129}$$

Here the block equations are

$$N_3 = 17 + 2b, \quad N_4 = 37 - 3b, \quad \sum_p \sigma_p = 18 + b.$$

Summing (129) gives (128). Since $b \le 8$, it gives $L_5 = 0$ and $L_4 \le 1$.

If $L_4 = 1$, let κ_p be the local slack in (129); then $\sum \kappa_p = b - 5$. The congruence $d_5 \equiv 1 - \ell_4 - \kappa \pmod 3$ now excludes $b = 5, 6$ as follows.

For $b = 5$, every slack vanishes. Points on the four-line have $d_5 \in \{0, 3\}$, and points off it have $d_5 \in \{1, 4\}$. If a counts off-line degree-four points and c on-line degree-three points, then

$$7 + 3a + 3c = 25, \qquad a + c = 6, \qquad a \geq 2.$$

Consequently

$$\sum_p \binom{d_5(p)}{2} = 6a + 3c = 18 + 3a \geq 24 > 20 = 2\binom{5}{2},$$

contrary to the two-point intersection bound for five-blocks.

For $b = 6$, exactly one pivot has slack one. If it lies on the four-line, its five-degree is two; with the same notation,

$$a + c = 7, \qquad c \leq 3, \qquad \sum_p \binom{d_5(p)}{2} = 1 + 6a + 3c = 22 + 3a > 30.$$

If the exceptional pivot lies off the line, its degree is $e \in \{0, 3\}$, and

$$3(a + c) + e = 24, \qquad c \leq 4, \qquad a \geq 3, \qquad \sum_p \binom{d_5(p)}{2} = 24 + 3a > 30.$$

Both contradict $2\binom{6}{2} = 30$.

If $L_4 = 0$, the restored eleven-point set has $t_2 = d_3 + 10 - 3\ell_3$. Kelly–Moser gives $t_2 \geq 5$, and summing the resulting bounds over the fourteen triple lines gives $42 \leq 28 + 2b$, hence $b \geq 7$. Thus only $b = 7, 8$ remain. $\square$

3.6 The eleven-point five-circle branch

Lemma 3.21 (The low layers)**.** *In the $n = 11$ maximum-circle-five branch one has $C \geq 39$. In particular $C = 36, 37, 38$ are impossible.*

Proof. Deleting a point preserves admissibility unless the other ten points are collinear or concyclic. In the first case the outsider pencil gives $\binom{10}{2} > 40$ circles; in the second it gives $1 + \binom{10}{2} - 5 = 41$. Thus deletion is legitimate below 41. By corollary 2.17, Lenchner gives at least three ordinary proper circles through every pivot, and proposition 3.10 gives $C \geq 36$.

For $C = 36$, the global rows force

$$B_3 = 23, \quad B_5 = 7, \quad L = 12, \quad L_4 = L_5 = 0, \quad C_3 = 11.$$

Equality in the deletion sum gives $c_3(p) = 3$ everywhere. The congruence in (125), Kelly–Moser, and the ray bound give $\ell_3(p) = 3$ everywhere, so $3L_3 = 33$ although $L_3 = 12$.

For $C = 37$, the only aggregate rows are

$$(B_5, B_3, L, K) = (7, 23, 11, 1), \qquad (8, 25, 11, 2).$$

They have respectively one and three pivots with $d_3 = 9$. Deletion forces $c_3(p) \leq C - f(10) = 4$ at every pivot. Hence at a pivot with $d_3 = 9$,

$$\ell_3(p) = d_3(p) - c_3(p) \geq 9 - 4 = 5.$$

The local added-centre row $3\ell_3 + 4\ell_4 + 5\ell_5 \leq 10 + \sigma$ then has left side at least 15, whereas $\sigma(p) \leq K \leq 2$ makes its right side at most 12.

At $C=38$, put $k=B_5$ and $h=L_4$. The exact reduction is

$$B_3=2k+13,\ B_4=38-3k,\ L_3=13-h,\ L_5=0,\quad k=5,6,7,8,\quad h=0 \text{ unless } k=8. \tag{130}$$

Every point has $d_3=6$ or 9. With $\kappa=10+\sigma-3\ell_3-4\ell_4$ one has $\sum\kappa=k-1-7h$. When $h=0$, the least possible κ at a high point ($d_3=9$) and a low point ($d_3=6$) is

d_5	0	1	2	3	4
high	1	0	2	1	0
low	1	0	2	1	forbidden.

For $k=5$ there is one high point and ten low points. The zero-cost baseline has degrees $(4,1^{10})$, of sum fourteen. Supplying the remaining eleven incidences costs at least five upgrades $1\to3$ and one upgrade $1\to2$, hence at least $5+2=7>k-1=4$. Thus $k=5$ is impossible. At $k=6,7$ equality in the same marginal-cost calculation forces, respectively, $(d_5)=(4^3,3^5,1^3)$ and $(4^5,3^4,2,1)$; the pair moments give

$$\sum\binom{d_5}{2}=33>2\binom{6}{2}=30,\qquad \sum\binom{d_5}{2}=43>2\binom{7}{2}=42.$$

At $k=8$, both $h=0,1$ force degrees $(4^7,3^4)$. Rebase at a five-circle Γ. Equations (126)–(127) force every outsider edge to be double-hosted. A one-Γ block with three outsiders is then forbidden by lemma 3.12; hence no four- or five-block meets Γ once. The seven other five-blocks and all fourteen four-blocks therefore meet Γ twice, so

$$H_\Gamma=n_{42}+3n_{52}=14+3\cdot7=35,$$

whereas the equality face of (127) gives $H_\Gamma=30$. Hence $C=38$ is impossible. □

Lemma 3.22 (The $C=39$ arithmetic front)**.** *Assume that eleven admissible points have* 39 *proper circles, contain a proper five-circle, and have no proper circle larger than five. Every generalized block has size at most five, and the number L of maximal lines belongs to* $\{12,15\}$*.*

Fix a proper five-circle Γ, with outsider set X. Let A_{ij} count nonselected blocks containing i points of Γ and j outsiders, and put

$$H=A_{22}+3A_{23},\quad x=A_{23},\quad r=A_{13},\quad s=A_{14},\quad q=A_{04}+3A_{05},\quad y=A_{05}.$$

If $L=3k$, and if

$$B_0=A_{03}+A_{04}+A_{05},\qquad T_0=A_{03}+4A_{04}+10A_{05},$$

then

$$\begin{aligned} A_{22}&=H-3x, & A_{21}&=60-2H+3x,\\ A_{12}&=75-2H-3r-6s, & B_0&=3H+L-97-x+2r+5s,\\ T_0&=20-x-r-4s, & q&=39-H-k-r-3s. \end{aligned}\tag{131}$$

Moreover

$$0\le q\le 3,\qquad y=1\Longrightarrow q=3,\tag{132}$$

and, if e_Γ,e_X count points with $d_3=9$ on the two sides,

$$e_\Gamma=55-2H+2x-r-2s,\qquad e_X=2k-r-2s-2q+2y.\tag{133}$$

Every point has $d_3\in\{6,9\}$ and $d_5\le4$.

Proof. Let A be a maximal a-line and put $b = 11 - a$. For each pair of outsiders, its pair-hosts on A form a matching, so the line pencil gives the uniform lower bound

$$C \geq b\binom{a}{2} - \binom{b}{2}\lfloor a/2 \rfloor .$$

For $a = 6, 7, 8, 9, 10$ its values are respectively $45, 66, 72, 68, 45$, so a line of size at least six is impossible. Let N_j be the number of generalized j-blocks. Counting blocks and triples gives

$$N_3 + N_4 + N_5 = 39 + L, \qquad N_3 + 4N_4 + 10N_5 = 165.$$

Consequently

$$L = 3k, \qquad N_4 + 3N_5 = 42 - k, \qquad N_3 = -3 + 4k + 2N_5. \tag{134}$$

At a pivot, inversion gives

$$d_3 + 3d_4 + 6d_5 = 45.$$

Kelly–Moser gives $d_3 \geq 5$. Deleting the pivot destroys at most $39 - f(10) = 6$ three-circles, and its three-lines use disjoint pairs of the other ten points, so $d_3 \leq 11$. The displayed congruence now yields $d_3 \in \{6, 9\}$.

If five four-point image lines existed at a pivot, their twenty incidences on ten points and $\sum_u \binom{r_u}{2} \leq \binom{5}{2}$ would force every image point to be the intersection of exactly two of the lines. A further image line distinct from those five contains at most two such pair-labelled points: three labels would either share an index, giving one of the five lines, or be three disjoint edges of K_5, which is impossible. The link equation nevertheless requires a further three-point line. Six four-point image lines are excluded already by $\sum_u \binom{r_u}{2} \geq 18 > \binom{6}{2}$. Hence $d_5 \leq 4$.

Melchior on the original lines gives

$$3\ell_3 + 7\ell_4 + 12\ell_5 \leq 52,$$

so $L \leq 15$. Summing the pivot slack and the deletion bound gives

$$\sum_p (d_3(p) - 3 - d_5(p)) = 4L + N_5 - 42 \geq 0, \qquad 2N_5 \leq 25 - k.$$

These inequalities exclude $L = 0, 3, 6$. For $L = 9$, put $E = \#\{p : d_3(p) = 9\} = 2N_5 - 13$. Then $7 \leq N_5 \leq 11$. Splitting the pivot slack between Γ and X gives

$$45 + 4y - 4N_5 \leq 3r + 7s + 6q \leq 39 + 4y - 3N_5, \qquad r + 3s + q \geq 6.$$

For $q = 0, 1, 2, 3$, these intervals leave only

$$(N_5, r, s) = (7, 6, 0), (7, 1, 2), (8, 0, 2).$$

The first and third rows have $H = 30$, hence no deficient outsider pair, although respectively an A_{13} or an A_{14} block would require one. The middle row has $H = 29$, whose sole deficient edge cannot meet all four outsider triples of an A_{14} block. Thus $L \neq 9$, proving $L \in \{12, 15\}$.

The three triple partitions relative to Γ give (131). Distinct outsider K_4's have disjoint complement pairs, while an outsider K_5 excludes every K_4. Therefore either $A_{05} = 1, A_{04} = 0$, or $A_{05} = 0, A_{04} \leq 3$, which is (132). Finally, summing three-block incidences on Γ and on X gives (133). □

Lemma 3.23 (Added-centre and singleton interfaces)**.** *Under the hypotheses of lemma 3.22, put $m = N_5$. Then*

$$L = 12 : \quad 7\ell_4 + 16\ell_5 \leq m + 8, \qquad L = 15 : \quad 7\ell_4 + 16\ell_5 \leq m - 7. \tag{135}$$

If two generalized five-blocks meet in the single point p, then

$$(d_3(p), d_5(p)) = (9, 2). \tag{136}$$

Moreover, every generalized five-block contains at most two points of type $(d_3, d_4, d_5) = (9, 4, 4)$.

Proof. Write $\sigma_p = d_3(p) - 3 - d_5(p)$. The inversion images are noncollinear: otherwise inversion back would put all eleven original points on one generalized block. Thus proposition 2.18 applies, gives $\sigma_p \geq 0$, and its restored-centre row is

$$3\ell_3(p) + \sum_{j\geq 4} j\ell_j(p) \leq 10 + \sigma_p.$$

After summing over the eleven pivots and using (134),

$$9\ell_3 + 16\ell_4 + 25\ell_5 \leq 110 + \sum_p \sigma_p, \qquad \sum_p \sigma_p = 4L + m - 42.$$

Since $L = \ell_3 + \ell_4 + \ell_5$, this is exactly (135).

For the singleton assertion, invert at its carrier p. The two five-blocks become disjoint four-point link lines, so $z = d_5(p) \geq 2$. The cap $z \leq 4$, pointwise $\sigma_p \geq 0$, and $d_3 \in \{6, 9\}$ leave only

$$(d_3, z) = (6, 2), (6, 3), (9, 2), (9, 3), (9, 4).$$

The two cases $z = 3$ are excluded by lemma 3.18(K3.1). In the case $(d_3, z) = (9, 4)$, the link equation gives $d_4 = 4$, so the complete-quadrangle conclusion (K3.2) says that every pair of base lines meets at a selected link point.

It remains to exclude $(d_3, z) = (6, 2)$. Dualizing the ten-point link turns the two disjoint four-point lines into two fourfold points. The ten dual lines split into families of sizes $4, 4, 2$; after removing pairs already meeting at the two fourfold points, the four pair classes have sizes

$$16, \qquad 8, \qquad 8, \qquad 1.$$

The link equation requires nine triple vertices. If the sole pair in the last class is unused, all nine triples require one pair from each eight-edge side class. If it is used, the other eight triples exhaust both side classes before the exceptional triple uses two further side edges. Both alternatives are impossible. This proves (136) for circles and for the possible five-line alike.

Finally, a point of type $(9, 4, 4)$ supplies a harmonic complement on every incident five-block by (K3.2). The real cross-ratio argument (K3.3) allows at most two such complements on five points, proving the last assertion. □

Lemma 3.24 (The $L = 15$ harmonic collapse)**.** *The $C = 39, L = 15$ branch is impossible.*

Proof. Let $z_p = d_5(p)$, and call p high when $d_3(p) = 9$. From (134),

$$E := \#\{p : p \text{ is high}\} = 2m - 5, \qquad \sum_p z_p = 5m.$$

Thus $3 \le m \le 8$. The second cut in (135) forces $m \in \{7,8\}$ and $\ell_4 = \ell_5 = 0$; in particular all fifteen maximal lines are triples and all generalized five-blocks are proper circles.

Define the nonnegative integral restored-centre slack

$$\kappa_p = 10 + \sigma_p - 3\ell_3(p).$$

Summing gives $\sum_p \kappa_p = m - 7$, while pointwise

$$3\ell_3(p) = \begin{cases} 13 - z_p - \kappa_p, & p \text{ low}, \\ 16 - z_p - \kappa_p, & p \text{ high}. \end{cases} \tag{137}$$

For $m = 7$, every κ_p vanishes. Pointwise $\sigma_p \ge 0$ and (137) make the two low points have $z = 1$, while the nine high points have $z \in \{1,4\}$. Since $\sum z = 35$, exactly eight high points have $z = 4$, and the link equation makes their type $(9,4,4)$. Counting their incidences with five-blocks gives $8 \cdot 4 \le 2m = 14$, contrary to lemma 3.23.

For $m = 8$, all eleven points are high and precisely one has $\kappa = 1$. The ten zero-slack points have $z \in \{1,4\}$, while the exceptional point has $z \in \{0,3\}$. The sum $\sum z = 40$ therefore forces respectively ten or nine points with $z = 4$. Their incidences give $40 \le 16$ or $36 \le 16$, again contradicting the harmonic cap. ☐

Lemma 3.25 (The $L = 12$ low-host moment)**.** *In the $C = 39, L = 12$ branch, some proper five-circle has host load $H \ge 26$.*

Proof. Suppose every proper five-circle has host load at most 25, and put

$$E = 2m - 9, \qquad z_p = d_5(p), \qquad G = 2\sum_p \binom{z_p}{2} - \sum_{p \text{ high}} z_p.$$

Here $m \ge 5$, $\sum_p z_p = 5m$, and the K1 moment is

$$\sum_\Delta H_\Delta = 20m + G + \sum_\Delta q_\Delta. \tag{138}$$

The successive costs of increasing z_p from zero to four are

	1	2	3	4
p low	0	2	4	6
p high	-1	1	3	5.

Filling these increasing marginal costs gives $G \ge 31$ at $m = 5$. For $m \ge 6$, Cauchy and $\sum_{p \text{ high}} z_p \le 4(2m-9)$ give

$$G \ge \frac{25m^2}{11} - 13m + 36 > 5m + 5.$$

The original-line Melchior row is $36 + 4\ell_4 + 9\ell_5 \le 52$, so there is at most one five-line. A proper five-circle contributes at most 25 to the left side of (138), and a possible five-line at most the universal host cap 30. Hence $\sum H_\Delta \le 25m + 5$, whereas the right side is at least 131 for $m = 5$ and is strictly larger than $25m + 5$ for $m \ge 6$. This contradiction proves the lemma. ☐

Lemma 3.26 (The $L = 12$ maximum-host router)**.** *In the $C = 39, L = 12$ branch, no proper five-circle of maximum host load has $26 \le H \le 30$.*

Proof. Let h be the maximum proper-circle host load, let $t = \ell_5 \in \{0, 1\}$, and use m, E, G as in the preceding proof. Besides $G \geq 31$ for $m = 5$, marginal filling gives

$$G \geq 45 \quad (m = 6), \qquad G \geq 61 \quad (m = 7). \tag{139}$$

and the Cauchy bound above is greater than $8m + 2$ for $m \geq 8$. Since the possible five-line has host at most thirty, K1 gives

$$20m + G \leq h(m - t) + 30t. \tag{140}$$

For $h = 26$, (139) and (140) leave only $(m, t) = (5, 1)$. For $h = 27$, they leave $(5, 0), (5, 1), (6, 1)$. But (135) gives $16t \leq m + 8$, removing every listed row with $t = 1$. Thus $h \neq 26$, and at $h = 27$ necessarily

$$m = 5, \qquad t = 0. \tag{141}$$

We record a consequence used twice. Rebase at a proper five-circle Δ, put $R = 30 - H_\Delta$, and let r, s, q, y have the front meaning. If $R \geq 3$ and $s = 0$, then

$$A_{12} = 3q - R, \qquad e_X = 3 - R - q + 2y. \tag{142}$$

If $y = 1$, then $q = 3$ and the second quantity is $2 - R < 0$. If $y = 0$, nonnegativity in (142) gives simultaneously $3q \geq R$ and $q \leq 3 - R$, again impossible. Hence every circle with $H \leq 27$ has

$$s_\Delta \geq 1. \tag{143}$$

In the face (141), form the graph whose five vertices are the five-blocks and whose edges are their singleton intersections. By (143) it has minimum degree one and hence at least three edges. But (136) injects its edges into the $E = 1$ high points: a carrier has $d_5 = 2$ and can carry only that one edge. This excludes $h = 27$.

Suppose next that $h = 28$. Equations (139) and (140) leave $m \in \{5, 6\}$, and the line cut gives $t = 0$. We claim that (143) still holds for a circle with $H = 28$. Otherwise its outsider deficiency $\delta(e) = 2 - h_e$ has total weight two. Its support is one zero edge Z, two adjacent single edges S_a, or two disjoint single edges S_d. Every A_{13} triangle contains a deficient edge by lemma 3.14(K2.1). A single edge has one host chord; (K2.1) places that chord in a unique page-colour class, and triple ownership allows at most one page of that colour through the edge. Thus a zero edge lies in four outsider triangles, while each single edge supports at most one clean page, with the one triangle on two adjacent singles as the only overlap. Therefore

$$r \leq 4, 3, 2 \quad \text{in types } Z, S_a, S_d. \tag{144}$$

With $s = 0$, the front says $r + q = 7$, so (144) and $q \leq 3$ force type Z and $(r, q) = (4, 3)$. If its zero edge is uv, the four pages are uvw, one for every other outsider w. The high split gives $e_X = -2 + 2y$, hence $y = 1$: there is one outsider K_5. It must omit u or v, since otherwise it shares an outsider triple uvw with a page, and it is consequently all-double. Suppose it omits u, and let γ_i be the colour of the page uvw. The adjacent double edges uw, vw determine two distinct vertices of D_i; triple ownership makes both host matchings omit i. The vertex coming from vw lies on the all-double K_5 trace and is its unique centre δ_i omitting i, whereas the page trace contains γ_i, δ_i and the second vertex coming from uw. This is exactly the tangent-separation obstruction in (K2.4). The case in which the K_5 omits v is symmetric. This proves the claim.

The singleton graph on the m proper five-circles consequently has minimum degree one. If $m = 5$, it has at least three singleton edges, whereas $E = 1$, a contradiction. Let $m = 6$.

Now it has at least three edges and (136) gives at most $E = 3$, so it is a perfect matching and all three high points have $z_p = 2$.

Let A_j count pairs of five-circles meeting in j points, and put $Q = \sum_\Delta q_\Delta$. Since $A_1 = 3$,

$$A_0 + A_1 + A_2 = 15, \qquad \sum_p \binom{z_p}{2} = A_1 + 2A_2 = 27 - 2A_0.$$

The three high points contribute six to the high incidence sum, hence

$$G = 48 - 4A_0. \tag{145}$$

Every disjoint pair contributes three to q_Δ at each endpoint, so $Q \geq 6A_0$. The global moment and $H_\Delta \leq 28$ give $G + Q \leq 48$. Together with (145), this forces

$$A_0 = 0, \qquad Q = 0, \qquad H_\Delta = 28 \quad \text{for every } \Delta. \tag{146}$$

For any Δ, the perfect matching gives $s = 1$, while (146) gives $q = 0$; the front therefore gives $r = 4$. But an A_{14} outsider four-set forces the two deficiency units to be the disjoint perfect matching on that set. Every A_{13} triangle then contains exactly one of its two single edges, and (K2.1) permits at most one clean page on each. Thus $r \leq 2$, the final contradiction for $h = 28$.

If $h = 29$, the deficiency graph of a maximum-host circle is one single edge uv. An A_{14} support is impossible, and every A_{13} triangle contains uv. The edge has one host chord; (K2.1) assigns that chord to a unique page colour, and triple ownership permits at most one page of that colour through u, v. Hence $r \leq 1$. The front, however, gives $r + q = 6$, so $q \leq 3$ forces $r \geq 3$.

Finally, if $h = 30$, every outsider edge is double. Clause (K2.1) forbids an A_{13} block, and any A_{14} block contains a forbidden all-double outsider triangle. Thus $r = s = 0$, while the front gives $q = 5 > 3$. This exhausts $26 \leq h \leq 30$. □

Lemma 3.27 (K2 routing of the $C = 39$ layer)**.** *There is no $n = 11$, maximum-circle-five configuration with $C = 39$.*

Proof. By lemma 3.22, all generalized blocks have size at most five and $L \in \{12, 15\}$. The value $L = 15$ is excluded directly by the added-centre slack and harmonic cap in lemma 3.24. If $L = 12$, lemma 3.25 supplies a proper five-circle of host load at least 26. Choose one of maximum load. The host matching gives $H \leq 30$, while lemma 3.26 excludes every integer in that interval. Hence $C = 39$ is impossible. □

Lemma 3.28 ($C = 40$ pivot spectrum)**.** *Suppose that an admissible eleven-point set has forty proper circles and no proper circle larger than five. At every point p,*

$$d_3(p) \in \{6, 9, 12\}, \qquad d_5(p) \leq 4, \qquad d_3(p) + 1 \geq 2d_5(p).$$

If, in addition, $L = 11$, then

$d_3(p)$	*possible* $d_5(p)$
6	$1, 2, 3$
9	$3, 4.$

(147)

Consequently, if $k = B_5$ and $n_i = \#\{p : d_3(p) = i\}$, then

$$n_9 = 2k - 9, \qquad n_6 = 20 - 2k. \tag{148}$$

Proof. Write $c_3(p)$ and $\ell_3(p)$ for the numbers of three-circles and three-lines through p. The set $P \setminus \{p\}$ is admissible: if its ten points were collinear, the remaining point would determine $\binom{10}{2} = 45$ distinct proper three-circles with pairs on that line; ten concyclic points would violate the assumed circle cap. Deleting p destroys only the three-circles through it, so the ten-point theorem gives

$$c_3(p) \le C(P) - f(10) = 40 - 33 = 7.$$

The three-lines through p use disjoint pairs of the other ten points, and hence $\ell_3(p) \le 5$. Thus $d_3(p) \le 12$.

Invert about p, and denote the image of $P \setminus \{p\}$ by Q_p. This set too is admissible. Indeed, a line or circle containing all of Q_p inverts either to a ten-point line in P, or to a circle containing all of P; both alternatives contradict the preceding caps. The ordinary image lines are precisely the three-blocks through p. Kelly–Moser therefore gives $d_3(p) \ge 5$, while the exact link equation

$$d_3(p) + 3d_4(p) + 6d_5(p) = 45$$

makes $d_3(p)$ divisible by three. Thus $d_3(p) \ge 6$, and this proves $d_3(p) \in \{6, 9, 12\}$.

By lemma 3.1, every original line has at most five selected points; together with the circle cap, every image line therefore has at most four of the ten points. Since $4 \le 2|Q_p|/3$, Langer's incidence inequality applies and yields

$$2d_3(p) + 3d_4(p) + 4d_5(p) \ge 44.$$

Subtracting the link equation gives

$$d_3(p) + 1 \ge 2d_5(p). \tag{149}$$

It remains to prove the absolute cap $d_5(p) \le 4$. Suppose first that five four-point image lines $L_1, \dots, L_5$ exist, and let r_x be the number of them through $x \in Q_p$. Then

$$\sum_x r_x = 20, \qquad \sum_x \binom{r_x}{2} \le \binom{5}{2} = 10.$$

Convexity gives the reverse inequality, with equality only when $r_x = 2$ at all ten points. Thus the points of Q_p are exactly the ten intersections $L_i \cap L_j$, labelled by the two-subsets $\{i, j\} \subset [5]$. A further line distinct from all L_i contains at most two of these points: among three two-subsets of a five-set, either two share an index, in which case their joining line is the corresponding L_i, or all three would have to be pairwise disjoint, which is impossible. Hence a sixth four-point image line cannot exist. If there are exactly five, the already proved range of d_3 and the link equation give

$$d_4(p) = 5 - \frac{d_3(p)}{3} \ge 1,$$

but the resulting additional three-point image line is equally impossible. Therefore $d_5(p) \le 4$.

Now assume $L = 11$. There are $B = C + L = 51$ generalized blocks. Under inversion, the circles of Q_p are in bijection with the blocks of P which avoid p; hence

$$C(Q_p) = 51 - \beta(p) \ge f(10) = 33, \qquad \beta(p) := d_3(p) + d_4(p) + d_5(p) \le 18.$$

Eliminating $d_4(p)$ by the link equation gives

$$\beta(p) = 15 + \frac{2d_3(p)}{3} - d_5(p). \tag{150}$$

For $d_3 = 12$, this forces $d_5 \geq 5$, already excluded. For $d_3 = 6$, (150) and (149) give $1 \leq d_5 \leq 3$; for $d_3 = 9$ they give $3 \leq d_5 \leq 4$. This is (147).

Finally, the block equations give $B_3 = 13 + 2k$. Double-counting incidences with three-blocks and using $n_6 + n_9 = 11$, we obtain

$$6n_6 + 9n_9 = 3B_3 = 39 + 6k, \qquad n_6 + n_9 = 11,$$

whose solution is (148). □

Lemma 3.29 ($C = 40, L = 11$ singleton exclusion)**.** *In the $C = 40, L = 11$ layer, two distinct five-blocks meet in zero or two points, never in exactly one.*

Proof. Suppose two five-blocks meet only at p. After inversion at p, their four-point link lines are disjoint. The spectrum (147) and $d_5(p) \geq 2$ leave

$$(d_3, d_5) = (6, 2), (6, 3), (9, 3), (9, 4).$$

Both degree-three cases are excluded by lemma 3.18(K3.1). In the last case the link equation gives $d_4 = 4$, and (K3.2) makes the four base lines a complete quadrangle, so no pair is disjoint.

It remains to exclude $(d_3, d_5) = (6, 2)$, whose link profile is $(6, 9, 2)$. Dualize the two disjoint four-point lines. The ten dual lines split into families of sizes $4, 4, 2$ through the two resulting fourfold points; after removing pairs already meeting there, the four pair classes have sizes

$$16, \qquad 8, \qquad 8, \qquad 1.$$

The nine three-point link lines become nine triple vertices. If none uses the sole pair in the last class, all nine require one edge from each eight-edge side class. If one uses that pair, the other eight triples exhaust both side classes before the exceptional triple uses two further edges on one side. Both alternatives are impossible. □

Lemma 3.30 (The $C = 40, L = 11$ defect–harmonic faces)**.** *In the $n = 11$, maximum-circle-five layer with $C = 40$ and $L = 11$, neither $B_5 = 5$ nor $B_5 = 6$ is possible.*

Proof. By lemma 3.29, two five-blocks meet in zero or two points. For a five-block E put

$$\delta_E = \sum_{F \neq E} (2 - |E \cap F|).$$

Thus every summand is zero or two and

$$D := \sum_{\{E,F\}} (2 - |E \cap F|) = 2\binom{B_5}{2} - \sum_p \binom{d_5(p)}{2}$$

is twice the number of disjoint block pairs.

Five blocks. There are ten low points with $d_3 = 6$ and one high point with $d_3 = 9$. Let $a \in \{0, 1\}$ indicate whether the high point has $d_5 = 4$, and let n_i count low points with $d_5 = i$ for $i = 1, 2$; all other low points have degree three. The incidence and defect sums are

$$n_2 + 2n_1 = a + 8, \qquad D = 3 - a - n_1.$$

Since D is nonnegative and even, the complete set of faces is

D	(a, n_1, n_2)
2	$(0,1,6),(1,0,9)$
0	$(0,3,2),(1,2,5).$

If c_E, d_E count in E the points of degrees two and one and a_E indicates the degree-four high point, then the row sum gives

$$\delta_E = c_E + 2d_E - a_E - 2. \tag{151}$$

On the face $(a, n_1, n_2) = (1, 0, 9)$, the unique disjoint pair F, G satisfies $c_E - a_E = 4$ at both ends. A block containing the high point has $c_E - a_E \le 3$, so the high point lies outside $F \cup G$. Each of the two disjoint blocks would then need four degree-two points and the unique degree-three point, impossible.

On the other $D = 2$ face the degrees are $(3^4, 2^6, 1)$. If U is the degree-one point and F, G the disjoint pair, (151) forces, after exchanging F, G,

$$F:\ U,\ 2^2,\ 3^2, \qquad G:\ 2^4,\ 3.$$

The fourth degree-three point q lies outside $F \cup G$ and hence belongs to all three remaining blocks A, B, C; their other pairwise intersections are the three degree-three points in $F \cup G$. Invert at q. The images of A, B, C are three four-lines, each split $2+2$ across the partition $F \sqcup G$. If q were low, the seven residual triple lines would each use two cross-pairs, but the three four-lines have already used twelve of the 25 cross-pairs, leaving only thirteen instead of fourteen. Thus q is the high point.

Rebase now at $\Gamma = F$. All its points are low, and (151) gives

$$\sum_{p\in F} d_5(p) = 11, \qquad \sum_{p\in F} d_4(p) = 5\cdot 13 - 2\cdot 11 = 43.$$

No four-block is disjoint from F, since four points of $G \cup \{q\}$ contain a triple of G. Hence, if n_{22}, n_{13} count four-blocks meeting F in two and one points,

$$n_{22} + n_{13} = 23, \qquad 2n_{22} + n_{13} = 43, \quad \text{so} \quad (n_{22}, n_{13}) = (20, 3).$$

The ten edges within G receive seventeen hosts from the $2F + 2G$ four-blocks and three from A, B, C, so every such edge is double-hosted. The five qG edges receive nine hosts; four are double-hosted. Each of the three $q+F+2G$ four-blocks therefore contains a double edge of G and an adjacent double qG edge. Both host matchings omit the colour of the page: a host chord through that point would repeat the triple already owned by the page. This contradicts the golden-separation clause lemma 3.14(K2.4).

It remains to dispose of the two $D = 0$ faces simultaneously. Choose a five-block Γ avoiding the high point. Every point on Γ is low, every other five-block meets it twice, and

$$\sum_{p\in\Gamma} d_5(p) = 13, \qquad \sum_{p\in\Gamma} d_4(p) = 39.$$

Let t count four-blocks disjoint from Γ, and let n_{22}, n_{13} count the other four-block types. Triple ownership gives

$$\begin{aligned} n_{22} &= 16 + t, & n_{13} &= 7 - 2t, \\ H = n_{22} + 3\cdot 4 &= 28 + t, & n_{12} &= 4t - 2, \qquad n_{03} = 9 - 2t. \end{aligned} \tag{152}$$

Here H is the host load on the fifteen outsider edges, and n_{12}, n_{03} count triple blocks of the indicated types. Host matching gives $H \le 30$, while $n_{12} \ge 0$, so $t = 1$ or 2. The graph of double-hosted outsider edges is then K_6 minus one edge, with sixteen triangles, or K_6, with twenty. By lemma 3.14(K2.1), every such triangle is owned either by one of the four $2\Gamma + 3X$ five-blocks or by a block disjoint from Γ. Their total triangle capacity is only

$$4 + (n_{03} + 4t) = 13 + 2t,$$

namely fifteen or seventeen. Both faces are impossible.

Six blocks. There are eight low and three high points. Let a count high points with $d_5 = 4$, and again let n_i count low points of degree i. Then

$$n_2 + 2n_1 = a + 3, \qquad D = 3 - a - n_1,$$

so the complete face list is

D	(a, n_1, n_2)
2	$(0,1,1), (1,0,4)$
0	$(1,2,0), (2,1,3), (3,0,6).$

If h_E, c_E, d_E count in E the points of degrees four, two, and one, respectively, then

$$\delta_E = c_E + 2d_E - h_E. \tag{153}$$

For $(0,1,1)$ the two ends of the disjoint pair require total defect four, but all degree-two and degree-one points have total weight only three in (153). For $(1,2,0)$, zero defect would give $2d_E = h_E$ in every block; with only one high point this forces $d_E = h_E = 0$, so no block could contain either degree-one point.

Consider $(1,0,4)$. Its degrees are $(4, 3^6, 2^4)$. Equality at the disjoint pair F, G places the high point h outside their union, all four degree-two points inside it, and two in each block. The other four five-blocks are precisely the blocks through h. After inversion at h they are four lines in complete-quadrangle position, with six pair-points v_{ij} of degree three and four private points u_i of degree two; F, G partition these ten points. Colour ij and i by F when v_{ij} and u_i lie in F. The equality

$$\deg_F(i) + \mathbf{1}_{\{u_i \in F\}} = 2$$

and $|F| = 5$ show that the F-edges form a Hamilton path and its two private points are the endpoints; the G-data are complementary. If ij is the endpoint pair and kl the internal pair of this path, then the residual triples $S_{ij} = \{u_i, u_j, v_{kl}\}$ and $S_{kl} = \{u_k, u_l, v_{ij}\}$ lie in F and G. The triangle-plus-pendant conclusion of lemma 3.18(K3.2) selects at least one of the two complementary indices. That triple would then belong both to F or G and to a distinct four-block through h, violating triple ownership.

For $(2,1,3)$ the degrees are $(4^2, 3^5, 2^3, 1)$. Let the high points be h_1, h_2 and the degree-one point be U. Equation (153) forces the block Z through U to contain both high points and no degree-two point. The cap lemma 3.18(K3.3), followed by the incidence and pair rows, forces one further common block W and four side blocks. Relabeling their supports gives

$$\begin{array}{c|l} Z & h_1, h_2, U, m, n \\ W & h_1, h_2, r, \ell_1, \ell_2 \\ A_1 & h_1, m, r, \ell_3, s \\ A_2 & h_1, n, \ell_1, s, t \\ B_1 & h_2, n, r, \ell_3, t \\ B_2 & h_2, m, \ell_2, s, t. \end{array} \tag{154}$$

For clarity, the support is forced without a catalogue: Z has the two high points; among the other five blocks their remaining six high incidences split as one double and four single incidences. Pair ownership then places the three degree-two incidences and successively the five degree-three incidences, yielding the displayed crossed pattern.

Invert first at h_1 and then at h_2. In either link, K3.2 gives the unique real triangle-plus-pendant orientation. On the common block W, the first orientation says that h_2 is the harmonic conjugate of r with respect to the unordered pair $\{\ell_1, \ell_2\}$; the second says the same of h_1. A harmonic conjugate is unique, so $h_1 = h_2$, a contradiction.

Finally, on $(3, 0, 6)$ the degrees are $(4^3, 3^2, 2^6)$. Zero row defect and K3.3 imply that every block contains exactly two high, two degree-two, and one degree-three point. Calling the highs h_1, h_2, h_3, each high pair labels two blocks:

$$X_0, X_1 \supset h_1h_2, \qquad Y_0, Y_1 \supset h_1h_3, \qquad Z_0, Z_1 \supset h_2h_3.$$

The two degree-three points choose, after relabeling, the triples $X_0Y_0Z_0$ and $X_1Y_1Z_1$; the six opposite-index block pairs own the six degree-two points. Invert at h_1 with bases X_0, X_1, Y_0, Y_1. The two complementary residual triples satisfy

$$S_{X_0Y_0} \subset Z_1, \qquad S_{X_1Y_1} \subset Z_0.$$

The complement of the triangle-plus-pendant graph in K3.2 is a pair of adjacent edges, so it cannot omit both complementary indices. One of these two triples is therefore also owned by a four-block through h_1, again contradicting triple ownership. All nine faces are excluded. □

Lemma 3.31 (Seven five-blocks)**.** *In the $n = 11$, maximum-circle-five layer with $C = 40$ and $L = 14$, the row $B_5 = 7$ is impossible.*

Proof. Let $F_1, \dots, F_7$ be the five-blocks and put $d(p) = d_5(p)$. Since two distinct maximal blocks meet in at most two points,

$$\sum_p d(p) = 35, \qquad I := \sum_p \binom{d(p)}{2} = \sum_{i<j} |F_i \cap F_j| \le 42.$$

With $x_p = d(p) - 3$, these identities become

$$\sum_p x_p = 2, \qquad I = 38 + \frac{1}{2}\sum_p x_p^2, \qquad \sum_p x_p^2 \le 8. \tag{155}$$

By lemma 3.28, $0 \le d(p) \le 4$; degree zero alone would contribute nine to the last sum, so $1 \le d(p) \le 4$. If a, b, c, e count the points of degrees $4, 3, 2, 1$, respectively, then

$$a - c - 2e = 2, \qquad c + 3e \le 3.$$

Thus either $e = 0$ and $c = 0, 1, 2, 3$, or $e = 1, c = 0$. This gives exactly

(156)

	$(d(p) : p \in P)$	$D := 42 - I$
A	$4^2\,3^9$	3
B	$4^3\,3^7\,2$	2
C	$4^4\,3^5\,2^2$	1
D	$4^5\,3^3\,2^3$	0
E	$4^4\,3^6\,1$	0.

This is an algebraic exhaustion, not a search over supports.

The triple partition gives $B_3 = 31$ and $B_4 = 16$. If $u = \#\{p : d_3(p) = 12\}$, the spectrum lemma and $\sum_p d_3(p) = 3B_3 = 93$ give

$$n_6 = u + 2, \qquad n_9 = 9 - 2u, \qquad n_{12} = u, \qquad 0 \le u \le 4. \tag{157}$$

The external-trace obstruction. Fix H with $d(H) = 4$. Remove H from its four five-blocks and, on the remaining ten points, let r_x be the number of the resulting four-sets through x. Then

$$\sum_x r_x = 16, \qquad \sum_x \binom{r_x}{2} \le 6.$$

Convexity gives the reverse inequality. Equality therefore holds: there are four private points u_i and six pair-points v_{ij}, one for each pair of the four bases.

Let G be one of the three five-blocks avoiding H, and suppose that it contains k pair-points. It contains $5-k$ private points, and its total intersection with the four bases is $5+k$. Each base meets G at most twice, while only four private points exist; hence $1 \le k \le 3$. Let $U \subset [4]$ index the private points in G, and let $E_G \subset \binom{[4]}{2}$ index its pair-points. Some $kl \in E_G$ has complementary pair $\{i, j\} \subset U$. Indeed, this is immediate for $k = 1$. For $k = 2$, otherwise the two edges of E_G would be a matching on the three indices in U; for $k = 3$, otherwise three edges would have to use the total residual capacity two of the two indices in U. Consequently

$$R_{ij} := \{u_i, u_j, v_{kl}\} \subset G, \qquad \{i, j, k, l\} = [4]. \tag{158}$$

The three external blocks give three distinct R_{ij}, since two maximal blocks cannot share a triple.

If H had $(d_3, d_5) = (9, 4)$, then lemma 3.18(K3.2) would select four of the six R_{ij}, in the triangle-plus-pendant pattern, as traces of four-blocks through H. A four-subset and a subset of at least three among six candidates intersect. The corresponding triple would belong both to such a four-block and to an external five-block, contrary to triple ownership. Thus no point has type $(9, 4)$. Since (149) forbids $(6, 4)$, every degree-four point has type $(12, 4)$, and there are at most four of them.

We shall also use the following two immediate carrier facts. A degree-four point cannot carry a singleton intersection: its four bases form the complete quadrangle just obtained, so every two share a second point. A degree-three point carries at most one singleton, since two singleton pairs among its three bases would make their three four-point traces occupy at least $4 + 4 + 4 - 1 = 11$ of the ten link points. Finally, lemma 3.18(K3.1) says that a degree-three singleton carrier cannot have $d_3 = 6$ or 9. Hence

$$d_5(p) = 3 \text{ and } p \text{ carries a singleton} \quad \Longrightarrow \quad d_3(p) = 12. \tag{159}$$

Three immediate profiles. A disjoint pair is impossible in profile A. Indeed, if a_F counts degree-four points on F, its row defect is

$$\delta_F := \sum_{G \ne F} (2 - |F \cap G|) = 2 - a_F.$$

The two ends of a disjoint pair would both have $a_F = 0$, although their ten-point union cannot omit both degree-four points. Thus $D = 3$ consists of three singleton pairs. By (159) their three distinct carriers have $d_3 = 12$; together with the two degree-four points this contradicts $u \le 4$.

In profile C, the unique defect is one singleton. Its carrier cannot have degree four, and by (159) it cannot have degree three, for the four degree-four points already exhaust $u \le 4$.

Call the carrier q, and call the other degree-two point r. If a_F counts the degree-four points of F, the row identity is

$$a_F = 2 + \mathbf{1}_{\{r\in F\}}.$$

The two blocks through r therefore contain r and three of the four degree-four points; they share at least three points, a contradiction. Profile D is still shorter: its five degree-four points would all have $d_3 = 12$, again contradicting $u \le 4$.

The path residue B. Here the row defect is $\delta_F = 2 - a_F + c_F$, where c_F records the unique degree-two point. The same ten-point-union argument excludes a disjoint pair. The two units of defect are therefore singleton pairs. At most one is carried by the degree-two point; (159) and the three degree-four points force equality in (157):

$$u = 4,$$

with three points of type $(12, 4)$, one singleton carrier of type $(12, 3)$, and the degree-two point carrying the other singleton.

Label the seven blocks by a set V. Let T_i be the three-element complement in V of the support of the i-th $(12, 4)$ point, and let $E, Q \in \binom{V}{2}$ be the two singleton pairs, carried respectively by the degree-three and degree-two points. Pair multiplicity and the row defect give, if $t_v = \#\{i : v \in T_i\}$,

$$t_v = 1 + \deg_{E\cup Q}(v) - \mathbf{1}_{\{v\in Q\}}. \tag{160}$$

Thus precisely the two ends of E have T-degree two and all other labels have degree one. Moreover $|T_i \cap T_j| \le 1$: at a degree-four pivot the six complete-quadrangle pair-points are distinct, so two complement triples cannot reuse a pair of block labels. The three linear triples consequently form a path; after relabelling,

$$T_1 = \{a, x_1, x_2\}, \qquad T_2 = \{a, b, y\}, \qquad T_3 = \{b, z_1, z_2\}, \qquad E = \{a, b\}.$$

Because Q is itself a singleton, it meets every T_i. The two-point transversals of this path, apart from E, allow us to take $Q = \{a, z_1\}$.

Writing w for the third block through the carrier of E, pair multiplicity leaves, up to the path reflections, $w = x_1, y, z_2$. The choice $w = y$ forces both ax_1z_2, ax_2z_2 and leaves the pair by uncovered; the choice $w = z_2$ forces az_2x_1, yz_2x_2 and then reuses x_2y. Hence $w = x_1$, and all positive residual pair degrees force the six supports

$$ayz_2, \quad ax_2z_2, \quad bx_1z_2, \quad x_1yz_1, \quad byx_2, \quad bx_2z_1. \tag{161}$$

At the middle pivot the four forbidden S-indices are

$$x_1x_2, \quad x_1z_1, \quad x_2z_2, \quad z_1z_2,$$

a four-cycle; the only safe indices are the matching $\{x_1z_2, x_2z_1\}$. A $(12, 4)$ pivot has three further local three-lines, each of type T or S, and at most one is of type T. With no T, three safe S's are required, but only two exist. With one T, the other two safe S's must meet its complementary vertex, whereas the safe matching has degree one at every vertex. Profile B is impossible.

The degree-one residue E. Let q be the degree-one point, E its five-block, and $D_1, \dots, D_4$ the four points of type $(12, 4)$. Zero defect gives

$$E = \{q, D_1, D_2, D_3, D_4\}.$$

Index the other six blocks by the edges of K_4. The residual pair multigraph on these six labels has multiplicity two on the three pairs of opposite edges and multiplicity one on the twelve adjacent pairs. Its six degree-three supports each contain exactly one opposite pair. Grouping the labels into the three opposite pairs A, B, C, all four cross-edges between any two groups must be assigned in one direction: assignments in both directions would repeat one cross-edge and omit another. Since every group anchors two triples, these directions form a directed three-cycle. Up to reversal and relabelling, the six supports are therefore

$$\begin{aligned}&\{12, 34, 13\},\ \{12, 34, 24\},\ \{13, 24, 14\},\ \{13, 24, 23\},\\ &\{14, 23, 12\},\ \{14, 23, 34\}.\end{aligned} \tag{162}$$

This derived orientation has two consequences visible directly in (162): at each D_i the forbidden S-indices form the triangle on the three bases other than E, and the complements of the six incidence triples cover all fifteen pairs of the six degree-three points.

Since $d_3(D_i) = 12$ and $d_5(D_i) = 4$, the link equation gives $d_4(D_i) = 3$. Avoiding the forbidden triangle forces its three four-blocks to be either the S-star centred at E, or one T and two edges of that star. Every such T/S completion contains exactly one of the four points D_i; hence it is counted at only one pivot. Thus each D_i supplies three distinct four-blocks, at least two through q. These are twelve distinct blocks and exhaust all four-block incidences with the D_i. Globally $B_4 = 16$, while the link equation gives $d_4(q) \leq 11$. Hence among the four remaining four-blocks at least one avoids both q and all D_i; it consists of four of the six degree-three points. Its complementary pair is contained in the complement of one incidence triple by the covering property above. Equivalently, one of the five-blocks in (162) shares three points with this four-block, contrary to triple ownership. Profile E, and hence every row in (156), is impossible. $\square$

Lemma 3.32 (The $C = 40$ endpoint dispatch)**.** *There is no $n = 11$, maximum-circle-five configuration with $C = 40$.*

Proof. Put $k = B_5$ and $B = C + L = 40 + L$. The block and triple partitions give

$$N_3 + N_4 + k = B, \qquad N_3 + 4N_4 + 10k = 165,$$

and hence

$$3N_4 + 9k = 125 - L, \qquad L \equiv 2 \pmod 3. \tag{163}$$

Moreover

$$\sum_p \beta(p) = 3N_3 + 4N_4 + 5k = \frac{8L + 485}{3} - k,$$

where $\beta(p)$ is the number of blocks through p. Inversion at p turns precisely the $B - \beta(p)$ blocks avoiding p into proper circles of the admissible ten-point image. Thus $B - \beta(p) \geq f(10) = 33$, so $\beta(p) \leq L + 7$. Summing over the eleven pivots yields

$$25L + 3k \geq 254. \tag{164}$$

By lemma 3.28, $5k = \sum_p d_5(p) \leq 44$, hence $k \leq 8$; therefore (164) and (163) give $L \geq 11$.

On the other hand, the line-pair partition and Melchior give

$$3L + 4L_4 + 9L_5 \leq 52,$$

so $L \leq 17$. The congruence in (163) leaves $L \in \{11, 14, 17\}$. If $L = 17$, the last inequality forces $L_4 = L_5 = 0$, hence $L_3 = 17$, contrary to the eleven-point ceiling $L_3 \leq 16$ in lemma 3.11. Consequently

$$L \in \{11, 14\}.$$

For $L = 11$ the four mutually exclusive rows are

$$\begin{array}{c|c|c} B_5 & (n_6, n_9, n_{12}) & (N_3, N_4, N_5) \\ \hline 5 & (10,1,0) & (23,23,5) \\ 6 & (8,3,0) & (25,20,6) \\ 7 & (6,5,0) & (27,17,7) \\ 8 & (4,7,0) & (29,14,8). \end{array} \tag{165}$$

The cases $B_5 = 5, 6$ are excluded by lemma 3.30. For $B_5 = 7$, let a be the number of points of type $(d_3, d_5) = (9, 4)$, and put

$$b = \sum_{p: d_3(p)=6} (3 - d_5(p)).$$

The domains in lemma 3.28 and $\sum_p d_5(p) = 35$ give $a - b = 2$. By K3.3, double-counting harmonic certificates over the seven five-blocks gives $4a \le 14$. Hence $(a, b) = (2, 0)$ or $(3, 1)$, with degree multisets

$$(4^2, 3^9) \quad \text{or} \quad (4^3, 3^7, 2).$$

Put

$$D = 2\binom{7}{2} - \sum_p \binom{d_5(p)}{2}.$$

The two displayed profiles give respectively $D = 3$ and $D = 2$. By lemma 3.29, the first profile cannot realize its odd defect.

In the second profile, the same lemma says that the defect must be one disjoint pair F, G. If a_E and c_E count the degree-four and degree-two points in a block E, its row defect is

$$\delta_E = \sum_{J \neq E} (2 - |E \cap J|) = 2 - a_E + c_E.$$

The disjoint pair exhausts the total defect, so $\delta_F = \delta_G = 2$ and $a_F = c_F$, $a_G = c_G$. Since F, G are disjoint and there is only one degree-two point, their union contains at most one of the three degree-four points. But $|F \cup G| = 10$ misses only one of the eleven points and therefore contains at least two, a contradiction.

For $B_5 = 8$, the profile is

$$7(9, 4, 4) + 4(6, 7, 3).$$

Thus

$$S_{55} = 7\binom{4}{2} + 4\binom{3}{2} = 54, \qquad I_\omega = 7 \cdot 4 = 28.$$

Since $58 - N_4 - 3B_5 = 20$, (127) becomes

$$\sum_\Delta H_\Delta = 240 + \sum_\Delta q_\Delta.$$

The bounds $H_\Delta \le 30$ and $q_\Delta \ge 0$ force $H_\Delta = 30$ and $q_\Delta = 0$ for every five-block Δ. Choose a proper five-circle Γ. By lemma 3.14(K2.1), the all-double triangle obstruction gives $n_{41} = n_{51} = 0$, while $q_\Gamma = 0$ gives $n_{40} = n_{50} = 0$. Hence $n_{42} = 14, n_{52} = 7$ and $H_\Gamma = 14 + 3 \cdot 7 = 35$, contradiction. This closes all of (165).

Now let $L = 14$. By lemma 3.20, only $B_5 = 7, 8$ remain. The seven-block row is excluded by lemma 3.31. For $B_5 = 8$, write $u = n_{12}$. The pivot equations give

$$(n_6, n_9, n_{12}) = (u, 11 - 2u, u).$$

Points with $d_3 = 6$ have $d_5 \le 3$, and all points have $d_5 \le 4$. If a counts the points of type $(d_3, d_5) = (9, 4)$, the degree sum $\sum d_5 = 40$ gives $a \ge 7 - u$. The harmonic cap K3.3 gives $4a \le 2B_5 = 16$, so $u \ge 3$; the degree maximum $44 - u \ge 40$ gives $u \le 4$. Equality in these bounds forces in both cases the common degree row

$$(d_5(p))_{p \in P} = (4^7, 3^4),$$

and every degree-three point has $d_3 \in \{6, 9\}$.

The total intersection defect is $2\binom{8}{2} - (7\binom{4}{2} + 4\binom{3}{2}) = 2$. A singleton pair is impossible. Indeed, if its carrier has degree four, delete that carrier from its four incident bases and let r_x be the resulting multiplicity of each of the other ten points. The sixteen incidences give

$$\binom{4}{2} \ge \sum_x \binom{r_x}{2} \ge \sum_x (r_x - 1)_+ \ge 16 - 10 = 6.$$

Thus equality holds throughout and every pair of those four bases has a second intersection point, contradicting singleton intersection. A degree-three carrier is excluded by K3.1. If instead the defect were one disjoint pair F, G, and a_E counted degree-four points in E, then

$$\delta_E = \sum_{J \ne E} (2 - |E \cap J|) = 4 - a_E.$$

The disjoint pair exhausts the defect, so $a_F = a_G = 2$. Their ten-point union would contain only four of the seven degree-four points although it can omit only one. This contradiction excludes $B_5 = 8$ and closes the $L = 14$ layer. ☐

3.7 The eleven-point six-circle branch

Lemma 3.33. *If eleven admissible points have a largest proper circle of size six, then* $C \ge 41$.

Proof. Fix the six-circle Γ and let X be the five outsiders. The rich-line bound excludes lines of size at least six. For a block type (g, x) write c_{gx}, ℓ_{gx}, and let N_s count blocks with s outsiders. Apart from Γ, the full type universe is

$$\mathcal{U} = \{(0,3), (0,4), (0,5), (1,2), (1,3), (1,4), (1,5), (2,1), (2,2), (2,3), (2,4)\},$$

with one circle and one line variable for each type (the already excluded six-line variables may harmlessly be retained as zeros). Triple ownership gives the exact rows

$$\begin{aligned} c_{60} &= 1, \\ \sum_{\mathcal{U}} \binom{x}{3} (c_{gx} + \ell_{gx}) &= 10, \\ \sum_{\mathcal{U}} g \binom{x}{2} (c_{gx} + \ell_{gx}) &= 60, \\ \sum_{\mathcal{U}} \binom{g}{2} x (c_{gx} + \ell_{gx}) &= 75. \end{aligned} \tag{166}$$

Each outsider triple has a unique owner block, whose footprint on X has size three, four, or five. The collections of three-subsets contained in distinct footprints are disjoint, by triple ownership. Two four-subsets of the five-set X share a triple, so at most one four-footprint occurs; a five-footprint contains all ten triples and excludes every other footprint. Consequently the ten outsider triples have exactly one of the three profiles

$$(N_3, N_4, N_5) = (10, 0, 0), (6, 1, 0), (0, 0, 1),$$

called A, B, C. Put

$$W = c_{22} + 3c_{23} + 6c_{24} = \sum_{e \in \binom{X}{2}} q_e.$$

The nonnegative slacks used below are, explicitly,

$$\begin{aligned}
\sigma_6 &= 30 - (2\ell_{12} + 3\ell_{13} + 4\ell_{14} + 5\ell_{15} + 2\ell_{21} + 4\ell_{22} + 6\ell_{23} + 8\ell_{24}),\\
\sigma_{10} &= 3c_{03} + 2c_{12} + c_{21} - 15,\\
\sigma_{17} &= 52 - (3\ell_{03} + 7\ell_{04} + 12\ell_{05} + 3\ell_{12} + 7\ell_{13} + 12\ell_{14} + 18\ell_{15}\\
&\qquad + 3\ell_{21} + 7\ell_{22} + 12\ell_{23} + 18\ell_{24}),\\
\sigma_{63} &= 28 - W,\\
\sigma_L &= 2 - (\ell_{03} + \ell_{13} + \ell_{23} + 2\ell_{04} + 2\ell_{05} + 2\ell_{14} + 2\ell_{15} + 2\ell_{24}).
\end{aligned} \tag{167}$$

Here σ_6 is the unused capacity among the thirty Γ–outsider pairs in line blocks, σ_{10} is the summed ordinary-block inequality over the outsider group, and σ_{17} is the global Melchior slack. More explicitly, the pivot bridge corollary 2.17 supplies at least three proper three-circles through each outsider. Summing over the five outsiders counts a c_{03} block three times, a c_{12} block twice, and a c_{21} block once. Hence

$$3c_{03} + 2c_{12} + c_{21} \geq 15,$$

which proves $\sigma_{10} \geq 0$.

For σ_L, consider the outsider footprints of maximal lines which contain at least three points of X. Distinct footprints meet in at most one point. On a five-set, a footprint of size four or five therefore excludes every other such footprint. If two three-footprints occur, they meet in one point and have union X; a third three-footprint would meet one of them in at least two points. Assigning weight one to a three-footprint and weight two to a footprint of size at least four gives total weight at most two, exactly the inequality $\sigma_L \geq 0$ printed above.

Finally U17 on each of the five four-subsets of X gives $3W \leq 5 \cdot 17$, hence the integral bound $W \leq 28$ and $\sigma_{63} \geq 0$.

The triple rows, Melchior rows, and U17 give the following exact nonnegative-slack identity:

$$\begin{aligned}
C - \left(32 + \frac{9}{10}N_3 + \frac{3}{5}N_4\right) &= \frac{\sigma_6}{2} + \sigma_{63} + \sigma_L + c_{12} + 2c_{24}\\
&\quad + \ell_{04} + \ell_{05} + \ell_{12} + \frac{3}{2}\ell_{13} + 3\ell_{14} + \frac{7}{2}\ell_{15} + \ell_{24}.
\end{aligned} \tag{168}$$

In footprint C, where $N_4 = 0$ and $N_5 = 1$, there is the second exact identity

$$\begin{aligned}
C - \frac{467}{12} &= \frac{\sigma_6}{8} + \frac{\sigma_{10}}{2} + \frac{\sigma_{17}}{12} + 2c_{04} + \frac{3}{2}c_{13} + 2c_{14}\\
&\quad + \frac{3}{4}\ell_{03} + \frac{19}{12}\ell_{04} + \frac{1}{2}\ell_{12} + \frac{35}{24}\ell_{13} + \frac{5}{2}\ell_{14} + \frac{9}{8}\ell_{15}\\
&\quad + \frac{1}{12}\ell_{22} + \frac{3}{4}\ell_{23} + \frac{3}{2}\ell_{24}.
\end{aligned} \tag{169}$$

For a coefficientwise certificate, denote the first and third partition rows in (166) by T_0, T_2. The complete multiplier data are

	inequality slacks			c_{60}	T_0	T_2	N_3	N_4
F1	$\frac{1}{2}\sigma_6$	σ_{63}	σ_L	1	$\frac{1}{10}$	1	$\frac{9}{10}$	$\frac{3}{5}$

For F2, after imposing the footprint-C row $N_4 + 6(N_5 - 1) = 0$, the multiplier data are

	inequality slacks			c_{60}	T_0	T_2	N_4	N_5
F2	$\frac{1}{8}\sigma_6$	$\frac{1}{2}\sigma_{10}$	$\frac{1}{12}\sigma_{17}$	1	$-\frac{1}{2}$	$\frac{1}{2}$	1	6.

Expanding these fixed linear combinations over the displayed universe $\mathcal{U}$ gives exactly the nonnegative residues printed in (168) and, in footprint C, (169); thus the identities are directly checkable without a type search. Therefore A gives $C \geq 41$, B gives $C \geq 38$, and C gives $C \geq 39$. At $C = 38$ equality in B would give, on writing $y = c_{13}, q = c_{14}, f = \ell_{22}, w = \ell_{23}$ and $z = c_{23}$,

$$3y + 6q + 2f + 6w = 4.$$

The remaining exact rows give

$$c_{04} = 1 - q, \quad c_{03} = 4 - y - z, \quad \ell_{03} = 2 - w,$$
$$c_{21} = 4 + 3z, \quad c_{22} = 28 - 3z, \quad \ell_{21} = 15 - 2f - 3w.$$

Substitution in (167) yields $\sigma_{10} = 1 - 3y \geq 0$ and $\sigma_{17} = 1 - f \geq 0$. Hence $y \leq 1/3$, $f \leq 1$, and the integer $r = q + w$ satisfies

$$\frac{1}{6} \leq r = \frac{4 - 3y - 2f}{6} \leq \frac{2}{3},$$

which is impossible.

It remains to treat $C = 39, 40$. In footprint B, (168), after substituting $\sigma_{63} = 28 - W$, reads exactly

$$C + W - 66 = \frac{\sigma_6}{2} + \sigma_L + c_{12} + 2c_{24} + \ell_{04} + \ell_{05} + \ell_{12} + \frac{3}{2}\ell_{13} + 3\ell_{14} + \frac{7}{2}\ell_{15} + \ell_{24}. \tag{170}$$

The unique four-host clause of lemma 3.15 gives $W \leq 26$. For $C = 39$, (170) instead requires $W \geq 27$. For $C = 40$, both inequalities are equalities. Thus $W = 26$ and every summand on the right of (170) vanishes; in particular,

$$\sigma_6 = \ell_{12} = \ell_{13} = \ell_{14} = \ell_{15} = \ell_{24} = 0.$$

The definition of σ_6 now reduces to

$$2\ell_{21} + 4\ell_{22} + 6\ell_{23} = 30.$$

Since $\ell_{24} = 0$, the left side divided by two is precisely the incidence number J of outsiders with original lines carrying a pair of points of Γ. Hence

$$J = \ell_{21} + 2\ell_{22} + 3\ell_{23} = 15,$$

contrary to $J \leq 14$ in lemma 3.15.

For footprint C at $C = 39$, multiplying (169) by 24 gives

$$2 = 3\sigma_6 + 12\sigma_{10} + 2\sigma_{17} + 12\ell_{12} + 27\ell_{15} + 2\ell_{22}.$$

All quantities are nonnegative integers, so

$$\sigma_6 = \sigma_{10} = \ell_{12} = \ell_{15} = 0, \qquad \sigma_{17} + \ell_{22} = 1.$$

There are no three- or four-outsider blocks in footprint C. Consequently the tight σ_6 row gives

$$\ell_{21} = 15 - 2\ell_{22},$$

and the definition of the Melchior slack becomes

$$\sigma_{17} = 52 - (12\ell_{05} + 3\ell_{21} + 7\ell_{22}) = 7 - 12\ell_{05} - \ell_{22}.$$

Adding ℓ_{22} and using $\sigma_{17} + \ell_{22} = 1$ yields $12\ell_{05} = 6$, impossible for an integer block count.

It remains to exclude footprint C at $C = 40$. Here the left side of $24\,(169)$ is 26. Its coefficient 27 excludes an ℓ_{15} host, so the unique five-outsider host has type c_{15}, ℓ_{05}, or c_{05}. Put

$$a = c_{12}, \quad b = c_{21}, \quad c = c_{22} = W, \qquad d = \ell_{12}, \quad e = \ell_{21}, \quad f = \ell_{22}.$$

Only these six inner variables can occur. The Γ–outsider line-pair capacity, namely $\sigma_6 \geq 0$, gives

$$d + e + 2f \leq 15. \tag{171}$$

For each outsider, invert at that point. The remaining ten points are noncollinear: otherwise restoring the pivot would put all eleven original points on one generalized block, contrary to the established size cap. Thus theorem 2.11 supplies at least five ordinary lines. Summing over the five outsiders counts each generalized three-block once for each outsider on it. Since footprint C has no three-outsider block, this gives the second structural row

$$2a + b + 2d + e \geq 25. \tag{172}$$

Suppose first that the host is c_{15}. Circle count and the two mixed triple rows in (166) become

$$a + b + c = 38, \qquad a + 2c + d + 2f = 50, \qquad b + 2c + e + 2f = 75. \tag{173}$$

Substitution in (172) and (171), respectively, gives

$$175 - 6c - 6f \geq 25, \qquad 87 - 3c - 2f \leq 15;$$

equivalently, $c + f \leq 25$ and $3c + 2f \geq 72$. Since $3c + 2f = c + 2(c + f)$, we get $W = c \geq 22$. But this host contains a point of Γ, so lemma 3.15 gives $W \leq 20$.

If the host is ℓ_{05}, it already owns all ten outsider pairs in the line-pair row, whence $d = f = 0$. The exact rows are

$$a + b + c = 39, \qquad a + 2c = 60, \qquad b + 2c + e = 75.$$

Thus $b = c - 21$ and

$$J = e = 96 - 3c = 96 - 3W.$$

The disjoint-host bound $W \leq 26$ now gives $J \geq 18$, contrary to the global bound $J \leq 14$ in lemma 3.15.

Finally suppose that the host is c_{05}. The three exact rows are

$$a+b+c=38, \qquad a+2c+d+2f=60, \qquad b+2c+e+2f=75. \tag{174}$$

Substitution in (172) and (171) gives

$$195-6c-6f \geq 25, \qquad 97-3c-2f \leq 15,$$

or $c+f \leq 85/3$ and $3c+2f \geq 82$. Hence $c \geq 76/3$, and integrality together with the disjoint-host bound forces $W=c=26$. The circle row gives $a+b=12$, while $\sigma_{10} \geq 0$ (here $c_{03}=0$) says $2a+b \geq 15$; hence $a \geq 3$. The last row of (174) now yields

$$J=e+2f=75-b-2c=11+a \geq 14.$$

All five outsiders lie on one fixed proper circle, so the localized clause of lemma 3.15 gives $J \leq 13$, the final contradiction. Thus all footprints and both boundary values are excluded. $\square$

Proposition 3.34. *One has $f(11)=41$.*

Proof. Assume $C \leq 40$. By lemma 3.1, the largest proper circle has size five or six. The five-circle branch is excluded by lemmas 3.21, 3.27 and 3.32, and the six-circle branch by lemma 3.33. Construction 2.5 attains 41. $\square$

3.8 Twelve points: the five-circle branch

We first isolate the scalar spine. Its role is only to route the proof; all geometric equality cases will be discharged explicitly afterwards.

Lemma 3.35 (Five-circle spine at twelve points)**.** *Suppose that $|P|=12$, the largest proper circle has size five, and $C \leq 50$. Put $B_s=C_s+L_s$ and*

$$\sigma_p=d_3(p)-3-d_5(p)-2d_6(p), \qquad K=\sum_p \sigma_p.$$

Then every block has size $3,4,5$, or 6, every six-block is a line, $B_5 \leq 12$, and

$$B_3+4B_4+10B_5+20B_6=220, \tag{175}$$
$$12B_4+35B_5+72B_6=624-K, \tag{176}$$
$$4L=256-4C-B_5-4B_6+K, \tag{177}$$
$$9B_5 \geq 1776-36C+5K+72B_6. \tag{178}$$

At every point,

$$d_3+3d_4+6d_5+10d_6=55, \qquad 3d_4+7d_5+12d_6=52-\sigma, \qquad d_5 \leq 6. \tag{179}$$

If $B_6=L_5=0$, write

$$d_3=7+3j, \quad \epsilon=\ell_4, \quad u=5-\ell_3-\ell_4, \quad q=u+2j+2-d_5. \tag{180}$$

Then $j,u,q \geq 0$. Moreover,

$$j=0 \Longrightarrow d_5 \leq 4, \qquad j=1 \Longrightarrow d_5 \leq 5, \qquad j \geq 2 \Longrightarrow d_5 \leq 6. \tag{181}$$

In addition, the two local types

$$(j,u,\epsilon,q,d_5)=(2,0,0,0,6), \qquad (2,1,0,1,6) \tag{182}$$

do not occur.

Proof. The line cap in lemma 3.1 leaves only blocks of the stated sizes, and a six-block cannot be a proper circle. Triple ownership gives (175). Summed Melchior after inversion gives

$$K = 3B_3 - 36 - 5B_5 - 12B_6.$$

Eliminating B_3 proves (176); adding $C + L = \sum_s B_s$ proves (177). Finally, the summed added-centre inequality is

$$5L_3 + 12L_4 + 20L_5 + 28L_6 \leq 4C - 124 + C_5.$$

Its left side is at least $5L + 23B_6$, while $C_5 \leq B_5$; substitution of (177) gives (178).

For a five-block F, let $y_F = \mathbf{1}_F - \frac{5}{12}\mathbf{1}$. These vectors lie in the eleven-dimensional space $\mathbf{1}^\perp$, and distinct ones satisfy

$$\langle y_F, y_G \rangle = |F \cap G| - \frac{25}{12} \leq -\frac{1}{12}.$$

A nonzero obtuse family in $\mathbb{R}^{11}$ has at most twelve members, proving $B_5 \leq 12$.

The first local equality in (179) partitions the pairs of the other eleven points; eliminating d_3 gives the second. If seven five-blocks passed through p, deleting p would leave seven four-sets on eleven points, pairwise meeting in at most one point. Their 28 incidences have pair moment at least

$$6\binom{3}{2} + 5\binom{2}{2} = 23 > \binom{7}{2},$$

which is impossible. Hence $d_5 \leq 6$.

Assume now $B_6 = L_5 = 0$. Kelly–Moser after inversion and the congruence in (179) give $d_3 = 7 + 3j$ with $j \geq 0$. The arms of the rich lines through p are disjoint, so $u \geq 0$. The added-centre slack is

$$\kappa = 3(u + j) - \epsilon - d_5 \geq 0. \tag{183}$$

If $j = 0$, then $\sigma \geq 0$ gives $d_5 \leq 4$. To prove the middle cap and record its two later equality adapters, suppose $d_5 = 6$ and invert at p. The six resulting four-lines have multiplicities m_x on the remaining eleven selected points. Pair ownership gives

$$\sum_x m_x = 24, \qquad \sum_x \binom{m_x}{2} \leq \binom{6}{2} = 15.$$

Writing $z_x = m_x - 2$, we have $\sum z_x = 2$ and

$$\sum_x \binom{m_x}{2} = 14 + \frac{1}{2}\sum_x z_x^2 \geq 15.$$

Thus equality holds: the profile is $(3, 3, 2^9)$ and every pair of the six lines meets at a selected point. On each four-line, $\sum_{x \text{ on the line}}(m_x - 1) = 5$, so it contains exactly one of the two triple points. The six lines are therefore two three-line pencils, and their other nine intersections form a 3×3 grid. Every further three-point image line avoids the pencil centres by pair ownership and is a grid transversal. Every image of an original line through p is a two-secant through the inversion centre o; apart from the possible line through the two pencil centres, it is an ordinary grid secant.

If $j = 1$, (179) gives $d_4 = 3$, hence three grid transversals, contrary to lemma 3.16. This proves $d_5 \leq 5$. For the first type in (182), the local equations give $(d_3, d_4, \ell_3) = (13, 2, 5)$; the two transversals normalize the grid, while at least four ordinary grid secants concur at

o. For the second they give $(13, 2, 4)$ and at least three such secants. Both contradict the final clause of lemma 3.16. The last cap in (181) is the already proved $d_5 \le 6$.

It remains to check $q \ge 0$. For $j = 0$, the cap settles $u \ge 2$, while (183) gives $d_5 \le 3u$ for $u \le 1$. For $j = 1$, the cap settles $u \ge 1$, and (183) settles $u = 0$. For $j \ge 2$, $d_5 \le 6 \le u + 2j + 2$. This proves every assertion. □

We shall use once the following parity consequence of the local line arrangement. After inversion and restoration of the centre, dualize the twelve-point configuration. The product of the twelve defining linear forms two-colours the faces. Counting edge–face incidences shows that the two colour sums of $|F| - 3$ are congruent modulo three. Consequently the total Melchior excess cannot equal one; in the notation above,

$$\kappa_p \ne 1 \qquad \text{for every } p. \tag{184}$$

Lemma 3.36 (The nonnegative q-front)**.** *Assume the hypotheses of lemma 3.35, and in addition $B_6 = L_5 = 0$ and $K = B_5 = k$. Put $h = L_4$. Then*

$$\sum j = 2k - 16, \qquad \sum u = 3C - 132 - h, \qquad \sum q = 3C - 140 - k - h. \tag{185}$$

In particular $k \ge 8$ and $k + h \le 3C - 140$.

Proof. Equations (175)–(177) give

$$B_3 = 12 + 2k, \qquad B_4 = 52 - 3k, \qquad L = 64 - C.$$

Now sum the four definitions in (180), using $\sum d_3 = 3B_3$, $\sum d_5 = 5k$, $\sum \ell_3 = 3(L - h)$, and $\sum \epsilon = 4h$. This gives (185). Its first and last members are sums of nonnegative integers. □

Lemma 3.37 (Twelve-point five-circle closure)**.** *If twelve admissible points contain a proper five-circle and no larger proper circle, then $C \ge 51$.*

Proof. Suppose $C \le 50$. From (178) and $B_5 \le 12$,

$$108 \ge 1776 - 36C + 5K + 72B_6.$$

Thus $C \ge 47$. Moreover (176) gives $B_5 \equiv K \pmod{12}$. At $C = 47$ the preceding inequality and $1 \le B_5 \le 12$ leave only $K = B_6 = 0, B_5 = 12$, and (177) then gives $L = 14$. For $r_p = \sum_{s=3}^{6} \ell_s(p)$, the pointwise added-centre row gives

$$r_p \le 3 + \left\lceil \frac{\sigma_p}{3} \right\rceil, \qquad \sum_p r_p \le 36 + K.$$

But $\sum_p r_p \ge 3L = 42$, a contradiction.

At $C = 48$, the inequality is $48 + 5K + 72B_6 \le 9B_5 \le 108$. Hence $B_6 = 0$, $K \le 12$, and the congruence $B_5 \equiv K \pmod{12}$ gives the complete list

$$(K, B_5, B_6, L) = (0, 12, 0, 13), \qquad (12, 12, 0, 16).$$

In the first row $\sigma = 0$ pointwise, so (179) modulo three and $d_5 \le 6$ give $d_5 \in \{1, 4\}$. Thus $\sum d_5 \le 48 < 60 = 5B_5$. In the second row the total added-centre slack is

$$Q = \sum_p \kappa_p = -7L_4 - 16L_5.$$

Hence $L_4 = L_5 = 0$, and lemma 3.36 gives $\sum q = 144 - 140 - 12 = -8$, impossible.

At $C = 49$, the corresponding inequality is $12 + 5K + 72B_6 \leq 9B_5 \leq 108$, so $B_6 \leq 1$. If $B_6 = 1$, the congruence leaves only $K = 0, B_5 = 12$; if $B_6 = 0$, it leaves that endpoint and $K = B_5 = k$, $3 \leq k \leq 12$. The two $K = 0, B_5 = 12$ endpoints are excluded by the same congruence and degree sum. Every remaining endpoint has

$$B_6 = 0, \qquad K = B_5 = k, \quad 3 \leq k \leq 12, \qquad L = 15,$$

and total added-centre slack

$$Q = k - 3 - 7L_4 - 16L_5 \geq 0.$$

Thus $L_5 = 0$. The two nonnegative sums in (185) say simultaneously $k \geq 8$ and $k + L_4 \leq 7$, a contradiction.

We turn to $C = 50$. Here $-24 + 5K + 72B_6 \leq 9B_5 \leq 108$, hence $B_6 \leq 1$. Splitting the single congruence $B_5 \equiv K \pmod{12}$ over $1 \leq B_5 \leq 12$ gives exactly five families:

B_6	K	B_5	L
0	k	k $(1 \leq k \leq 12)$	14
0	0	12	11
0	$B_5 + 12$	$9 \leq B_5 \leq 12$	17
1	0	12	10
1	12	12	13.

(186)

The two $K = 0$ rows again fail the congruence degree sum. In the last row, $Q = -7L_4 - 16L_5$, so $L_3 = 12, L_6 = 1$ and $\kappa = 0$ pointwise. It follows that $\sigma = 1$ at every point. Equation (179) then gives $d_5 \in \{0, 3\}$ on the six-line and $d_5 \in \{0, 3, 6\}$ off it, whence $\sum d_5 \leq 6 \cdot 3 + 6 \cdot 6 = 54 < 60$.

In the $L = 17$ row, write $k = B_5$. The total slack forces $L_4 = L_5 = 0$. The definitions in (180), with the corresponding shifted value of K, give

$$\sum j = 2k - 12, \qquad \sum u = 9, \qquad \sum q = 9 - k.$$

Thus $k = 9$, $Q = 0$, all q and κ vanish, and exactly three points have $(j, u, d_5) = (2, 0, 6)$; all others have $(j, u, d_5) = (0, 1, 3)$. At any of the three former points $\epsilon = q = 0$, so this is the first forbidden type in (182). We are left with

$$B_6 = 0, \qquad K = B_5 = k, \qquad L = 14. \tag{187}$$

We next remove a possible five-line. The total slack in (187) is $6 + k - 7L_4 - 16L_5$. Hence $L_5 > 0$ would force $L_5 = 1, L_4 = 0$ and $k \in \{10, 11, 12\}$. Put

$$z_p = \ell_5(p), \qquad d_3(p) = 7 + 3j_p, \qquad u_p = 5 - \ell_3(p) - 2z_p, \qquad \widehat{q}_p = u_p + 2j_p + 2 + z_p - d_5(p).$$

Here $z_p \in \{0, 1\}$. The eleven-point inversion link is noncollinear by the established block caps, so Kelly–Moser and the link congruence give $j_p \geq 0$; the line-pencil partition gives $u_p \geq 0$. The degree caps in (181) require only $B_6 = 0$, not $L_5 = 0$. Indeed, for $j_p = 0$ they follow from $\sigma_p = 4 - d_5(p) \geq 0$, and for $j_p \geq 2$ from the universal bound $d_5(p) \leq 6$. If $j_p = 1$ and $d_5(p) = 6$, the six four-point lines in the inversion link have equality profile $(3, 3, 2^9)$, exactly as in the proof of (181); they form two three-line pencils and a real 3×3 grid. The local equations give $d_4(p) = 3$, hence three grid transversals, contrary to lemma 3.16. This argument is unchanged if one of the six base lines is the image of the unique five-line. Thus the three caps are valid throughout the present branch.

Restored-centre Melchior gives the nonnegative slack

$$\widehat{\kappa}_p = 3(u_p + j_p) + z_p - d_5(p) \geq 0.$$

If $j_p = 0$, the degree-four cap settles $u_p \geq 2$, while $\widehat{\kappa}_p \geq 0$ settles $u_p \leq 1$. If $j_p = 1$, the degree-five cap settles every case except $u_p = z_p = 0$, which is again covered by $\widehat{\kappa}_p \geq 0$. If $j_p \geq 2$, use $d_5(p) \leq 6$. Consequently $\widehat{q}_p \geq 0$ for every point. On the other hand,

$$\sum_p z_p = 5, \quad \sum_p u_p = 11, \quad \sum_p j_p = 2k - 16, \quad \sum_p d_5(p) = 5k,$$

and therefore

$$\sum_p \widehat{q}_p = 11 + 2(2k - 16) + 24 + 5 - 5k = 8 - k < 0.$$

This contradiction proves $L_5 = 0$.

Set $h = L_4$ and retain q from (180). The exact sums now are

$$\sum j = 2k-16, \quad \sum u = 18-h, \quad \sum q = 10-k-h, \quad \sum \epsilon = 4h, \quad \sum \kappa = 6+k-7h. \tag{188}$$

Their nonnegativity leaves precisely

$$(h, k) = (0, 8), (0, 9), (0, 10), (1, 8), (1, 9), (2, 8).$$

Eliminating d_5 between q and (183) gives

$$d_5 = u + 2j + 2 - q, \qquad \kappa = 2u + j + q - \epsilon - 2. \tag{189}$$

Using (184), the caps (181), and the two exclusions (182), these identities imply

local condition	forced conclusion
$\epsilon = 0, q = 0$	$j = 0$
$\epsilon = 0, q = 1$	$j \leq 1$
$q = 0, \epsilon = 1$	$(j, u, d_5, \kappa) = (1, 1, 5, 0)$
$q = 0, \epsilon = 2$	$(j, u, d_5, \kappa) = (0, 2, 4, 0).$

(190)

For completeness, in the first two rows $j \geq 1$ reduces either to $\kappa = 1$ or to $(j, u, q, d_5) = (2, 0, 0, 6)$ or $(2, 1, 1, 6)$; the latter two are (182). In the last two rows, (189) and $\kappa \geq 0$ leave the displayed value directly.

Now $(0, 10)$ contradicts $\sum j = 4$ and the first row of (190); $(0, 9)$ contradicts $\sum j = 2$ because only the unique point with $q = 1$ can have $j > 0$; $(1, 9)$ puts $j = 1$ at all four points of the four-line although $\sum j = 2$; and $(1, 8)$ would require positive q at all four line points although $\sum q = 1$. At $(2, 8)$, $j = q = 0$ pointwise, so every point of either four-line must have $\epsilon = 2$; the two distinct lines would have the same support.

Only $(h, k) = (0, 8)$ remains. Here $j = \epsilon = 0$ pointwise, $\sum q = 2$, and $\sum u = 18$. At least ten points have $q = 0$; at each of them (189) and $d_5 \leq 4$ force $u \in \{1, 2\}$. If all had $u = 2$, their contribution to $\sum u$ would already be twenty. Hence some point has

$$(j, u, \epsilon, q, d_5, \kappa) = (0, 1, 0, 0, 3, 0).$$

Inversion, restoration of the centre, and duality turn this point into a twelve-line arrangement with multiplicity vector $(t_2, t_3, t_4) = (6, 14, 3)$ and distinguished word $3D + 4T$. This is the first forbidden completion in lemma 3.17. The last endpoint, and therefore the entire five-circle branch, is impossible. $\square$

3.9 Twelve points: the six-circle branch

Lemma 3.38 (Universal six-circle spine). *Assume $|P| = 12$, $C \le 50$, and the largest proper circle has size six. Put*

$$m = B_6, \quad k = B_5, \quad h = L_4, \quad g = L_5, \quad r = L_6, \quad e = 50 - C,$$

and define

$$\sigma_p = d_3(p) - 3 - d_5(p) - 2d_6(p), \qquad \kappa_p = 11 + \sigma_p - 3\ell_3(p) - 4\ell_4(p) - 5\ell_5(p) - 6\ell_6(p).$$

If $K = \sum_p \sigma_p$ and $Q = \sum_p \kappa_p$, then, for an integer $s \ge 0$,

$$K = k + 12s, \tag{191}$$

$$B_3 = 12 + 4m + 2k + 4s, \qquad B_4 = 52 - 6m - 3k - s,$$

$$B = 64 - m + 3s, \qquad L = 14 - m + 3s + e, \tag{192}$$

$$Q = 6 + 9m + k - 15s - 9e - 7h - 16g - 27r. \tag{193}$$

Moreover $1 \le m \le 4$, $m + k \le 12$, and

$$m = 4 \Rightarrow k = 0, \qquad m = 3 \Rightarrow k \le 3, \qquad m = 2 \Rightarrow k \le 8, \qquad m = 1 \Rightarrow k \le 11. \tag{194}$$

Pointwise,

$$3d_4 + 7d_5 + 12d_6 = 52 - \sigma, \qquad \sigma \equiv 1 - d_5 \pmod 3, \qquad \kappa \equiv \ell_5 - d_5 - \ell_4 \pmod 3, \tag{195}$$

where $\sigma, \kappa \ge 0$ and $\kappa \ne 1$.

Proof. The selected circle gives $m \ge 1$, and the outer router bounds every block by six. For a six-block A and a five-block F, set

$$y_A = \mathbf{1}_A - \frac{1}{2}\mathbf{1}, \qquad z_F = \mathbf{1}_F - \frac{5}{12}\mathbf{1}.$$

Distinct maximal blocks meet in at most two points, so

	$\lVert\cdot\rVert^2$	same-size inner product	mixed inner product
y	3	≤ -1	$\le -\frac{1}{2}$
z	$\frac{35}{12}$	$\le -\frac{1}{12}$	$\le -\frac{1}{2}$.

Thus all $m+k$ vectors form an obtuse family in $\mathbf{1}^\perp$, giving $m+k \le 12$. In particular $k \le 11$. Triple ownership and summed Melchior give $K \equiv k \pmod{12}$; hence $K = k + 12s$ with $s \ge 0$. Substitution in the triple equation and the circle–line count gives (192); summing the definition of κ gives (193).

For the six-block vectors alone,

$$0 \le \left\lVert \sum_A y_A \right\rVert^2 \le m(4 - m),$$

so $m \le 4$. More sharply, applying constant positive coefficients to the two size classes majorizes their Gram form by

$$G(m,k) = \begin{pmatrix} m(4-m) & -mk/2 \\ -mk/2 & k(36-k)/12 \end{pmatrix}. \tag{196}$$

Because the off-diagonal entries are nonpositive, copositivity forces the relevant determinant to be nonnegative. For $m = 3, 2, 1$ the determinants factor, respectively, as

$$\frac{k(36-10k)}{4}, \qquad \frac{k(36-4k)}{3}, \qquad \frac{k(36-2k)}{4}.$$

Together with $m+k \leq 12$ this gives all bounds in (194), except that $m = 2$ initially permits $k = 9$. At that null endpoint equality forces every rich-block intersection to have size two and

$$9(y_{A_1} + y_{A_2}) + 4\sum_F z_F = 0, \qquad 4d_5(p) + 9d_6(p) = 24.$$

The two six-blocks meet in two points, leaving eight points with $d_6 = 1$; there $4d_5 = 15$, impossible. When $m = 4$, the first diagonal entry of (196) is zero, so its negative mixed entry forces $k = 0$.

Finally, the pair partition at a pivot and the definitions of the slacks give (195). The assertion $\kappa \neq 1$ is precisely the even-cell argument preceding lemma 3.36; it depends only on the parity of the twelve-line dual arrangement, not on the selected circle size. ☐

Lemma 3.39 (Ordinary-direction inequality)**.** *Under the hypotheses of lemma 3.38, suppose $L_6 = 0$. At every point lying on a six-block,*

$$\sigma_p \geq 2d_5(p) + 6d_6(p) - 8. \tag{197}$$

If also $d_5(p) = \ell_4(p) = \ell_5(p) = \ell_6(p) = \kappa_p = 0$ and the branch under consideration has $d_6(p) \leq 2$, then $d_6(p) = 2$, $\sigma_p \geq 4$, and $\ell_3(p) \geq 5$.

Proof. Invert at p, restore the inversion centre, and dualize the resulting twelve-point configuration. A chosen proper six-block through p becomes a five-point line not containing the restored centre, hence supplies a fivefold vertex v of the dual twelve-line arrangement. Let t_i be the number of its i-fold vertices, let

$$\delta = \sum_{w \notin \mathrm{star}(v)} (\mathrm{mult}(w) - 1) = t_2 + 2t_3 + 3t_4 + 4(t_5 - 1) - 35,$$

and let

$$R = \#\{\text{vertices on the distinguished line}\} = \ell_2(p) + \ell_3(p) + \ell_4(p) + \ell_5(p).$$

The distinguished line misses v. Its intersections with the five lines of $\mathrm{star}(v)$ are therefore distinct, and every further vertex on it contributes at least one unit to the off-star defect. Thus $\delta \geq R - 5$.

For completeness, the inversion dictionary at p is

$$\begin{aligned} t_2 &= d_3 - \ell_3 + \ell_2, \quad t_3 = d_4 - \ell_4 + \ell_3, \\ t_4 &= d_5 - \ell_5 + \ell_4, \quad t_5 = d_6 + \ell_5, \end{aligned}$$

while the pointwise pair partitions are

$$\ell_2 + 2\ell_3 + 3\ell_4 + 4\ell_5 = 11, \qquad d_3 + 3d_4 + 6d_5 + 10d_6 = 55.$$

(All local quantities in these two displays are evaluated at p.) Substituting these identities and $\sigma_p = d_3(p) - 3 - d_5(p) - 2d_6(p)$ gives the exact integral identity

$$\delta - (R-5) = \frac{11 + \sigma_p - 2d_5(p) - 6d_6(p)}{3}. \tag{198}$$

Equality $\delta = R - 5$ is impossible. Indeed, equality says that every off-star vertex is ordinary and lies on the distinguished line. Send v to infinity and dualize back: the five lines in its star become five marked directions, and every remaining join passes through the point dual to the distinguished line. Deleting that point leaves six distinct real points whose joins use only those five directions. They are noncollinear. A sixfold vertex on the distinguished line would correspond before inversion to a six-line, excluded by $L_6 = 0$; a sixfold vertex away from it would correspond to a proper seven-circle, excluded by maximality of the selected six-circle. This contradicts theorem 2.14. Hence the integral left-hand side of (198) is at least one, and (197) follows.

For the final assertion, the same inversion dictionary gives

$$t_2 = d_3 - \ell_3 + \ell_2 = 3 + d_5 + 2d_6 + \kappa + \ell_4 + \ell_5 + \ell_6.$$

Under the clean hypotheses this is $t_2 = 3 + 2d_6$. Kelly–Moser gives $t_2 \geq 6$, so $d_6(p) \geq 2$; the branch hypothesis makes $d_6(p) = 2$. Now (197) gives $\sigma_p \geq 4$, and $\kappa_p = 0$ reads $11 + \sigma_p = 3\ell_3(p)$. □

Lemma 3.40 (Four six-blocks)**.** *The branch $B_6 = 4$ is impossible when $C \leq 50$.*

Proof. By lemma 3.38, $k = 0$. Equality in $\|\sum_A y_A\|^2 \leq m(4-m) = 0$ says that every pair of six-blocks meets in two points and $\sum_A y_A = 0$; coordinatewise, $d_6(p) = 2$ at all twelve points. Equation (195) gives $\sigma_p = 1 + 3j_p$, so $s \geq 1$. Two distinct six-lines meet in only one point, whereas every pair of the four six-blocks meets in two; consequently $r \leq 1$.

If $r = 0$, lemma 3.39 gives $\sigma_p \geq 4$ everywhere and therefore $s \geq 4$. On the other hand $Q = 42 - 15s - 9e - 7h - 16g \geq 0$ gives $s \leq 2$.

If $r = 1$, nonnegativity in (193) forces

$$s = 1, \quad e = h = g = Q = 0, \quad C = 50, \quad L_3 = 12, \quad L_6 = 1.$$

The residues force $\sigma_p = 1$ and $\kappa_p = 0$ at every point. At a point off the six-line the exact local data are

$$(d_3, d_4, d_5, d_6) = (8, 9, 0, 2), \qquad (\ell_2, \ell_3, \ell_4, \ell_5, \ell_6) = (3, 4, 0, 0, 0).$$

After inversion and duality this is a twelve-line arrangement with $(t_2, t_3, t_5) = (7, 13, 2)$ and distinguished word $3D + 4T$, the second forbidden completion in lemma 3.17. □

Lemma 3.41 (Three six-blocks)**.** *The branch $B_6 = 3$ is impossible when $C \leq 50$.*

Proof. Write $k = B_5$. The Gram cap gives $0 \leq k \leq 3$. The three six-blocks have total incidence eighteen. Since their union has at most twelve points,

$$\sum_p (d_6(p) - 1)_+ \geq 6,$$

whereas pairwise intersection at most two gives $\sum_p \binom{d_6(p)}{2} \leq 6$. Since $\binom{d}{2} \geq (d-1)_+$, equality holds throughout and

$$d_6 = (2^6, 1^6). \tag{199}$$

The scalar equation is

$$K = k + 12s, \qquad Q = 33 + k - 15s - 9e - 7h - 16g - 27r,$$

so $s \in \{0, 1, 2\}$.

For $s = 0$, the degree rows follow from one scalar defect. For $k = 2, 3$ let T be the total deficit from intersection size two among all rich-block pairs. Expanding the centred norm gives

$$0 \le 3 - \frac{k^2}{12} - 2T,$$

so the integer T is at most one. If $r_p = d_5(p) + d_6(p)$, then $\sum r_p = 18 + 5k$ and

$$\sum r_p^2 = \sum r_p + 2(2\binom{3+k}{2} - T).$$

For $k = 2$, the value $T = 1$ would give 66, below the balanced minimum 68 for total degree 28 on twelve points; for $k = 3$ it would give $91 < 93$, the balanced minimum for total degree 33. Thus $T = 0$. Taking first and second moments now gives the following degree rows and minimum residue costs:

k	$(d_5(p))_{p \in P}$	$\min \sum_p \sigma_p$
0	(0^{12})	12
1	$(1^5, 0^7)$	7
2	$(2^2, 1^6, 0^4)$	8
3	$(2^6, 1^3, 0^3)$	15.

The first two rows are immediate. In the last column we used the least nonnegative value in the class $1 - d_5 \pmod 3$; every entry is larger than $K = k$, excluding $s = 0$.

For $s = 1, 2$, nonnegativity of Q forces $r = 0$. Summing (197) and using $\sum d_5 = 5k$, $\sum d_6 = 18$ gives

$$K \ge 10k + 12. \tag{200}$$

At $s = 1$ this forces $k = 0$; then (199), the direction bound, and the residue give $K \ge 6 \cdot 4 + 6 \cdot 1 = 30 > 12$. At $s = 2$, (200) gives $k \le 1$. The case $k = 0$ again costs at least thirty. If $k = 1$, the minimum sigma cost is twenty five and equality forces the unique five-block to use five of the six points with $d_6 = 1$. Here $e = h = g = r = 0$ and $Q = 4$; those five points have $\kappa \equiv -1 \pmod 3$, hence $Q \ge 10$, a contradiction. □

Lemma 3.42 (Two six-blocks)**.** *The branch $B_6 = 2$ is impossible when $C \le 50$.*

Proof. Put $k = B_5$; by lemma 3.38, $0 \le k \le 8$ and $s \in \{0, 1, 2\}$. Let $I \le 2$ be the intersection size of the two six-blocks.

First take $s = 0$. If $r = 0$, sum (197) over the union of the six-blocks. If A, B, D denote the sums of d_5 on the classes $d_6 = 2, 1, 0$, respectively, then $D \le Ik$ and

$$k = K \ge 2(A + B) + 8I - 24.$$

Since $A + B = 5k - D$, this gives, for $I = 0, 1, 2$ respectively,

$$k \ge 10k - 24, \qquad k \ge 8k - 16, \qquad k \ge 6k - 8. \tag{201}$$

All three fail for $k \ge 3$. If $k = 0$, the sigma residue costs at least twelve; if $k = 1$, $(d_5) = (1^5, 0^7)$ costs at least seven. For $k = 2$, the five-block pair moment and the sigma budget force

$$d_5 = (1^{10}, 0^2), \qquad \sigma = (0^{10}, 1^2).$$

A point with $d_6 = 2$ violates (197), so $I = 0$; then $\sum d_5 d_6 = 10 > 4k$, although the four mixed block pairs meet in at most two points each.

If $s = 0$ and $r = 1$, (193) gives $e = h = g = 0$, $k \ge 3$, and $Q = k - 3$. Counting triples with two points on the six-line, while (192) gives

$$(C_3, C_4, C_5, C_6) = (9 + 2k, 40 - 3k, k, 1), \qquad L_3 = 11.$$

Hence

$$90 \le C_3 + 2C_4 + 3C_5 + 4C_6 = 93 - k,$$

so $k = 3$ and equality holds. Every proper circle then meets the six-line twice. Among the twenty triples off the line, the three five-circles cover at most three, the proper six-circle at most four, and the eleven triple-lines at most eleven. Thus at most $3 + 4 + 11 = 18$ can be owned, a contradiction.

Next let $s = 1$. Here $r = 0$ and

$$K = k + 12, \qquad Q = k + 9 - 9e - 7h - 16g. \tag{202}$$

When $g = 0$, the residue for κ implies pointwise

$$\kappa_p + 2\binom{d_5(p)}{2} \ge 2d_5(p) - 2\ell_4(p). \tag{203}$$

It is immediate for $d_5 \ge 3$ and follows from the three residues for $d_5 = 0, 1, 2$. Since $\sum_p \binom{d_5(p)}{2} \le k(k-1)$, summing gives

$$2k^2 - 11k + 9 - 9e + h \ge 0. \tag{204}$$

This removes $k = 2, 3, 4$ in the core $e = g = 0$ for every allowed h, and $k = 1, \dots, 5$ when $(e, h, g) = (1, 0, 0)$.

Here is the complete scalar routing, recorded to make exhaustiveness transparent. Since $k \le 8$, (202) gives

nonzero parameter	only values not already killed by (204)
$g = 1$	$(e, h, k) = (0, 0, 7)$
$e = 1, h = 1$	$(g, k) = (0, 7)$
$e = 1, h = 0$	$g = 0,\ k \in \{0, 6, 7, 8\}$
$e = g = 0$	$h \in \{0, 1, 2\},\ 0 \le k \le 8.$

Indeed $g \ge 2$, $e \ge 2$, or $h \ge 3$ makes $Q < 0$; in the first two rows $k = 8$ would give the forbidden value $Q = 1$. Thus $k = 6, 7, 8$ below cover every rich scalar row, $k = 5$ covers the sole remaining middle gap, and the clean argument for $k = 0, 1$ covers all tiny rows.

For $k \ge 5$, the two six-blocks must meet twice. Indeed $I = 0$ would give $\sum d_5 d_6 = 5k > 4k$. If $I = 1$, the three degree-class sums satisfy

$$A + B + D = 5k, \qquad 2A + B \le 4k, \qquad D \le k.$$

Hence $D - A \ge k$, so necessarily $D = k, A = 0, B = 4k$. Summing the direction inequality on the ten middle points gives $k + 12 \ge 2B - 20$, which forces $k \le 4$. Thus $I = 2$ for $k \ge 5$. On the classes of sizes $2, 8, 2$ we then have

$$A + B + D = 5k, \qquad D - A \ge k, \qquad B \le \frac{k + 28}{2}. \tag{205}$$

Let $\Phi_t(S)$ be the balanced minimum of $\sum_{i=1}^{t} \binom{x_i}{2}$ over nonnegative integers of sum S. Writing $S = qt + r$, $0 \le r < t$, balancing gives

$$\Phi_t(S) = t\binom{q}{2} + rq, \qquad \Phi_t(S+1) - \Phi_t(S) = \lfloor S/t \rfloor .$$

Convexity and (205) give

k	(A,B,D)	$\Phi_2(A)+\Phi_8(B)+\Phi_2(D)$	$k(k-1)$
8	$(7,18,15)$	70	56
7	$(5,17,13)$ or $(6,16,13)$	50	42
6	$(4,16,10)$	30	30.

Here is a supporting-slope certificate for the three minima. Write $(A,B,D)=(A_0+a,B_0+b,D_0+c)$ at the displayed row, so $a+b+c=0$. For $k=8$, the active constraints are $b\le 0$ and $b+2c\ge 0$; the three supporting slopes are $3,2,7$, and the increase is at least

$$3a+2b+7c=-b+4c\ge 0.$$

For $k=7$, use the first displayed minimizer. The constraints are $b\le 0$ and $b+2c\ge -1$, while slopes $2,2,6$ give an increase at least $4c$. Thus $c\ge 0$; when $c=0$, only $b=0,-1$ are possible, giving exactly the two displayed minimizers. For $k=6$, the constraints are $b\le 1$ and $b+2c\ge 0$. Slopes $2,2,5$ give an increase at least $3c$. If $c=0$, the only point besides the displayed minimizer has $(a,b,c)=(-1,1,0)$ and objective 31; if $c<0$, the two constraints conflict. This proves the table without an integer search. The first two rows are impossible. Equality at $k=6$ forces $d_5=(2^{10},5^2)$; every point then has minimum sigma two, so $K\ge 24>18$. This also disposes of the only scalar five-line case $(g,k)=(1,7)$ and the only positive-e or positive-h cases not already removed by (204).

For $k=5$, necessarily $e=g=0$. Set

$$G_p=\kappa_p+2\binom{d_5(p)}{2}-2d_5(p)+2\ell_4(p)\ge 0.$$

Equations (202) and the pair moment give $\sum G_p\le 4+h\le 6$. If $h\le 1$, any $d_5\ge 4$ has $G_p\ge 6$, exceeding the relevant bound; if $h=2$, then $Q=0$ and the residue improves this to $G_p\ge 8$. Hence $d_5\le 3$ everywhere. In (205), $D\le 6$ and $D-A\ge 5$, so $A+D\le 7$ and $B\ge 18>(5+28)/2$, impossible.

Only $k=0,1$ remain. If $h=1$, every point of the four-line contributes at least two to Q, whereas (202) gives $Q=2$ or 3. In the clean core $e=g=h=0$, the $k=0$ row has at least nine points with $d_5=\kappa=0$, and the $k=1$ row has seven; the clean clause of lemma 3.39 would put all of them in the at most two-point intersection of the six-blocks. The sole extra row $(e,h,g,k)=(1,0,0,0)$ has all twelve points clean and is identical. This exhausts $s=1$.

Finally let $s=2$. The scalar formula forces

$$e=h=g=r=0,\qquad k\in\{6,7,8\},\qquad Q=k-6.$$

The value $k=7$ has forbidden total slack one. For $k=8$, summing (197) on the six-block union gives, according as $I=1$ or 2, $K\ge 8k-16=48$ or $K\ge 6k-8=40$, while $K=32$; $I=0$ already violates the mixed moment. For $k=6$, $Q=0$ makes every d_5 a multiple of three. Writing $d_5=3a$ and using the pair moment,

$$\sum a=10,\qquad \frac{9\sum a^2-30}{2}\le 30,$$

so $d_5=(3^{10},0^2)$. The direction and residue costs are made explicit by

	$d_6=0$	$d_6=1$	$d_6=2$
$d_5=3$	1	4	10
$d_5=0$	1	1	4.

If the two six-blocks meet in $I = 0, 1, 2$ points, the d_6 classes have sizes $I, 12 - 2I, I$ in the order $2, 1, 0$. Were all twelve points of five-degree three, their minimum sigma cost would be $48 + 3I$. Placing the two zero-degree points optimally saves at most $6I + 3(2 - I) = 6 + 3I$. Thus the actual cost is at least $42 > 30 = K$, and $s = 2$ is impossible as well. $\square$

Lemma 3.43 (One six-block)**.** *The branch $B_6 = 1$ is impossible when $C \leq 50$.*

Proof. Let Γ be the unique six-block and put $k = B_5$. It is a proper circle, so $r = 0$. The scalar spine becomes

$$K = k + 12s, \qquad Q = 15 + k - 15s - 9e - 7h - 16g, \qquad s \in \{0, 1\}. \tag{206}$$

Write $A = \sum_{p \in \Gamma} d_5(p)$. Every five-block has an outsider footprint of size three, four, or five. Distinct footprint families own disjoint outsider triples. A size-five footprint is unique and excludes all size-four footprints; otherwise the complements of the size-four footprints are disjoint pairs, hence form a matching of size at most three. Consequently there is $u \in \{0, 1, 2, 3\}$ such that

$$A = 2k - u, \qquad \sum_{p \notin \Gamma} d_5(p) = 3k + u. \tag{207}$$

At a point of Γ, the direction inequality and the sigma residue give the lower envelope

$$\sum_{p \in \Gamma} \sigma_p \geq f(A), \qquad f(A) = \begin{cases} 6 - A, & A \leq 6, \\ 2A - 12, & A \geq 6. \end{cases} \tag{208}$$

For $k \geq 5$, equations (207)– (208) give $K \geq 4k - 18$. Hence

$$s = 0 \Rightarrow k \leq 6, \qquad s = 1 \Rightarrow k \leq 10. \tag{209}$$

The two boundary values are already contradictory. At $(s, k) = (0, 6)$, equality forces $A = 9, u = 3$ and uses the entire sigma budget on Γ. The six outsider degrees, all 1 modulo three, sum to 21 and have pair moment at least 30; the six positive degrees on Γ, of sum nine, add at least three. This exceeds $k(k-1) = 30$. At $(s, k) = (1, 10)$ one has $A = 17, u = 3$, and discrete convexity gives

$$\sum_p \binom{d_5(p)}{2} \geq \Phi_6(17) + \Phi_6(33) = 16 + 75 > 90 = k(k-1).$$

We next remove five-lines. The local ordinary-point identity is

$$t_2(p) = 3 + d_5(p) + 2\mathbf{1}_{p \in \Gamma} + \kappa_p + \ell_4(p) + \ell_5(p). \tag{210}$$

If $g = 1$, necessarily $s = 0$. Set

$$b_p = \frac{d_5(p) + \ell_4(p) + \kappa_p - \ell_5(p)}{3}.$$

The residue makes b_p integral, and Kelly–Moser applied to (210) gives $b_p \geq 1$, except possibly at the at most two points of Γ on the five-line. Thus $\sum b_p \geq 10$, while

$$\sum b_p = 2k - 2 - 3e - h.$$

Together with $k \leq 6$, this forces $k = 6, e = h = 0$, the boundary value already excluded. Two five-lines make $Q < 0$. Hence

$$g = 0. \tag{211}$$

Define the integer and the convex energy

$$a_p = \frac{d_5(p) + \ell_4(p) + \kappa_p}{3}, \qquad H(d) = \frac{(d-4)(d-5)}{2} = \binom{d}{2} - 4d + 10.$$

By (210) and Kelly–Moser,

$$a_p \geq 1, \qquad \sum_p a_p = 2k + 5 - 5s - 3e - h. \tag{212}$$

If $N_1 = |\{p : a_p = 1\}|$, then

$$N_1 \geq 19 + 5s + 3e + h - 2k. \tag{213}$$

The five-block pair moment gives

$$E := \sum_p (\sigma_p + H(d_5(p))) \leq k^2 - 20k + 120 + 12s. \tag{214}$$

There is one apparent exception to the useful pointwise lower bound. If $p \notin \Gamma$ and

$$(d_5, \ell_4, \ell_5, \kappa, \sigma) = (3, 0, 0, 0, 1), \tag{215}$$

then inversion and duality give $(t_2, t_3, t_4) = (6, 14, 3)$ with distinguished word $3D+4T$. The first part of lemma 3.17 excludes this Type-A pivot. In its absence, $a_p = 1$ implies

$$\sigma_p + H(d_5(p)) \geq 5. \tag{216}$$

Indeed $d_5 \leq 1$ gives $H \geq 6$; $d_5 = 2$ forces $(\ell_4, \kappa) = (1, 0)$ and $\sigma \geq 2$; and $d_5 = 3$ forces $\ell_4 = \kappa = 0$, where lemma 3.39 gives $\sigma \geq 4$ on Γ and (215) is the only smaller off-circle residue. Combining (213)– (216) yields the single inequality

$$(k-5)^2 \geq 13s + 15e + 5h. \tag{217}$$

For $s = 0$, (212) gives $2k \geq 7 + 3e + h$. With the boundary $k = 6$ gone, (217) leaves only $k \in \{4, 5\}$ and $e = h = 0$. Let z count the $a = 1$ points with $(d_5, \kappa) = (3, 0)$. Every other $a = 1$ point has energy at least six, so

$$z \geq 6N_1 - E.$$

For $k = 4$, (213)–(214) give $N_1 \geq 11, E \leq 56, z \geq 10$; for $k = 5$ they give $N_1 \geq 9, E \leq 45, z \geq 9$. Each of these z points has $\sigma \geq 4$ by the exclusion of Type A, contradicting $K = k$.

For $s = 1$, (212) gives $k \geq 6$, and the boundary $k = 10$ is gone. Inequality (217) leaves only

$$k = 9, \qquad e = h = g = 0, \qquad K = 21, \qquad Q = 9.$$

Here $\sum a_p = 18$. Equations (213)– (214) force exactly six points with $a = 1$ and six with $a = 2$. Let q_i be the sum of κ on the class $a = i$; then $q_1 + q_2 = 9$. Since $d_5 = 3i - \kappa$, one checks

$$a = 1 : \ 2H(d_5) \geq 2 + 5\kappa, \qquad a = 2 : \ 2H(d_5) \geq 2 - \kappa;$$

the difference in either inequality is $\kappa(\kappa - 2) \geq 0$. Consequently

$$2 \sum H(d_5) \geq 15 + 6q_1, \qquad \sum H(d_5) \leq k(k-1) - 60 = 12.$$

If $q_1 > 0$, then $q_1 \geq 2$ and these bounds conflict. Thus $q_1 = 0$; all six $a = 1$ points have $(d_5, \kappa) = (3, 0)$ and, with Type A excluded, contribute at least $24 > K$ to the sigma sum. This closes $s = 1$ and the one-six-block branch. ☐

Lemma 3.44 (Twelve-point six-circle closure)**.** *If twelve admissible points have largest proper circle of size six, then* $C \geq 51$*.*

Proof. Assume $C \leq 50$. By lemma 3.38, $B_6 \in \{1, 2, 3, 4\}$. These four values are excluded respectively by lemmas 3.40 to 3.43. □

Proposition 3.45. *One has* $f(12) = 51$*.*

Proof. Assume $C \leq 50$. By lemma 3.1, the largest proper circle has size five or six. These alternatives are excluded by lemmas 3.37 and 3.44. Construction 2.5 attains 51. □

Theorem 3.46. *The three remaining finite values are*

$$f(10) = 33, \qquad f(11) = 41, \qquad f(12) = 51.$$

Proof. Combine propositions 3.10, 3.34 and 3.45. □

4 The thirteen-point case and the direct large range

Write

$$F(n) = 1 + \binom{n-1}{2} - \left\lfloor \frac{n-1}{2} \right\rfloor .$$

We finish the finite range at $n = 13$ and then prove the lower bound $C \geq F(n)$ directly for every $n \geq 14$. The large-range argument is not recursive: it uses neither a value at a preceding integer nor a distinguished base value.

For a point set P, let C_s and L_s be the numbers of maximal proper circle blocks and maximal line blocks of size s, respectively, and put $B_s = C_s + L_s$ and $C = \sum_s C_s$. Every triple has a unique maximal generalized-block owner. We shall repeatedly use

$$\mathsf{T} := \sum_{s \geq 3} \binom{s}{3} B_s = \binom{n}{3}, \tag{218}$$

$$\mathsf{P} := \sum_{s \geq 3} s(4 - s) B_s \geq 3n, \tag{219}$$

$$\mathsf{D} := \sum_{s \geq 3} s(s-4) C_s + \sum_{s \geq 3} 2s(s-2) L_s \leq n(n-4). \tag{220}$$

Indeed, inversion at a selected point followed by Melchior's inequality gives

$$\sigma_p = d_3(p) - 3 - \sum_{s \geq 5} (s-4) d_s(p) \geq 0.$$

Summing this inequality gives (219). Applying Melchior once more after restoring the centre of inversion gives $\sum_s s\ell_s(p) \leq n - 1 + \sigma_p$; summing over p and eliminating the σ_p gives (220). These are geometric consequences of the real projective-plane inequality of Melchior [Mel40], not merely formal identities.

4.1 The complete thirteen-point dispatch

Suppose, towards a contradiction, that $|P| = 13$ and $C \leq 60$. Let m be the largest size of a proper circle block.

Lemma 4.1 (Outer routing at thirteen points)**.** *Under the preceding assumptions,* $m \in \{5, 6\}$*. Moreover every line block has size at most six.*

Proof. If a line has s selected points and $q = 13 - s$, the rich-line pencil gives

$$q\binom{s}{2} - \binom{q}{2}\left\lfloor \frac{s}{2} \right\rfloor. \tag{221}$$

For $7 \le s \le 12$ its minimum is 66. Thus no such line occurs, while $s = 13$ is excluded by noncollinearity. For a proper m-circle, the corresponding Bonferroni pencil is

$$1 + q\left(\binom{m}{2} - \left\lfloor \frac{m}{2} \right\rfloor\right) - \binom{q}{2}\left\lfloor \frac{m}{2} \right\rfloor, \quad q = 13 - m. \tag{222}$$

For $m = 7, 8, 9, 10, 11, 12$ this equals, respectively,

$$64, 81, 105, 106, 96, 61,$$

so $m \ge 7$ is impossible. The case $m = 13$ is excluded by the standing non-concyclicity hypothesis.

It remains to exclude $m \le 4$. Besides (218) and (219), global Melchior for line blocks gives

$$\mathsf{G} := 3L_3 + \sum_{s\ge 4}\left(\binom{s}{2} + s - 3\right)L_s \le \binom{13}{2} - 3.$$

Since line sizes are at most six, the fixed combination

$$\frac{1}{4}\mathsf{T} + \frac{7}{60}\mathsf{P} - \frac{1}{5}\mathsf{G}$$

has coefficients

C_3	C_4	L_3	L_4	L_5	L_6
$3/5$	1	0	$-2/5$	$-29/60$	$0.$

It is therefore at most C, whereas its numerical lower bound is

$$\frac{1}{4}\binom{13}{3} + \frac{7}{60}(3\cdot 13) - \frac{1}{5}\left(\binom{13}{2} - 3\right) = \frac{1221}{20} > 60.$$

Thus $m \le 4$ is impossible as well. $\square$

4.1.1 The six-circle branch

Fix a six-point circle Γ and let $X = P \setminus \Gamma$, so $|X| = 7$. If $e = xy \in \binom{X}{2}$, let w_e be the number of proper circles through x, y and two points of Γ, and put

$$W = \sum_{e\in\binom{X}{2}} w_e = \sum_j \binom{j}{2} c(2, j). \tag{223}$$

Here $c(i, j)$ counts nonselected circle blocks meeting Γ in i points and X in j points; $l(i, j)$ denotes the analogous number of line blocks. For either class $Y \in \{\Gamma, X\}$, define its pivot row by

$$M_Y = \sum_B |B \cap Y|(4 - |B|), \tag{224}$$

where the sum is over all maximal blocks, including Γ. Summing the pointwise Melchior slack over $p \in Y$ gives

$$M_Y = 3|Y| + \sum_{p\in Y}\sigma_p.$$

Consequently

$$M_\Gamma \ge 18, \quad M_X \ge 21. \tag{225}$$

Lemma 4.2 (The opening inequality)**.** *If $C \le 60$, then $W \ge 48$.*

Proof. For thirteen points, (220) reads $\mathsf{D} \le 117$. Let $O = 3C_3$ be ordinary-circle incidence. For every pivot p, the inverted arrangement in corollary 2.17 has $q = 13 - 1 = 12 \ne 6$ lines. Thus at least three ordinary proper circles pass through each pivot. Summing over the thirteen pivots, and using (225), gives

$$O \ge 39, \quad M_X \ge 21.$$

Weight the four triple partitions, in order of their number of Γ-points, by

$$(141, 340, 539, 141).$$

Add $79O + 199M_X$ and subtract $1344C + 414W$. For a nonselected block write $g = |B \cap \Gamma|$, $x = |B \cap X|$, and $s = g + x$. Its coefficient is

$$\begin{aligned} A(g,x) = 141\binom{x}{3} + 340g\binom{x}{2} + 539\binom{g}{2}x + 199x(4-s) \\ + 237\mathbf{1}_{\{B \text{ is a 3-circle}\}} - 1344\mathbf{1}_{\{B \text{ is a circle}\}} \\ - 414\mathbf{1}_{\{B \text{ is a circle and } g=2\}}\binom{x}{2}. \end{aligned}$$

Direct factorisation, recorded in section 4.4, gives $A(g,x) \le 123D_B$ for every geometrically possible block, including equality at the selected circle. Since the four triple totals are $35, 126, 105, 20$, summing gives

$$123\mathsf{D} \ge 33810 - 414W. \tag{226}$$

Together with $\mathsf{D} \le 117$, this first yields $W \ge 47$.

At $W = 47$, retaining every nonnegative slack in the same calculation gives

$$R + 79(O - 39) + 199(M_X - 21) + 1344(60 - C) \le 39,$$

where every positive local residual contributing to R is at least 96. Thus all four terms vanish. For the zero-residual circle types

$$c(0,3), c(1,2), c(1,5), c(2,1), c(2,2), c(2,3),$$

write their multiplicities, in order, as (a, b, q, d, e, f). The three zero-residual line types and their multiplicities are

type	$l(0,3)$	$l(1,2)$	$l(2,1)$
multiplicity	p_0	p_1	p_2.

The circle count, ordinary-circle count, W, and the three triple partitions reduce to

$$\begin{gathered} a + b + d = 13, \quad q + e + f = 46, \quad e + 3f = 47, \quad q = 2f - 1, \\ p_0 = 45 - a - 21f, \quad p_1 = 42 - b - 20f, \quad p_2 = 11 - d + 3f. \end{gathered}$$

Consequently

$$M_X = 3a + 2b + d + 3p_0 + 2p_1 + p_2 - 10q - 3f = 240 - 123f.$$

The equality $M_X = 21$ would force $f = 73/41$, a contradiction. □

We now invoke the six-marked-conic interface lemma 3.15. In the present notation, its first clause gives $w_e \leq 3$ and at most four projective involution signatures for the *full* edges $w_e = 3$. Each signature support is a matching in X, and at most three full-signature centres lie on one trace line. A pair of support edges of the same signature is a *repetition event*. The trace dictionary lemma 3.12 and the second clause of lemma 3.15 give every event a unique maximal line-or-circle host. If that host has b outsider points, its event capacity is

$$k_0(b) = \min\left\{2\binom{b}{4}, 3\binom{\lfloor b/2 \rfloor}{2}\right\}; \quad k_0(4), k_0(5), k_0(6) = 2, 3, 9. \tag{227}$$

This is exactly the capacity (118). The host cannot be a line meeting Γ or a circle meeting Γ once. A host circle meeting Γ in two points is admissible only for the signature containing that marked pair. These assertions follow by uniqueness of the lift plane in lemma 3.12 and by the fact that two proper circles sharing three points coincide.

Lemma 4.3 (Capacity-gap lemma)**.** *Put $r = W - 48$. Then $0 \leq r \leq 6$. If R_{rep} is the number of repetition events, there is a nonnegative integer*

$$E = 2h + 3k + 9A_6 + 2\ell + 2g, \quad (h, k, A_6, \ell, g) = (c(0,4), c(0,5), c(0,6), l(0,4), c(2,4)),$$

such that

$$R_{\text{rep}} \leq E, \quad 4E \leq 6r + 7 + 8A_6, \quad A_6 = 0. \tag{228}$$

Proof. By lemma 4.2, $W \geq 48$. If F_3 denotes the number of full edges, then

$$W \leq 2\binom{7}{2} + F_3.$$

The four signature supports are matchings of size at most three, so $F_3 \leq 12$ and hence $W \leq 54$. This proves $0 \leq r \leq 6$.

Two possible host types must first be removed explicitly. Write $\lambda_5 = l(0,5)$ and $\lambda_6 = l(0,6)$. On a line of type $l(0,x)$, the second-dual residual is $x(x-3)(14-x)$, hence it is 90 at $x = 5$ and 144 at $x = 6$. Before forced-zero variables are suppressed, the full second added-centre slack inequality therefore contains

$$90\lambda_5 + 144\lambda_6 + \text{other nonnegative terms} \leq 12 + 6r \leq 48.$$

Hence $\lambda_5 = \lambda_6 = 0$. Host uniqueness and admissibility now leave only

$$c(0,4),\ c(0,5),\ c(0,6),\ l(0,4),\ c(2,4).$$

Their capacities from (227) are respectively $2, 3, 9, 2, 2$, so this exhaustive list gives $R_{\text{rep}} \leq E$.

For the numerical inequality, introduce

$$\begin{gathered} y = c(1,3),\ z = c(1,4),\ f = c(2,3), \quad w = l(1,3),\ t = l(2,2),\ p = l(2,3), \\ d = 60 - C, \quad j = C_3 - 13, \quad u = M_X - 21, \quad v = M_\Gamma - 18, \end{gathered}$$

all nonnegative integers. The second added-centre calculation has the exact retained-slack budget

$$12h + 26k + 36A_6 + 3y + 5z + 2f + 40\ell + 31w + 22t + 48p + 6u + 3v + 36d \leq 12 + 6r. \tag{229}$$

Let $N = j + y + z$, $a = y + z$, and define

$$K_0 = N + 2u + v + 4h + 8k + 12A_6 + 10\ell + 8d + 4t + 7w,$$
$$Q_0 = N + u + 4h + 8k + 12A_6 + 8\ell + 8d + 2t + 5w + 7p.$$

The eight conservation rows displayed in section 4.4 give an integer $\mu \geq 0$ and $\delta = 7\mu - 4j$ satisfying

$$K_0 + 7j - 2r = 14\mu, \qquad Q_0 + 4\mu - 2j \leq 3 + 2r, \tag{230}$$
$$3\delta \leq 6 - k - 5\ell - 6d - 5t - 5w - 24p. \tag{231}$$

They also give, with

$$J = 21 + 18r - (24g + 24h + 36k + 84A_6 + 24\ell),$$

the exact identity

$$7J = 49f + 12j - 60\delta + 14a + 7y + 63u + 14v + 105k + 168\ell + 252d - 161p + 42t + 189w. \tag{232}$$

Its right side is nonnegative. If $p > 0$, (231) gives

$$\delta \leq 2 - 8p, \quad -60\delta - 161p \geq 319p - 120 \geq 199 > 0.$$

Thus the only apparently negative pair of terms is already positive before the remaining nonnegative terms are restored. If $p = 0$ and $\delta \leq 0$, this is immediate. For $\delta = 1$, the congruence $\delta = 7\mu - 4j$ gives $j \equiv 5 \pmod 7$, hence $12j - 60 \geq 0$. For $\delta = 2$, (231) forces $k = \ell = d = t = w = 0$ and $j \equiv 3 \pmod 7$. If $j \geq 10$, then $12j - 120 \geq 0$. For $j = 3$ one has $\mu = 2$, and (230) gives

$$a + u + 4h + 12A_6 \leq 2r - 2.$$

The two eliminated conservation identities then read $v \geq 6 - u$ and

$$f = (y - a) - 2(v - (6 - u)).$$

Since $f \geq 0$ and $y \leq a$, equality holds throughout: $f = 0$, $y = a$, and $v = 6 - u$. The right side of (232) is therefore $21a + 49u \geq 0$. Hence $J \geq 0$ in every case.

Since

$$12E - 24A_6 = 24g + 24h + 36k + 84A_6 + 24\ell,$$

$J \geq 0$ is precisely the middle inequality of (228).

It remains to remove A_6. A first fixed added-centre row gives

$$2388A_6 \leq 453 + 414r,$$

and hence $A_6 = 0$ for $0 \leq r \leq 4$. For $r = 5, 6$, the number of full edges is at least $6 + r \geq 11$. Four matching supports, each of size at most three, must therefore all have size at least two. If a six-outsider circle existed, every other event host would share at least three of its six outsider points and hence would coincide with it. All four signature centres would then lie on its trace line, contradicting the three-centre cap. Thus $A_6 = 0$ also for $r = 5, 6$. □

The number F_3 of full outsider edges satisfies $W \leq 2\binom{7}{2} + F_3$, so $F_3 \geq 6 + r$. Balancing these edges among four matching supports of capacity three gives

$$R_{\mathrm{rep}} \geq \rho(r), \qquad \begin{array}{c|ccccccc} r & 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline \rho(r) & 2 & 3 & 4 & 6 & 8 & 10 & 12. \end{array} \tag{233}$$

Together, (228) and (233) immediately exclude $r = 0, 4, 5, 6$. At $r = 1$, (229) forces $k = A_6 = \ell = 0$, so $E = 2(h + g)$ is even; the capacity gap gives $E \leq 3$, and hence $E \leq 2 < \rho(1)$. Only $r = 2, 3$ remain.

Lemma 4.4 (Three-residue terminal)**.** *The equality layers $r = 2$ and $r = 3$ are empty.*

Proof. For $r = 2$, (233) and the capacity gap give

$$4 \le E \le \left\lfloor \frac{19}{4} \right\rfloor = 4;$$

for $r = 3$ they give

$$6 \le E \le \left\lfloor \frac{25}{4} \right\rfloor = 6.$$

Thus $E = 4, 6$, respectively. Since $A_6 = 0$, the definition of J becomes $J = 21 + 18r - 12E$, and hence $J = 9$ for $r = 2$ and $J = 3$ for $r = 3$.

We next justify the reduction of variables. At $r = 2$, the right side of (229) is 24; its coefficients force

$$k = \ell = d = w = p = 0.$$

At $r = 3$, its right side is 30, so $\ell = d = w = p = 0$ and $k \le 1$. But here

$$E - 3k = 2(h + g)$$

is even, whereas $E = 6$; hence k cannot equal one and again $k = 0$. Together with $A_6 = 0$, this leaves precisely the nonnegative integers $a, y, u, v, h, t, f, j, \mu$, with $0 \le y \le a$. They satisfy

$$\begin{aligned}
\delta &= 7\mu - 4j, && (234)\\
v &= 2\delta + 2r - a - 2u - 4h - 4t, &&\\
f &= 2j + \mu + y - v + 4h - 5t - 2r, && (235)\\
a + u + 4h + 2t + 4\mu - j &\le 3 + 2r, && (236)\\
J &= 7a + 48h + 90j - 123\mu - 24r - 9t + 19u + 8y. && (237)
\end{aligned}$$

The capacity inequality gives $\delta \le 2$. Negative δ is impossible: for $r = 3$, the term -60δ in the uneliminated identity (232) already exceeds $7J = 21$; for $r = 2$, the value $\delta = -1$ forces $j \equiv 2 \pmod 7$, and then $-60\delta + 12j \ge 84 > 63 = 7J$. More negative values only increase this lower bound. Thus $\delta \in \{0, 1, 2\}$.

Moreover,

$$4\mu - j = \frac{4\delta + 9j}{7}.$$

Dropping the other nonnegative terms in (236) gives $4\delta + 9j \le 49$ for $r = 2$ and at most 63 for $r = 3$. The congruence $j \equiv 5\delta \pmod 7$ therefore leaves

$$(\delta, j, \mu) \in \{(0, 0, 0), (1, 5, 3), (2, 3, 2)\}. \tag{238}$$

For completeness, the sole apparent extra triple is $(0, 7, 4)$ when $r = 3$, but then the nonnegative right side of (232) contains $12j = 84 > 21$.

For $(0, 0, 0)$, the two inequalities $v, f \ge 0$ give

$$a + 2u + 4h + 4t \le 2r, \quad a + y + 2u + 8h - t \ge 4r.$$

Since $y \le a$, subtraction yields

$$2r \le y + 4h - 5t \le a + 4h - 5t \le 2r - 2u - 9t.$$

Hence $u = t = 0$, $y = a = 2r - 4h$, and $J = 6r - 12h$. The pairs $(r, J) = (2, 9), (3, 3)$ would respectively make $h = 1/4, 5/4$.

For $(1,5,3)$, the two relevant rows become

$$\begin{aligned} a+u+4h+2t &\leq 2r-4, \\ J+24r-81 &= 7a+48h-9t+19u+8y. \end{aligned}$$

When $r=2$, the budget makes every variable zero, whereas the left side of the identity is -24. When $r=3$, the budget is two: for $t=0$ the right side is nonnegative, and for $t=1$ it equals -9, while the left side equals -6.

Finally, for $(2,3,2)$,

$$\begin{aligned} a+u+4h+2t &\leq 2r-2, \\ J+24r-24 &= 7a+48h-9t+19u+8y. \end{aligned}$$

For $r=2$ the left side is 33 and the budget is two. Thus $h=0$. If $t=1$, the right side is -9; if $t=0$, parity forces $a+u=1$, and the right side is at most 19.

For $r=3$ the left side is 51 and the budget is four. The value $h=1$ gives 48, while $t=2$ gives -18 and $t=1$ gives at most 29. Thus $h=t=0$. Parity leaves $a+u \in \{1,3\}$. The first value gives at most 19; for the second, substituting $u=3-a$ gives $4(2y-3a)=-6$. This is impossible. ☐

Proposition 4.5. *If a thirteen-point set has a largest proper circle of size six, then $C \geq 61$.*

Proof. Assume $C \leq 60$. The capacity-gap lemma places $r = W-48$ in $\{0,\ldots,6\}$ and excludes $r=0,1,4,5,6$, while lemma 4.4 excludes $r=2,3$. ☐

It remains to close the five-circle value left by lemma 4.1.

Lemma 4.6 (The five-circle transfer)**.** *If a thirteen-point set has no proper circle larger than five, then $C \geq 61$.*

Proof. Assume $C \leq 60$. By (221), every line has size at most six, so only B_3, B_4, B_5, B_6 occur and $B_6 = L_6$. Set

$$\begin{aligned} K &= \sum_{p\in P} \sigma_p, \quad Q = 3B_3 - 5B_5 - 12B_6 = 39+K, \\ S &= 3L_3 + 4L_4 + 10L_5 + 12L_6, \\ A &= 5L_3 + 12L_4 + 20L_5 + 28L_6. \end{aligned}$$

The triple partition and the two applications of Melchior described above give

$$B_3 + 4B_4 + 10B_5 + 20B_6 = 286, \tag{239}$$

$$12B_4 + 35B_5 + 72B_6 = 819 - K, \tag{240}$$

$$A \leq 4C - 169 + C_5, \quad A \geq \frac{5}{3}S, \tag{241}$$

$$24 \cdot 286 + 27Q \leq 35(S+3C). \tag{242}$$

The last inequality is coefficientwise:

s	3	4	5	6
left side	105	96	105	156
right side, circle	105	105	105	$--$
right side, line	105	140	350	420.

The dash records $C_6 = 0$ in this branch. Thus, on every type that can occur, no separation or optimisation theorem is hidden in (242).

From (240), $C_5 \le B_5 \le 23$. Equations (241)–(242) now give

$$377 - 5C \le A \le 4C - 146,$$

and hence $C \ge 523/9$, so only $C = 59, 60$ remain.

For these two endpoints let $L = \sum_{s=3}^{6} L_s$. Exact elimination from (239)–(241) gives

$$4L = 325 - 4C - B_5 - 4B_6 + K, \tag{243}$$

$$9B_5 \ge 2301 - 36C + 5K + 72B_6. \tag{244}$$

Combining (240), $B_4 \ge 0$, and (244) yields

$$\begin{aligned} C = 59 &: \quad 23K + 396B_6 \le 147, \\ C = 60 &: \quad 46K + 792B_6 \le 609. \end{aligned}$$

Thus $B_6 = 0$. Reducing (240) modulo 12, then using (244) and $B_4 \ge 0$, gives exactly

$$(B_3, B_4, B_5, B_6) = (48 + 2K, 7 - 3K, 21 + K, 0), \quad K \in \{0, 1, 2\}. \tag{245}$$

At every point p, pair partition and the definition of σ_p give

$$7d_5(p) + 3d_4(p) = 63 - \sigma_p. \tag{246}$$

If $K = 0$, (246) and $d_4(p) \le B_4 = 7$ force $d_4(p) \in \{0, 7\}$. Since $\sum_p d_4(p) = 4B_4 = 28$, four points lie in all seven four-blocks, making two distinct maximal blocks share four points. If $K = 1$, twelve zero-slack points have $(d_5, d_4) = (9, 0)$ and the remaining point has $(8, 2)$; hence $\sum_p d_5(p) = 116$, rather than $5B_5 = 110$. Finally, if $K = 2$, any point with positive slack must have $d_4(p) \ge 2$, while (245) gives $B_4 = 1$. All three cases are impossible. □

Theorem 4.7. *The thirteen-point value is*

$$f(13) = 61.$$

Proof. For a hypothetical set with $C \le 60$, lemma 4.1 leaves only $m = 5$ and $m = 6$. These are excluded by lemma 4.6 and proposition 4.5, respectively. Hence $C \ge 61 = F(13)$. The circle-with-centre construction from the foundational section has exactly $F(13)$ proper circles, proving equality. □

4.2 A uniform half-cap and one master for $n \ge 15$

Lemma 4.8 (Uniform half-cap)**.** *Let $n \ge 15$ and suppose $C \le F(n) - 1$. Every maximal line or proper circle block then has size at most*

$$H = \left\lfloor \frac{n}{2} \right\rfloor.$$

Proof. Let a block have size s, put $q = n - s \ge 1$, and write $h = \lfloor s/2 \rfloor$. The rich-circle and rich-line pencils are

$$R(n, s) = 1 + q\left(\binom{s}{2} - h\right) - \binom{q}{2}h, \tag{247}$$

$$Q(n, s) = q\binom{s}{2} - \binom{q}{2}h = R(n, s) - 1 + qh. \tag{248}$$

We show $R(n,s) \geq F(n)$ whenever $s \geq H+1$. The following four rows cover all parity choices; in each row $c \geq 0$.

s	q	parameter relation	$R(n,s) - F(n)$
$2a$	$2b$	$a = b+1+c$	$(4b-2)c^2 + (6b^2-3b)c + 2b^3 - 5b^2 + b$
$2a$	$2b+1$	$a = b+1+c$	$b\{4c^2 + (6b-1)c + 2b^2 - 5b - 3\}$
$2a+1$	$2b$	$a = b+c$	$(4b-2)c^2 + (6b^2-7b+2)c + 2b^3 - 7b^2 + 4b$
$2a+1$	$2b+1$	$a = b+1+c$	$b\{4c^2 + (6b+3)c + 2b^2 - b - 1\}$.

(*)

The condition $n \geq 15$ is, row by row, $2b + c \geq 7, 6, 7, 6$. For $b \geq 3$ all relevant coefficients in the first three rows are nonnegative, and the fourth row is nonnegative for every $b \geq 1$. The remaining possibilities reduce to

	$b=1$	$b=2$
row 1	$2c^2 + 3c - 2$ $(c \geq 5)$	$6c^2 + 18c - 2$ $(c \geq 3)$
row 2	$4c^2 + 5c - 6$ $(c \geq 4)$	$2(4c^2 + 11c - 5)$ $(c \geq 2)$
row 3	$2c^2 + c - 1$ $(c \geq 5)$	$6c^2 + 12c - 4$ $(c \geq 3)$,

all strictly positive. In the two odd-q rows, $b = 0$ gives equality $R = F$. Thus a circle of size $s \geq H+1$ already determines at least $F(n)$ circles. Equation (248) and $qh \geq 1$ give the same conclusion for a line. Either conclusion contradicts $C \leq F(n) - 1$. □

Under the cap $s \leq M$, inversion leaves at most $M-1$ collinear images. For $M = \lfloor n/2 \rfloor$,

$$M - 1 \leq \frac{2(n-1)}{3}.$$

The Langer–de Zeeuw inequality theorem 2.12, applied after each inversion and summed over all centres, therefore gives

$$\mathsf{L} := \sum_{s \geq 3} s(s-1)B_s \geq \frac{n(n-1)(n+2)}{3}. \tag{249}$$

Lemma 4.9 (The S_M master)**.** *Assume every maximal block has size at most $M \geq 3$ and that (249) holds. Then*

$$C \geq A(n,M) := \frac{\binom{n}{3}}{2M} + \frac{n(M+3)}{6M} + \frac{(M-2)n(n-1)(n+2)}{36M} - \frac{n(n-4)}{9}. \tag{250}$$

Proof. Consider the single functional

$$\mathsf{S}_M = \frac{\mathsf{T}}{2M} + \frac{M+3}{18M}\mathsf{P} + \frac{M-2}{12M}\mathsf{L} - \frac{1}{9}\mathsf{D}. \tag{251}$$

If $k_C(s)$ and $k_L(s)$ are its coefficients on an s-circle and an s-line, respectively, direct factorisation gives

$$1 - k_C(s) = \frac{(M-s)(s-4)(s-3)}{12M}, \tag{252}$$

$$k_L(s) = \frac{s(s-3)(-7M+3s-12)}{36M}. \tag{253}$$

Thus $k_C(3) = k_C(4) = 1$, $k_C(s) \leq 1$ for $5 \leq s \leq M$, and $k_L(s) \leq 0$ for every $3 \leq s \leq M$. Hence $\mathsf{S}_M \leq C$. Substitution of (218), (219), (220), and (249), with attention to the negative coefficient of D, gives (250). □

Taking $M = \lfloor n/2 \rfloor$ in (250) yields

$$A(2r, r) - F(2r) = \frac{(2r-1)(r^2 - 9r + 13)}{9}, \tag{254}$$

$$A(2r+1, r) - F(2r+1) = \frac{4r^4 - 32r^3 + 37r^2 + 9}{18r}. \tag{255}$$

The first is positive for $r \geq 8$; after $r = 8 + t$ its quadratic factor is $t^2 + 7t + 5$. In the second, after $r = 7 + t$, the numerator is

$$4t^4 + 80t^3 + 541t^2 + 1302t + 450,$$

so it is positive for $r \geq 7$. Together with lemmas 4.8 and 4.9, this proves $C \geq F(n)$ for every $n \geq 15$.

4.3 The exact dichotomy at fourteen points

Proposition 4.10 (Seven-circle dichotomy)**.** *Every admissible fourteen-point set determines at least* 73 *proper circles.*

Proof. Assume $C \leq 72$. The pencil bounds (247)–(248) show that proper circle blocks have size at most seven and line blocks have size at most six. If $C_7 = 0$, all blocks have size at most six. Every inverted line then contains at most five of the thirteen image points, and $5 \leq 2 \cdot 13/3$, so theorem 2.12 supplies (249). The S_M master with $M = 6$ gives

$$C \geq A(14, 6) = \frac{3899}{54} > 72.$$

It remains to consider a selected seven-circle Γ with seven-point complement X. For every block put $g = |B \cap \Gamma|$, $x = |B \cap X|$, and $s = g + x$, and define

$$W = \sum_{\substack{B \text{ a proper circle} \\ g=2}} \binom{x}{2}. \tag{256}$$

The four triple totals according to their number of Γ-points are $\binom{7}{i}\binom{7}{3-i}$. Put

$$M_\Gamma = \sum_B g(4-s), \quad M_X = \sum_B x(4-s),$$

where the sums include the selected block. Summed pivot Melchior over each of the two seven-point classes gives $M_\Gamma, M_X \geq 21$, while (220) gives $\mathsf{D} \leq 140$.

For every block set

$$A_B = 6\binom{x}{3} + 9g\binom{x}{2} + 12\binom{g}{2}x + 6\binom{g}{3} + (3g + 6x)(4 - s)$$
$$- 36\mathbf{1}_{\{B \text{ is a circle}\}} - 6\mathbf{1}_{\{B \text{ is a circle and } g=2\}}\binom{x}{2}.$$

For a nonselected block one has $g \leq 2$, and the residual $4D_B - A_B$ factors as

	$g=0$	$g=1$	$g=2$
circle	$(x-3)(-x^2+10x-12)$	$(5-x)(x-2)(2x-3)/2$	$(4-x)(x-2)(x-1)$
line	$x(14-x)(x-3)$	$(x-2)(-2x^2+21x+17)/2$	$(x-1)(-x^2+7x+12)$.

(257)

Under the block caps every line residual is nonnegative; a negative circle residual has magnitude at most $3\binom{x}{3}$. Outsider triples have unique owners, so the total correction is at most $3\binom{7}{3}$. The selected seven-circle contributes the remaining defect 27. Summing the block inequality and using $C \le 72$ gives

$$\begin{aligned} 6W \ge {} & 3\binom{7}{3} + 9 \cdot 7\binom{7}{2} + 12\binom{7}{2} \cdot 7 + 6\binom{7}{3} + 3 \cdot 21 + 6 \cdot 21 \\ & - 36 \cdot 72 - 4 \cdot 140 - 27 = 412. \end{aligned}$$

Hence $W \ge 69$. On the other hand, for a fixed outsider pair the Γ-pairs used by circles counted in W are disjoint. It contributes at most $\lfloor 7/2 \rfloor = 3$, and therefore

$$W \le 3\binom{7}{2} = 63,$$

a contradiction. □

Theorem 4.11. *For every integer $n \ge 14$,*

$$f(n) = F(n) = 1 + \binom{n-1}{2} - \left\lfloor \frac{n-1}{2} \right\rfloor.$$

Proof. The lower bound for $n = 14$ is proposition 4.10. For $n \ge 15$, lemma 4.8 supplies the common block cap, Langer supplies (249), and the one functional in lemma 4.9 gives the strict contradictions (254) and (255) under $C \le F(n) - 1$. The foundational circle-with-centre construction attains $F(n)$ for every $n \ge 14$. □

4.4 Fixed-certificate appendix and verification boundary

For clarity, we record which parts of this section are fixed arithmetic certificates and which parts are mathematical inputs.

The opening certificate. In lemma 4.2, the residual $123D_B - A(g, x)$ has the following line residuals:

$$\begin{aligned} R_L(0, x) &= \frac{x(x-3)(890-47x)}{2}, \\ R_L(1, x) &= \frac{(x-2)(-47x^2+597x+246)}{2}, \\ R_L(2, x) &= \frac{x(x-1)(304-47x)}{2}. \end{aligned}$$

The circle residuals are

$$\begin{aligned} R_C(0, x) &= 237 + \frac{(x-3)(-47x^2+644x-738)}{2} - 237\mathbf{1}_{\{x=3\}}, \\ R_C(1, x) &= \frac{(5-x)(47x^2-210x+390)}{2} - 237\mathbf{1}_{\{x=2\}}, \\ R_C(2, x) &= \frac{(x-2)(x-3)(284-47x)}{2} - 237\mathbf{1}_{\{x=1\}}. \end{aligned}$$

It is enough—and slightly stronger than the outer cap requires—to use the intervals

	$g = 0$	$g = 1$	$g = 2$
line	$3 \le x \le 7$	$2 \le x \le 6$	$1 \le x \le 5$
circle	$3 \le x \le 6$	$2 \le x \le 5$	$1 \le x \le 4$.

Each displayed expression is nonnegative on its indicated interval. Thus the opening is a coefficientwise proof, not a conclusion inferred from sampled incidence patterns.

Affine certificate. For reference, retain the zero- and positive-slack variables

$$c_{03}, c_{12}, c_{15}, c_{21}, c_{22}, l_{03}, l_{12}, l_{21}$$

in addition to the variables in lemma 4.3. With every row written as left side minus right side, the eight conservation equations are

$$\begin{aligned}
R_C &= 1 + c_{03} + h + k + A_6 + c_{12} + y + z + c_{15} + c_{21} + c_{22} + f + g - (60 - d),\\
R_O &= c_{03} + c_{12} + c_{21} - (13 + j),\\
R_0 &= c_{03} + 4h + 10k + 20A_6 + y + 4z + 10c_{15} + f + 4g + l_{03} + 4\ell + w + p - 35,\\
R_1 &= c_{12} + 3y + 6z + 10c_{15} + 2c_{22} + 6f + 12g + l_{12} + 3w + 2t + 6p - 126,\\
R_2 &= c_{21} + 2c_{22} + 3f + 4g + l_{21} + 2t + 3p - 105,\\
R_W &= c_{22} + 3f + 6g - (48 + r),\\
R_X &= 3c_{03} - 5k - 12A_6 + 2c_{12} - 4z - 10c_{15} + c_{21} - 3f - 8g\\
&\qquad + 3l_{03} + 2l_{12} + l_{21} - 3p - (21 + u),\\
R_\Gamma &= -12 + c_{12} - z - 2c_{15} + 2c_{21} - 2f - 4g + l_{12} + 2l_{21} - 2p - (18 + v).
\end{aligned}$$

All eight equal zero.

The second-dual inequality. For a nonselected block B, put $g = |B \cap \Gamma|$, $x = |B \cap X|$ and $s = g + x$. Let $D_B = s(s-4)$ for a circle and $D_B = 2s(s-2)$ for a line. Weight the four triple rows by $6, 9, 12, 6$, add $6M_X + 3M_\Gamma$, and subtract $36C + 6W$. The coefficient of B is

$$\begin{aligned}
A_2(g,x) = 6\binom{x}{3} + 9g\binom{x}{2} + 12\binom{g}{2}x + (6x+3g)(4-g-x)\\
- 36\mathbf{1}_{\{B \text{ is a circle}\}} - 6\mathbf{1}_{\{B \text{ is a circle and } g=2\}}\binom{x}{2}.
\end{aligned}$$

For $g = 0, 1, 2$, the residual $4D_B - A_2(g,x)$ is

	$g=0$	$g=1$	$g=2$
line	$x(x-3)(14-x)$	$(x-2)(-2x^2+21x+17)/2$	$(x-1)(-x^2+7x+12)$
circle	$(x-3)(-x^2+10x-12)$	$(5-x)(x-2)(2x-3)/2$	$(4-x)(x-2)(x-1).$

The line ranges are respectively $3 \le x \le 7$, $2 \le x \le 6$, $1 \le x \le 5$, and the circle ranges are $3 \le x \le 6$, $2 \le x \le 5$, $1 \le x \le 4$. Every displayed factor is nonnegative; the selected circle separately has equality $48 = 4 \cdot 12$. Summing and using $\mathsf{D} \le 117$ gives exactly

$$\begin{aligned}
B_0 := 12h + 26k + 36A_6 + 3y + 5z + 2f + 40\ell + 31w + 22t + 48p\\
+ 6u + 3v + 36d - (12 + 6r) \le 0. \qquad (258)
\end{aligned}$$

The zero-residual types are precisely $c_{03}, c_{12}, c_{15}, c_{21}, c_{22}, l_{03}, l_{12}, l_{21}$ and $g = c(2,4)$; every omitted type has nonnegative residual and may safely be discarded from the retained budget. This proves the budget used in (229) coefficientwise.

The integral coordinate. In the row order $(R_C, R_O, R_0, R_1, R_2, R_W, R_X, R_\Gamma)$, define

$$\begin{aligned}
F_0 &= 14f - \big(180A_6 + 24\ell - 30r + 36d + 60h + 36j + 106k - 140p\\
&\qquad - 66t + 2u - 13v - 21w + 15y + z\big),\\
G_0 &= 336g - \big(294 + 342r - 1716A_6 - 240\ell - 276d - 516h - 276j\\
&\qquad - 976k + 1302p + 618t - 20u + 123v + 189w - 87y + 25z\big).
\end{aligned}$$

Their fixed row certificates are

$$F_0 = -36R_C + 36R_O - 6R_0 + 9R_1 + 24R_2 - 30R_W + 2R_X - 13R_\Gamma = 0, \tag{259}$$
$$G_0 = 276R_C - 276R_O + 60R_0 - 83R_1 - 226R_2 + 342R_W - 20R_X + 123R_\Gamma = 0. \tag{260}$$

Put $N = j + y + z$ and

$$\begin{aligned} B_1 &= 9N + 36h + 72k + 108A_6 + 76\ell + 49w + 22t + 49p \\ &\quad + 72d + 11u + 2v - (21 + 18r), \\ H &= 2j + y - v + 4h + 7k + 12A_6 + \ell + 2d - 10p - 5t - 2w - 2r. \end{aligned}$$

The right side subtracted from $14f$ in (259) has the exact decomposition

$$(K_0 + 7j - 2r) + 14H.$$

Consequently

$$K_0 + 7j - 2r = 14(f - H).$$

Set $\mu = f - H$ and $E_\mu = K_0 + 7j - 2r - 14\mu$. Then μ is an integer, $E_\mu = 0$, and $\mu \geq 0$: indeed the left side above is at least $-2r \geq -12$, so it cannot be a negative multiple of 14.

The remaining budget transformations are the fixed identities

$$7B_0 - 4B_1 - F_0 = 0, \tag{261}$$
$$B_1 + 21 + 18r = 2K_0 + 7Q_0, \tag{262}$$
$$B_1 = 7\{Q_0 + 4\mu - 2j - (3 + 2r)\} + 2E_\mu. \tag{263}$$

Thus $B_0 \leq 0$ and $F_0 = E_\mu = 0$ imply $B_1 \leq 0$ and precisely the two relations in (230).

The δ and J certificates. Let

$$\delta = 7\mu - 4j, \qquad L = 3 + 2r - (Q_0 + 4\mu - 2j) \geq 0.$$

Direct substitution in the displayed definitions gives the fixed identity

$$\begin{aligned} S &:= 6 - k - 5\ell - 6d - 5t - 5w - 24p - 21\mu + 12j \\ &= f + 2L + z \geq 0, \end{aligned} \tag{264}$$

and hence

$$3\delta = 6 - k - 5\ell - 6d - 5t - 5w - 24p - S.$$

This is (231). The same substitution gives the two exact equalities

$$v = 2\delta + 2r - a - 2u - 4h - 8k - 12A_6 - 10\ell - 8d - 4t - 7w, \tag{265}$$
$$f = 2j + \mu + y - v + 4h + 7k + 12A_6 + \ell + 2d - 10p - 5t - 2w - 2r. \tag{266}$$

For the last elimination set $G = 24g$ and

$$\begin{aligned} G_* &= 21 + 42r + 123\mu - 228A_6 - 105\ell - 90d - 72h - 83j - 140k \\ &\quad + 93p + 9t - 19u - 48w - 8y - 7N. \end{aligned}$$

Then the first fixed certificate is

$$G_0 - 14(G - G_*) + 123E_\mu = 0, \tag{267}$$

so $G = G_*$. Let $E_\delta = \delta - (7\mu - 4j)$, let E_v be the left side minus the right side of (265), and put

$$E_{7f} = 7f - \bigl(18j + 15\delta + 7(y-a) - 14v - 7k - 63\ell - 42d \\ - 70p - 63t - 14u - 63w\bigr).$$

Equation (266) gives $E_{7f} = 0$. Finally, let E_J be $7J$ minus the right side of (232). The terminal fixed identity is

$$E_J + 7(G - G_*) - 84E_v + 7E_{7f} - 123E_\delta = 0. \tag{268}$$

Every term except E_J has just been shown to vanish, proving (232). This explicit chain establishes (230)–(232); no multiplier is chosen by optimisation or incidence search.

What was checked arithmetically. During preparation, exact-arithmetic checks independently replayed the opening residuals, the $W = 47$ identities, all eight affine rows and their fixed eliminations, the seven values of ρ, the capacity-gap conversion, the three-residue calculations, the four half-cap polynomials, the coefficients of S_M, and the fourteen-point residual table. These checks manipulate integers, rationals, and symbolic polynomials only.

Such arithmetic checks do not prove Melchior's theorem, Langer's inequality, the pivot form of Lenchner's theorem [Len07], the inversion dictionary, or the real six-marked-conic signature and host lemmas. Those are the stated geometric inputs to the written proofs. The classification-free Γ_5 transfer in lemma 4.6 is checked directly through its displayed blockwise coefficients and endpoint equations. The distributed exact-arithmetic supplement `supplement/verify_n13_gamma5_global_closure.py` checks an independent global closure of that branch. In every case the arithmetic is corroborative, not a substitute for the theorem-level arguments above.

Proof of theorem 1.1. The exact values for $4 \le n \le 9$ are theorem 2.28. The values for $n = 10, 11, 12$ are theorem 3.46, and the value at $n = 13$ is theorem 4.7. Finally, theorem 4.11 proves the stated formula for every $n \ge 14$. The constructions in section 2 give equality in every case. □

Reproducibility statement

The distributed supplementary checker uses exact integer, rational, and symbolic arithmetic. Its logical role is secondary verification of one independent closure whose direct proof is already printed in the manuscript. No optimizer, satisfiability result, configuration catalogue, support search, or orbit search is a premise of the proof. This supplementary replay is not a Lean formalization.

References

[dZ18] Frank de Zeeuw. Spanned lines and Langer's inequality. arXiv:1802.08015, 2018.

[Ell67] P. D. T. A. Elliott. On the number of circles determined by n points. *Acta Math. Acad. Sci. Hungar.*, 18:181–188, 1967.

[Erd61] Paul Erdős. Some unsolved problems. *Magyar Tud. Akad. Mat. Kutató Int. Közl.*, 6(1–2):221–254, 1961.

[Hal35] Philip Hall. On representatives of subsets. *J. London Math. Soc.*, s1-10(1):26–30, 1935.

[KM58] L. M. Kelly and W. O. J. Moser. On the number of ordinary lines determined by n points. *Canad. J. Math.*, 10:210–219, 1958.

[Lan03] Adrian Langer. Logarithmic orbifold euler numbers of surfaces with applications. *Proc. London Math. Soc.*, 86:358–396, 2003.

[Len07] Jonathan Lenchner. An improved bound for the affine Sylvester problem. In *Proceedings of the 19th Canadian Conference on Computational Geometry*, pages 57–60, 2007.

[LMM+18] Aaron Lin, Mehdi Makhul, Hossein Nassajian Mojarrad, Josef Schicho, Konrad Swanepoel, and Frank de Zeeuw. On sets defining few ordinary circles. *Discrete Comput. Geom.*, 59:59–87, 2018.

[Mel40] E. Melchior. über vielseite der projektiven ebene. *Deutsche Mathematik*, 5:461–475, 1940.

[PS10] George B. Purdy and Justin W. Smith. Lines, circles, planes and spheres. *Discrete Comput. Geom.*, 44:860–882, 2010.

[Ung82] Peter Ungar. $2N$ Noncollinear Points Determine at Least $2N$ Directions. *J. Combin. Theory Ser. A*, 33:343–347, 1982.